

Normalization by the Matching Number for Weighted Matching-Support Polynomials

Small ranks, unit and one-shore counterexamples, rank six, and eventual scaling

Research notes prepared for Vladimir Reshetnikov with OpenAI
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Abstract

For a finite bipartite graph G with positive vertex activities, the matching-support polynomial $p_G(t) = \sum_k a_k t^k$ counts each pair of endpoint sets of a k -edge matching once, weighted by the product of its activities. Its degree is the matching number $\nu(G)$. This report collects batch-72 research notes on the question of *preorder-root-polytopes* (Parts XI–XIII) whether p_G is ultra-log-concave of order $\nu(G)$. Part I (sources 74, 72, 76 and 75) proves it for arbitrary positive activities on both shores whenever some minimum vertex cover meets one shore in at most two vertices, through an explicit transversal-matroid lift; in particular for every graph with $\nu(G) \leq 5$. With the rank-six Hall counterexample of *preorder-root-polytopes*, the least matching number of a weighted failure is exactly six, and there are failures at every larger matching number. Equality in any one normalized inequality occurs only for balanced star forests; graphs with a vertex cover of two vertices on each shore have real-stable basis polynomials and real-rooted support polynomials; and the full Hall family is ultra-log-concave for all activities exactly when one core has at most two vertices. Part II (sources 71, 73, 55, 47 and 51) answers the unit-weight and one-shore questions of *preorder-root-polytopes* negatively: a connected all-unit graph with 6916 vertices and matching number 3450 fails, and so does a 125-vertex graph whose left activities are all one and whose right activities are one except for activity 10000 on its 42 right-core vertices; for one common activity on the right core of a full Hall graph the least failing matching number is exactly 38. Nothing here is formalized in Lean or Rocq, and nothing is refereed.

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1 Guide to this report

1.1 Sources, status and plan

This report is a merge of research manuscripts delivered to ProveIt’s incoming-report intake on 1 October 2026 (batch 72, arrival commit 1512ef835, placement commit 8bb543f0f). The sources are numbered by their batch-72 manuscript numbers. Twenty-six archives and one manuscript delivered only inside archive 08 (called 08s here) are sources; a twenty-eighth archive (78) is an exact duplicate of the leaf-compression manuscript already printed as Part XIII of *preorder-root-polytopes* and is not a source. The full record, archive by archive, is Appendix A. At this stage of the write-up the report prints Parts I and II (nine sources); Parts III and IV, built from the other eighteen sources, are added in the following commits of the same write.

Every source is printed in full: statements, proofs, examples, remarks, limitations and verification notes. A result proved by several sources is printed once, under the source named in the Part’s introduction, and the other proofs follow as marked second routes. A result that *preorder-root-polytopes* already proves is cited by its number there and is not presented as new; where a source re-proves it, the source’s proof is either printed as a marked second route or replaced by a pointer, as recorded at the place and in Appendix A.

Status. All sources are AI-assisted research notes (their title pages read “Research note prepared for Vladimir Reshetnikov with OpenAI”, except source 72’s “AI-assisted mathematical research”). Their proofs are ordinary mathematical arguments supported by exact finite checks; they are not refereed, and nothing in them is formalized in Lean or Rocq. No source claims global priority. Text in square brackets marked *Merge note* was added in this merge; everything else is the sources’ text, with labels prefixed and section levels shifted (Section 1.5).

1.2 What *preorder-root-polytopes* already proves

The neighbouring report `SetTheory/Cardinals/docs/reports/enumerative-combinatorics/preorder-root-polytopes` ([8], cited here as *preorder-root-polytopes*) introduced the question and proves the following. They are cited by their printed numbers and labels in that report and are not reprinted here.

- Part V: Theorem 70.1 (`lor:thm:weightedULC`), the two-shore inequality: p_G is $\text{ULC}_{\min(|X|,|Y|)}$ for arbitrary positive activities (normalization by the smaller shore); its Research question 37 (`lor:q:first`) asks for normalization by the matching number.
- Part VI: the stability tools, Theorem 85.3 (`ade:mat:thm:stable-bridge`, the reciprocal bridge between the signed support polynomial and the basis polynomial), Theorem 91.2 (`ade:thm:expected`, independent-column expectations are real stable), Theorem 93.2 (`ade:mat:thm:vertex-stable`, stability under one-vertex sums), Theorem 94.1 (`ade:mat:thm:biclique`, complete bipartite blocks are stable) and Corollary 94.3 (`ade:mat:cor:block-family`).
- Part XI (Hall bottlenecks): the family $H_{s,n}$ and its exact coefficient formula, Theorem 175.3 (`hb:thm:coefficients`); the 30-vertex counterexample $H_{4,11}$ with weight 500 on both cores, Theorem 176.1 (`hb:thm:witness`); the complete classification of the core-three family $H_{3,n}$, Theorem 177.1 (`hb:thm:rank6phase`), whose last order-six margin fails at $n = 33$ (the rank-six counterexample $H_{3,33}$), with its asymptotic Corollary 177.2 (`hb:cor:asymptoticweight`); the endpoint threshold, Theorem 178.2 (`hb:thm:endpoint`); and the first-margin defect identity with its equality case, Theorem 180.2 (`hb:thm:firstidentity`) and Corollary 180.3 (`hb:cor:equality`): $(r-1)a_1^2 \geq 2ra_2$ for every positively weighted graph of matching number $r \geq 2$, with equality exactly for r disjoint stars of equal edge totals.
- Part XII (rank-three matching supports): the first rank inequality, Theorem 190.2 (`mr:thm:first`), and its unweighted equality case, Theorem 190.3 (`mr:thm:equality`); normalization by the matching number at unit weights for $\nu \leq 3$, Theorem 191.1 (`mr:thm:rankthree`); the explicit rank-three bimatroid obstruction, Theorem 192.1 (`mr:thm:matrix`); and a connected rank-four family, Theorem 196.1 (`mr:thm:family`).
- Part XIII (leaf compression): the weighted smaller-shore inequality, Lemma 205.1 (`lc:lem:ambient`); weighted leaf compression, Lemma 206.1 (`lc:lem:compression`); normalization by the matching number for every minimum cover with a singleton shore or on one shore, Theorem 206.2 (`lc:thm:singleton`); and weighted rank three, Corollary 206.3 (`lc:cor:rankthree`).

Archive 78 of this batch is byte-identical to the batch-71 archive printed as Part XIII; it is not printed again.

1.3 Notions and notation

The sources share one setting but not one vocabulary. The following conventions hold throughout; where a source uses another word or letter, it is listed here and kept in the source’s text.

Notion	Meaning in this report, and the readings to avoid
Support polynomial $p_G(t) = \sum_k a_k t^k$	$a_k = \sum u_I v_J$ over pairs $I \subseteq X, J \subseteq Y$ with $ I = J = k$ for which $G[I, J]$ has a perfect matching. Each pair of endpoint sets is counted <i>once</i> , not once per matching: this is not the matching (rook) polynomial. At unit activities it is the perfectly matchable set polynomial of Davis–Kohl [5]. Source 75 writes $P_H(t) = \sum p_k t^k$; <i>preorder-root-polytopes</i> writes $p_G(t; u, v)$ (Part XIII) and $A_H(t; u, v)$ (Part V).
Matching number $\nu(G)$, also r	The maximum size of a matching. The sources call it the <i>matching rank</i> ; <i>preorder-root-polytopes</i> calls it the <i>matching number</i> . Both words are used here with one meaning. For positive activities $\deg p_G = \nu(G)$ and every a_k with $0 \leq k \leq \nu(G)$ is positive.
ULC _D and the gaps $\Delta_k^{(D)}$	p is ULC _D if $a_k / \binom{D}{k}$ (padded by zeros) is log-concave without internal zeros; for support polynomials this is $\Delta_k^{(D)} = k(D-k)a_k^2 - (k+1)(D-k+1)a_{k-1}a_{k+1} \geq 0$ for $1 \leq k \leq D-1$ (display (1)). “Normalization by the matching number” means $D = \nu(G)$. The sources call Δ_k a <i>gap</i> , <i>margin</i> , <i>normalized gap</i> or <i>Newton gap</i> ; the <i>first</i> gap is $k = 1$, the <i>last</i> is $k = D - 1$. <i>Strict</i> ULC _D means all $D - 1$ gaps are positive.
Weightings	<i>Two-shore</i> (or simply <i>weighted</i>): arbitrary positive activities u on X and v on Y (Part I). <i>Unit</i> : $u \equiv v \equiv 1$ (Research question 84 of <i>preorder-root-polytopes</i> ; Part II). <i>One-shore</i> : $u \equiv 1$ on the left shore, v arbitrary positive on the right shore (Research question 85 of <i>preorder-root-polytopes</i> ; Parts II–IV). <i>One common right-core activity</i> : activity w on the right core of a full Hall graph and 1 elsewhere, a special one-shore weighting (sources 73, 55, 47, 51).
Displayed cover $A \sqcup B$	A vertex cover with $A \subseteq X$ on the left and $B \subseteq Y$ on the right; $L = X \setminus A$ and $R = Y \setminus B$ are the exterior sets and there are no L – R edges; $G[A, B]$ is the <i>core</i> . Source 74 writes a $2 + q$ cover for $ A = 2, B = q$; a cover is <i>genuine</i> when $\nu(G) = A + B $, that is, when it is minimum.
Full Hall graph $H_{a,b;n,m}$	Core A (a vertices) and B (b), exterior L (n) and R (m), edges $(A \times B) \cup (L \times B) \cup (A \times R)$ (sources 75, 73, 55, 47, 51; source 74 writes $H_{a,b;N,M}$, source 76 $H_{2,s;n,m}$). <i>preorder-root-polytopes</i> ’s $H_{s,n}$ is $H_{s,s;n,n}$ with weight w on <i>both</i> cores. The weighting differs by source: both cores (74, 75, <i>preorder-root-polytopes</i>), the right core only (73, 55, 47, 51).
Scalings	Two different scalings occur. Source 74 (Section 6) multiplies the activities of a <i>whole</i> minimum cover, on both shores, by a common factor; then failures persist at every matching number ≥ 6 . The sources of Part IV multiply only the <i>right</i> part of a fixed minimum cover, with unit left activities; there no failure occurs through matching number eight. These statements do not conflict.

Notion	Meaning in this report, and the readings to avoid
Real stability, real-rootedness, Lorentzian	A polynomial is real stable if it has no zero with all variables in the upper half-plane $\mathbb{H}$; a real-stable univariate polynomial is real-rooted. A homogeneous bivariate polynomial is Lorentzian exactly when its coefficients are ULC of its degree. Real-rootedness of p_G implies $\text{ULC}_{\deg p_G}$ by Newton's inequalities; the converse fails (source 75's rank-five example, Section 16).
Letters local to a source	Letters are local to the source (often to the section) that defines them. In particular A, B are cover parts in source 74 but numerical gap factors in Section 3 of source 75 and <i>preorder-root-polytopes</i> 's core in Part XI; a, b, c are right-class activity sums in Section 5 but core sizes in Section 6; q is $ B $ in source 74 and $2^{-1/2}$ in source 76; $Q = ab + ac + bc + d$ in source 74; H is $G[L, B]$ in source 74, a full Hall graph in sources 75 and 76, and $H_{s,n}$ in <i>preorder-root-polytopes</i> ; h is a residual sector in source 74 and a population offset in source 75.

Renamed symbols. No printed symbol of a source was renamed.

1.4 Research questions of *preorder-root-polytopes* answered here

Question of <i>preorder-root-polytopes</i>	Status after this report
RQ 86 hb:q:rank (first possible weighted rank)	Answered: every weighted graph with $\nu \leq 5$ is ULC_ν (Corollary 2.2); Part XI's $H_{3,33}$ fails at $\nu = 6$. The least weighted failure rank is exactly six.
RQ 87 hb:q:order (a universal rank-only order)	In part: $\mathcal{D}(4) = 4$ and $\mathcal{D}(5) = 5$ (Corollary 2.2); the question beyond rank five is open.
RQ 93 mr:q:rankfour (finish rank four)	Answered affirmatively, for unit and arbitrary two-shore weights (Corollary 2.2; second route by real stability, Corollary 15.2).
RQ 89 hb:q:asymmetry (asymmetric bottlenecks)	In large part: full Hall graphs are ULC for all activities exactly when one core has at most two vertices (Theorem 16.1); explicit failures for all unequal cores $a, b \geq 3$ (Sections 6 and 16.5). Unequal core activities: Part II (one core at activity one).
RQ 90 hb:q:defects (higher-index defect identities)	Bears on it: under the two-vertex-shore cover hypothesis, equality in any one gap holds only for balanced star forests (Theorem 14.1); no higher defect identity is given.
RQ 84 hb:q:unit (unit weights)	Answered negatively (source 71, Theorem 17.1): a connected all-unit graph with $\nu = 3450$ is not ULC_ν ; infinitely many deficiency-one examples. The least matching number of a unit counterexample is open.
RQ 85 hb:q:oneshore (one-shore fields)	Answered negatively (source 73: $\nu = 50$, 125 vertices; $\nu = 38$; source 55: for one common right-core activity on full Hall graphs the least failing matching number is exactly 38).

Question of <i>preorder-root-polytopes</i>	Status after this report
RQ 88 hb:q:phase (Hall phase diagrams, $s \geq 4$)	Not answered (it concerns weight on both cores). Its one-shore analogue is answered by sources 47 and 51: the last gap controls all gaps through $\nu = 38$ but not at $\nu = 49$.
RQ 94 mr:q:allrank (a structural all-rank proof)	Answered negatively for unit weights (source 71): no all-rank proof can exist.
RQ 37 lor:q:first (Part V)	Weighted: holds through $\nu = 5$, fails from $\nu = 6$ (Part I). Unit and one-shore: fail (Part II).

1.5 Labels and editorial marks

Every label carries the prefix **mrn:**. Labels written in this merge are **mrn:sec:...**, **mrn:src:NN** (the start of source NN) and **mrn:w:tab:...**; a delivered label of source NN carries a sub-prefix naming its Part (**w** for Part I, **u** for Part II, **s** for Part III, **e** for Part IV) and the source, for example **mrn:w:rf:thm:main** for source 74's **thm:main**. The table of sub-prefixes is in Appendix A. Source 74, the base, keeps its sections as sections of Part I, and its appendices are ordinary sections there; every other source is one section, whose sections are subsections. Section and theorem numbers therefore differ from the delivered PDFs, which are not shipped.

Part I

Weighted activities on both shores: normalization through matching number five

Contents of Part I. Part I prints four sources. Source 74, *Weighted Rank Five and Covers with a Two Vertex Shore*, is the base of the whole report; its sections are Sections 2–13, and its appendices are printed as ordinary sections (Sections 8–13). Source 72 (Section 14) classifies equality under the same cover hypothesis; source 76 (Section 15) proves real stability for covers with two vertices on each shore; source 75 (Section 16) classifies the full Hall family.

What is new here and what is cited. New relative to *preorder-root-polytopes*: the transversal lift of Theorem 2.1 and its consequences (normalization by the matching number for every minimum cover meeting one shore in at most two vertices, hence for $\nu \leq 5$, and the cover obstruction Corollary 5.1); the forced-sector and scaling refinements; the asymmetric negative shapes; the equality classification at every gap (source 72); real stability for $2 + 2$ covers, the $2 \times s$ stability threshold and the 17-vertex example (source 76); the full Hall classification, its strictness and real-rootedness refinements and the nonreal rank-five example (source 75). Cited, not new: the smaller-shore inequality and leaf compression (Parts V and XIII of *preorder-root-polytopes*; source 74’s Lemmas 3.1 and 3.2 and the corresponding steps of sources 72 and 76 are pointers); the first-gap inequality with its equality case (Part XI’s Theorem 180.2 and Corollary 180.3, Part XII’s Theorem 190.2; the Motzkin–Straus proofs of sources 74 and 72 are printed as second routes); the rank-six and rank-eight witnesses $H_{3,33}$ and $H_{4,11}$ (Part XI, restated by source 74); and the stability tools of source 76 (Part VI).

Same theorem in several sources. Source 74’s Corollary 2.2 contains source 76’s Corollary 15.2 ($\nu \leq 4$) and Part XIII’s Corollary 206.3 ($\nu \leq 3$); source 76’s real-stability route is printed as a second route. The positive half of source 75’s Theorem 16.1 is the full Hall case of source 74’s Theorem 2.1; source 75’s comparison-polynomial proof is printed as a second route. For equal cores and populations the negative half of Theorem 16.1 is Part XI’s endpoint theorem.

Non-claims of Part I. Source 74’s theorem asserts neither real-rootedness nor stability of the support polynomial (remark after Corollary 5.1); at higher rank it does not address the unrestricted unit-weight and one-shore problems (Section 13); finite computations are regression checks, not proofs (sources 72 and 74); source 76 does not assert real-rootedness for every graph of matching number four; source 75’s classification concerns the full Hall family only and its negative assertion says nothing about unit or one-shore weights.

Abstract of source 74, as delivered. Every bipartite matching-support polynomial with a minimum vertex cover meeting one shore in at most two vertices is ultra-log-concave after normalization by its matching rank, for arbitrary positive activities on both shores. The main construction replaces a displayed $2 + q$ cover by a rank- $(q + 2)$ represented matroid whose Lorentzian basis polynomial specializes exactly to the homogenized support polynomial. Consequently weighted rank normalization holds through rank five, and known rank-six Hall obstructions show that the first weighted failure rank is exactly six. We also prove a sharp all-core boundary inequality with constant three, give independent Lorentzian transfer proofs

for star cores, and derive quantitative eventual strictness under two natural activity scalings. Full proofs, finite exact coefficient certificates, sum-of-squares identities and independent verification scripts are included.

Scope and status. The rank-five weighted question posed in the inspected ProveIt draft is resolved here. At higher rank the theorem covers every minimum cover with a shore of size at most two; the unrestricted unit-weight and one-shore-weighted problems at higher rank are not addressed here. These are proofs in ordinary mathematics, not Lean-certified or independently refereed results. The work extends the inspected rank manuscript and the separate rank-four and full-Hall classification reports [10, 14, 13]; global priority is not claimed. [*Merge note:* The “inspected ProveIt draft” and “rank manuscript” [10] is the manuscript printed as Part XII of *preorder-root-polytopes* (pinned at [1b3960d8a](#)); the companions [14] and [13] are sources 76 and 75, printed in Sections 15 and 16; [11] and [9] are Parts XIII and XI of *preorder-root-polytopes*. The unit-weight and one-shore problems that this paragraph leaves aside are the subject of Parts II–IV (batch-72 sources 71, 73 and later).]

2 Definitions and main conclusions

For a finite bipartite graph $G = (X, Y; E)$ and positive vertex activities u_x, v_y , write

$$p_G(t) = \sum_{k=0}^r a_k t^k, \quad a_k = \sum_{\substack{I \subseteq X, J \subseteq Y \\ |I|=|J|=k \\ G[I,J] \text{ perfectly matchable}}} u_I v_J, \quad r = \nu(G).$$

Each endpoint pair is counted once. In particular $p_{K_{2,2}}(t) = 1 + 4t + t^2$ at unit activities. The polynomial is ULC_D , for $D \geq \deg p_G$, if its zero-padded sequence $a_k / \binom{D}{k}$ is log-concave and has no internal zeros. Equivalently, for these support polynomials,

$$\Delta_k^{(D)} := k(D-k)a_k^2 - (k+1)(D-k+1)a_{k-1}a_{k+1} \geq 0. \quad (1)$$

Strict ULC_D means all $D-1$ displayed gaps are positive.

A displayed $2+q$ cover is $A \sqcup B$, where $A = \{A_1, A_2\} \subseteq X$ and $|B| = q$, $B \subseteq Y$. Put $L = X \setminus A$ and $R = Y \setminus B$. There are no L – R edges. The induced graph $G[A, B]$ is called the core. We call the displayed cover *genuine* when $\nu(G) = q+2$.

Theorem 2.1 (A two-vertex cover shore). *If G has a displayed $2+q$ cover, fix positive activities v on its right shore. The homogeneous left marginal*

$$\sum_{\substack{I \subseteq X \\ |I| \leq q+2}} \left(\sum_{\substack{J \subseteq Y \\ (I,J) \text{ feasible}}} v_J \right) z_I s^{q+2-|I|}$$

is Lorentzian. In particular, the degree- $(q+2)$ homogenization of its weighted matching-support polynomial is Lorentzian. Thus p_G is ULC_{q+2} for every positive activity assignment. If the displayed cover is minimum, the normalization is by the actual matching rank.

Corollary 2.2 (Weighted rank five and the first failure rank). *Every bipartite matching-support polynomial of matching rank $r \leq 5$ is ULC_r for arbitrary positive activities on both shores. The smallest rank admitting a weighted counterexample is exactly six. At every rank $r \geq 6$ there is a weighted counterexample.*

The positive conclusion follows from Theorem 2.1, the pure and singleton-side cover cases proved below, and König’s theorem. The negative construction is given in Section 6. For nonnegative activities, delete zero-activity vertices first; the same conclusion holds with r interpreted as the surviving degree. [Merge note: Corollary 2.2 contains Part XIII’s Corollary 206.3 of *preorder-root-polytopes* ($\nu \leq 3$, `lc:cor:rankthree`) and source 76’s Corollary 15.2 ($\nu \leq 4$), which is printed in Section 15 as a second route by real stability. The rank-six counterexample it relies on is Part XI’s $H_{3,33}$ (Theorem 177.1 of *preorder-root-polytopes*, restated in Section 6). It answers Research questions 86 and 93 of *preorder-root-polytopes* and gives $\mathcal{D}(4) = 4$, $\mathcal{D}(5) = 5$ in its Research question 87 (Section 1.4).]

Theorem 2.3 (Forced-sector and strict-scaling refinements). *For any displayed $2 + q$ cover with $q \geq 1$, the coefficient of the product of the two A activities in p_G is $t^2 C(t)$, where C is ULC_q . For a genuine cover, common scaling of only the two A activities makes p_G strictly ULC_{q+2} for all sufficiently large scale factors.*

For a rank-five $2 + 3$ cover whose core has matching rank two, common scaling of all five cover activities also makes all four gaps strictly positive eventually. An explicit coefficient bound gives a sufficient threshold for both scalings.

Every fixed minimum-cover scaling at rank at most five is therefore ULC at *every* positive scale factor. At each rank at least six, some fixed graph and minimum cover fail for every sufficiently large common cover scale, so the universal scaling threshold is sharp as well. [Merge note: Here the whole minimum cover is scaled, on both shores. Sources 08s, 08, 03, 02 and 01 (Part IV) scale only the right part of a minimum cover, with unit left activities, and there no failure occurs below matching number nine. See the entry “Scalings” in Section 1.3.]

3 Lorentzian input and singleton covers

We use the following published facts from Brändén–Huh [1]: matroid basis polynomials are Lorentzian (Theorem 3.10); partial derivatives, nonnegative linear substitutions, products and coefficientwise limits preserve this class (Corollary 2.11, Theorem 2.10, Corollary 2.32, and closedness); homogeneous stability preservers that preserve nonnegative coefficients preserve Lorentzian polynomials (Theorem 3.4); and multiaffine truncation and polarization preserve the class (Corollary 3.5 and Proposition 3.1). A homogeneous bivariate polynomial of degree D is Lorentzian exactly when its coefficients form a ULC_D sequence (Example 2.26). The zero polynomial is included by closure.

Lemma 3.1 (Weighted smaller-shore inequality). *Every bipartite support polynomial is $\text{ULC}_{\min(|X|, |Y|)}$.*

Proof (pointer). This is Theorem 70.1 of *preorder-root-polytopes* (Part V, `lor:thm:weighted ULC`), restated as Lemma 205.1 of its Part XIII (`lc:lem:ambient`). [Merge note: Source 74 proves it by the same specialization of the basis polynomial of the transversal matroid on the private copies $\bar{X} \sqcup Y$, at s/u_x and tv_y , followed by the bivariate Lorentzian characterization and transposition, and calls it the standard bimatroid specialization [7]. That proof is Part XIII’s proof of Lemma 205.1 and is not reprinted.] $\square$

Lemma 3.2 (Singleton-side minimum covers). *If a rank- r graph has a minimum cover entirely on one shore or split as $1 + (r - 1)$, its support polynomial is ULC_r for all positive activities.*

Proof (pointer). This is Theorem 206.2 of *preorder-root-polytopes* (Part XIII, `lc:thm:singleton`), proved there by weighted leaf compression, Lemma 206.1 (`lc:lem:compression`). [Merge note: Source 74’s proof is the same argument (delete isolated vertices for a pure cover;

for a cover $\{x_0\} \sqcup B$ merge the leaves at x_0 into one leaf carrying their total activity, which leaves p_G unchanged, and apply Lemma 3.1); source 74 credits it as “the exact leaf-compression argument of” [11], which is Part XIII. It is not reprinted.] $\square$

4 The exact two-row formula

Let the two A activities be x, y , and let $H = G[L, B]$. For a virtual left vertex with neighborhood $S \subseteq B$ and activity one, define g_S by

$$p_H = f, \quad p_{H+S} = f + tg_S.$$

If S_1, S_2 are the actual core-row neighborhoods, define h by

$$p_{H+A} = f + t(xg_1 + yg_2) + xyt^2h, \quad g_i = g_{S_i}, \quad g_U = g_{S_1 \cup S_2}.$$

The feasible support set for g_U is the union of those for g_1, g_2 : in a matching the virtual row uses one edge, and that edge belongs to at least one of the original neighborhoods.

For R , let a, b, c be the sums of activities in classes with neighborhoods $\{A_1\}$, $\{A_2\}$ and A , respectively. If the individual common-class activities are $r_1, \dots, r_m$, put

$$d = \sum_{i < j} r_i r_j, \quad Q = ab + ac + bc + d, \quad 0 \leq d \leq c^2/2.$$

Then the exact full polynomial is

$$\begin{aligned} p_G = f[1 + (x(a+c) + y(b+c))t + xyQt^2] + t(xg_1 + yg_2) \\ + xyt^2(h + ag_2 + bg_1 + cg_U). \end{aligned} \tag{2}$$

To prove this, condition on the selected A vertices and then on the number of selected R vertices. With both A selected, two selected R contribute Qf ; one singleton-class vertex forces the corresponding A and gives ag_2 or bg_1 ; one common-class vertex leaves the union-neighborhood alternative and gives cg_U ; no selected R gives h . These cases have disjoint endpoint sets. In particular the common term is cg_U , not $c(g_1 + g_2)$.

5 The full rank compression matroid lift

We prove Theorem 2.1. First suppose $Q > 0$. The same construction includes $q = 0$, with no original B slots. Form a transversal matroid matched into the $q + 2$ slots

$$B \sqcup \{e_1, e_2\}.$$

Its elements and allowed slots are as follows:

- a private element $\bar{b}$ for each $b \in B$, allowed only in slot b ;
- an element for each $l \in L$, allowed in its original neighborhood in B ;
- two augmented core elements $\hat{A}_i$, allowed in $S_i \cup \{e_i\}$;
- two plane elements P_1, P_2 , allowed only in e_1, e_2 , respectively;
- one common element Z_j for each common right-exterior vertex, allowed in both e_1, e_2 .

The rank is $q + 2$, since all private B elements together with P_1, P_2 form a basis. Every basis is counted once, regardless of how many injections into the slots it admits. We call P_1, P_2, Z_j the plane elements, as they use only the two additional slots.

Give the private B variables the values s/v_b , and the exterior-left variables tu_l . Keep core variables X, Y , and specialize the plane variables to

$$P_1 : \frac{bs}{Q}, \quad P_2 : \frac{as}{Q}, \quad Z_j : \frac{r_js}{Q}. \quad (3)$$

Multiply this basis specialization by $Q \prod_{b \in B} v_b$. It is Lorentzian by the matroid and nonnegative-substitution theorems.

We now compute it exactly. Put

$$f_h = s^q f(t/s), \quad G_i = s^{q-1} g_i(t/s), \quad G_U = s^{q-1} g_U(t/s), \quad H_h = s^{q-2} h(t/s),$$

where $G_i = G_U = 0$ is defined directly for $q = 0$, and $H_h = 0$ is defined directly for $q < 2$; the displayed homogenizing formulas are used only for nonnegative exponents. Classify a basis by the number k of core elements and ℓ of plane elements. Other elements use only the q original slots, so $k + \ell \geq 2$ and $0 \leq k, \ell \leq 2$. There are six cases.

1. $(k, \ell) = (0, 2)$. Any pair of distinct plane elements can fill the two extra slots, and the other q elements fill the B slots. The sum of plane-pair activities is

$$\frac{s^2}{Q^2} \left(ab + (a + b)c + \sum_{i < j} r_i r_j \right) = \frac{s^2}{Q}.$$

After the exterior factor Q , the contribution is $s^2 f_h$.

2. $(k, \ell) = (1, 1)$. With core $\hat{A}_1$, the plane element must be able to occupy e_2 , hence is P_2 or a Z_j . Its total activity is $(a + c)s/Q$. The remaining q elements fill the B slots, giving $Xs(a + c)f_h$. With $\hat{A}_2$, the contribution is $Ys(b + c)f_h$.
3. $(k, \ell) = (1, 2)$. The plane pair fills both extra slots, forcing the core element into B . The remaining support uses row S_i and $q - 1$ ordinary elements, with profile G_i . The contribution is $Xs^2 G_1 + Ys^2 G_2$.
4. $(k, \ell) = (2, 0)$. The two core elements occupy their dedicated extra slots. The other q elements fill B , contributing $QXY f_h$.
5. $(k, \ell) = (2, 1)$. The element P_1 occupies e_1 , forcing $\hat{A}_2$ into e_2 and $\hat{A}_1$ into B ; hence its profile is G_1 and its activity coefficient is b . Similarly P_2 gives G_2 with coefficient a . A common Z_j can occupy either extra slot, so feasibility holds exactly when either core row completes the remaining B -slot support. This is the union profile G_U , counted once even if both alternatives work. Thus the contribution is $XYs(bG_1 + aG_2 + cG_U)$.
6. $(k, \ell) = (2, 2)$. The two plane elements fill the extra slots, forcing both core elements into B together with $q - 2$ other elements. The contribution is $XYs^2 H_h$; this case is absent for $q < 2$.

Consequently the constructed Lorentzian polynomial is

$$\begin{aligned} \mathcal{L}(s, t, X, Y) &= s^2 f_h + sX[(a + c)f_h + sG_1] + sY[(b + c)f_h + sG_2] \\ &\quad + XY[Qf_h + s(bG_1 + aG_2 + cG_U) + s^2 H_h]. \end{aligned} \quad (4)$$

Set $X = xt$ and $Y = yt$. Equation (2) identifies (4) with $s^{q+2}p_G(t/s)$, proving the bivariate assertion for $Q > 0$. More generally, keep each exterior-left element variable as its own z_l rather than tu_l , and set $X = z_{A_1}$, $Y = z_{A_2}$. The same six basis cases give exactly the homogeneous left marginal in Theorem 2.1. Every plane and private variable remains a nonnegative multiple of s , so this full marginal is Lorentzian. No analogous opposite-marginal assertion is inferred when $q > 2$.

The six cases are disjoint classes of matroid bases. In particular the union case does not count injections or matching allocations. The construction also has a generic matrix representation obtained by giving each allowed slot incidence its own indeterminate. Its six block-determinant expansions provide a second interpretation of the same identity, including dependent or zero original core-row vectors.

If $Q = 0$, add a singleton right vertex at each A_i with activity $\varepsilon > 0$. The perturbed graph still has a displayed $2 + q$ cover, its Q is positive, and its coefficients converge to those of the original graph as $\varepsilon \downarrow 0$. Closedness gives the result. Other zero activities are handled by continuity at fixed order $q + 2$; private reciprocals are used only before taking the positive-activity limit.

Corollary 5.1. *Any failure of weighted rank-normalized ultra-log-concavity must have every minimum vertex cover meet both shores in at least three vertices.*

Proof. A pure or singleton-side minimum cover is covered by Lemma 3.2; a cover meeting one shore in exactly two vertices is covered by Theorem 2.1. $\square$

Remark 5.2. The new rank- $(q + 2)$ transversal matroid is used through its Lorentzian basis polynomial. The theorem asserts neither real-rootedness of p_G nor stability of the original complemented-row polynomial. The support identity, and not the values of nonzero determinants, determines all coefficients.

6 Exact failure at rank six and at every higher rank

We include the elementary obstruction calculation to make the sharpness conclusion self-contained. It is the negative construction from the Hall reports [13, 9]. [*Merge note:* [13] is source 75 (Section 16.5) and [9] is Part XI of *preorder-root-polytopes*. For $a = b$ and $N = M$ the graph $H_{a,b;N,M}$ with weight w on both cores is Part XI's $H_{s,n}$, and formula (5) is Part XI's Theorem 175.3 (`hb:thm:coefficients`) extended to unequal cores and populations. The endpoint quadratic below is the asymmetric form of Part XI's endpoint analysis (Theorem 178.2, `hb:thm:endpoint`).]

Let $H_{a,b;N,M}$ have a complete $a \times b$ core, N exterior-left vertices adjacent to every right-core vertex, and M exterior-right vertices adjacent to every left-core vertex. Give core vertices activity w and exterior vertices activity one. If $N \geq b$ and $M \geq a$, the core is a minimum cover of size $r = a + b$: match its two shores independently into the opposite exterior sets.

A size- k endpoint support with i selected left-core and j selected right-core vertices is feasible exactly when $i + j \geq k$, subject to the cardinality bounds. Match all exterior vertices to the core first, then use the complete core for the remainder. Thus

$$a_k = \sum_{\substack{0 \leq i \leq a, 0 \leq j \leq b \\ i+j \geq k}} \binom{a}{i} \binom{b}{j} \binom{N}{k-i} \binom{M}{k-j} w^{i+j}. \quad (5)$$

Put $A_i = \binom{N}{b-i}$, $B_i = \binom{M}{a-i}$ for $i = 0, 1, 2$, and

$$X = aA_0B_1 + bA_1B_0, \quad Y = aA_1B_2 + bA_2B_1, \quad Z = \binom{a}{2}A_0B_2 + abA_1B_1 + \binom{b}{2}A_2B_0.$$

Counting at most two omitted core vertices gives

$$a_r = A_0B_0w^r, \quad a_{r-1} = A_1B_1w^r + Xw^{r-1}, \quad a_{r-2} = A_2B_2w^r + Yw^{r-1} + Zw^{r-2}.$$

Hence

$$\Delta_{r-1}^{(r)} = w^{2r-2}(\alpha w^2 + \beta w + \gamma), \tag{6}$$

where

$$\begin{aligned} \alpha &= (r-1)A_1^2B_1^2 - 2rA_0A_2B_0B_2, \\ \beta &= 2(r-1)A_1B_1X - 2rA_0B_0Y, \\ \gamma &= (r-1)X^2 - 2rA_0B_0Z. \end{aligned}$$

6.1 A concrete rank-six witness

Take $a = b = 3$ and $N = M = 33$. Here $A_0 = B_0 = 5456$, $A_1 = B_1 = 528$, $A_2 = B_2 = 33$, and exact simplification of (6) gives

$$\Delta_5^{(6)} = 404794368 w^{10}(-w^2 + 26784w + 522784).$$

At $w = 26804$, the parenthesis equals -13296 . This is an exact positive-integer-activity counterexample on 72 vertices and 207 edges, with minimum cover and matching rank six. Combined with Corollary 2.2, the first weighted failure rank is exactly six. [*Merge note*: This witness is Part XI's $H_{3,33}$: Theorem 177.1 of *preorder-root-polytopes* (`hb:thm:rank6phase`) gives the last order-six margin of $H_{3,n}$ at $n = 33$ as 16 times the same quadratic $-w^2 + 26784w + 522784$. Source 74 restates and verifies it (its bibliography says so); it is not new here.]

6.2 Permanent scaling failure at every rank at least six

Let $a, b \geq 3$. Write

$$K = (r-1)ab, \quad J = (r+1)ab - 2r(r-1).$$

Because $ab \geq 3(r-3)$ and $r \geq 6$,

$$J \geq r^2 - 4r - 9 > 0.$$

Choose $N = b + 30$, $M = a + 30$, so that

$$\frac{A_1^2B_1^2}{A_0A_2B_0B_2} = \frac{ab}{(a-1)(b-1)} \left(\frac{32}{31} \right)^2.$$

The inequality $\alpha < 0$ is equivalent to

$$K(2 \cdot 31 + 1) < J \cdot 31^2.$$

Indeed

$$15J - K = (14r + 16)ab - 30r(r-1) \geq 12(r-6)(r+2) \geq 0,$$

and $63/961 < 1/15 \leq J/K$. Thus (6) is negative for every sufficiently large w . An explicit sufficient integer is

$$w \geq 1 + \left\lfloor \frac{|\beta| + |\gamma|}{-\alpha} \right\rfloor,$$

since for $w \geq 1$ the bracket is at most $w(\alpha w + |\beta| + |\gamma|) < 0$. Taking $a = 3$, $b = r - 3$ supplies every $r \geq 6$. In contrast, every graph of rank at most five remains ULC at every positive minimum-cover scale by Corollary 2.2. This proves the sharp universal rank threshold for minimum-cover scaling.

6.3 The shapes recorded by source 74's verifier

[*Merge note:* This subsection was written in the merge from source 74's recorded output `data/74-weighted-rank-five-negative_shapes_verification.json`; its numbers were recomputed independently from (5). Source 74's text does not print them.] Source 74's negative-family verifier records seven exact witnesses, listed in Table 3. Each has exterior activity one and one common integer activity w on all $a + b$ core vertices; the populations are $N = b - 1 + h$ and $M = a - 1 + h$ with source 75's offset $h = 1 + \lceil 2K/H \rceil$ (Section 16.5), and w is the least integer at which the last gap $\Delta_{r-1}^{(r)}$ is negative for these populations: the primitive endpoint quadratic (the bracket of (6) divided by its content) is negative at w and positive at $w - 1$. The (4, 4) row is Part XI's 30-vertex counterexample (Theorem 176.1 of

(a, b)	r	(N, M)	vertices	edges	w	primitive quadratic	in <i>preorder-root-polytopes</i>
(3, 3)	6	(33, 33)	72	207	26804	$-w^2 + 26784w + 522784$	Part XI, $H_{3,33}$
(3, 4)	7	(16, 15)	38	121	2012	$-3w^2 + 6006w + 55601$	new
(3, 5)	8	(15, 13)	36	129	418	$-115w^2 + 46860w + 469964$	new
(4, 4)	8	(11, 11)	30	104	485	$-w^2 + 480w + 2304$	Part XI, $H_{4,11}$
(4, 5)	9	(11, 10)	30	115	203	$-215w^2 + 42560w + 192472$	new
(5, 5)	10	(10, 10)	30	125	131	$-11w^2 + 1400w + 4928$	new
(3, 8)	11	(16, 11)	38	185	235	$-158w^2 + 34860w + 518355$	new

Table 3: Exact last-gap failures of $H_{a,b,N,M}$ recorded by source 74's verifier (merge table).

preorder-root-polytopes, `hb:thm:witness`, which uses $w = 500$); the (3, 3) row is its rank-six witness. The other five rows are asymmetric or larger-core shapes, which bear on Research question 89 of *preorder-root-polytopes*; none has fewer than 30 vertices, and none has matching number below six, as Corollary 2.2 requires.

7 An established geometric interpretation

For unit activities, p_G is the *perfectly matchable set polynomial* of Davis–Kohl [5]. It is distinct from the ordinary matching or rook polynomial, which counts matchings with multiplicity. Their Theorem 3.10, restating results of Ohsugi–Tsuchiya, gives the following precise Ehrhart interpretation.

Given bipartite $G = (X, Y; E)$, form $\hat{G}$ by adjoining x_0 on the left and y_0 on the right, with edges from x_0 to every old right vertex, from y_0 to every old left vertex, and $x_0 y_0$. For the edge polytope

$$Q_{\hat{G}} = \text{conv}\{e_x + e_y : xy \in E(\hat{G})\},$$

one has

$$h^*(Q_{\hat{G}}; t) = p_G(t) = h^*(\text{STAB}(\overline{G}); t).$$

The second identity is their Lemma 3.11; $\text{STAB}(\overline{G})$ is the stable-set polytope of the complement.

Corollary 7.1. *If G has matching rank $r \leq 5$, or a minimum vertex cover meeting one shore in at most two vertices, these two h^* -polynomials are ULC_r , normalized by their degree $r = \nu(G)$.*

Only the unit-activity interpretation is used here. No assertion is made about arbitrary Ehrhart h^* -polynomials or about the h^* -polynomial of an unrelated preorder polytope. [*Merge note:* At unit activities without the cover or rank hypothesis the conclusion fails: source 71 (Part II) gives a connected graph with $\nu = 3450$ whose two h^* -polynomials are not ULC_{3450} .] [*Merge note:* Sections 8–13 were the appendices of source 74; they are printed here as ordinary sections of Part I.]

8 A generic-line matroid model for the forced-left sector

By (2), the polynomial in Theorem 2.3 is

$$C = Qf + ag_2 + bg_1 + cg_U + h. \quad (7)$$

Write $f_h = s^q f(t/s)$, $G_i = s^{q-1} g_i(t/s)$, $G_U = s^{q-1} g_U(t/s)$ and $H_h = s^{q-2} h(t/s)$. For $q = 1$ the last term is zero. The right-complement transversal matroid of $H + A$ gives a degree- q basis specialization

$$F = f_h + z_1 G_1 + z_2 G_2 + z_1 z_2 H_h.$$

The private B variables are s/v_b , the exterior-left variables are tu_l , the two distinguished elements have variables z_1, z_2 , and we multiply by $\prod_b v_b$.

Represent this matroid by a generic matrix: use an independent indeterminate for each allowed adjacency and private standard basis columns. A square determinant is nonzero exactly when its support admits a matching, since different matchings give distinct monomials. Denote the two distinguished columns by v_1, v_2 . Adjoin columns

$$U_j = \alpha_j v_1 + \beta_j v_2 \quad (1 \leq j \leq m),$$

with generic coefficients, algebraically independent over the original matrix field. For a basis containing just one U_j from this distinguished set, its determinant is $\alpha_j D_1 + \beta_j D_2$ and is nonzero iff at least one of D_1, D_2 is nonzero. Thus its remaining-element profile is exactly G_U . Any pair of distinct distinguished columns has determinant a nonzero scalar multiple of the determinant using v_1, v_2 , hence profile H_h . Three distinguished columns are dependent. These statements also hold when v_1, v_2 are dependent or zero: pair profiles then vanish, and the single profile remains the union.

The resulting matroid basis specialization is therefore

$$\begin{aligned} F_{\text{ext}} = f_h + z_1 G_1 + z_2 G_2 + \left(\sum_j Z_j \right) G_U \\ + \left[z_1 z_2 + (z_1 + z_2) \sum_j Z_j + \sum_{i < j} Z_i Z_j \right] H_h. \end{aligned} \quad (8)$$

It is Lorentzian. For $Q > 0$, substitute

$$z_1 = bs/Q, \quad z_2 = as/Q, \quad Z_j = r_j s/Q.$$

The bracket becomes s^2/Q , and

$$QF_{\text{ext}} = Qf_h + s(ag_2 + bG_1 + cG_U) + s^2 H_h = s^q C(t/s).$$

This proves the first part of Theorem 2.3. For $Q = 0$, add two singleton right vertices with positive activities tending to zero and use coefficientwise closedness.

Corollary 8.1 (Scaling only the two cover vertices). *For a genuine $2 + q$ cover with $q \geq 1$, scaling only the two A activities by a common sufficiently large factor makes every ULC_{q+2} gap strictly positive.*

Proof. Every matching of size $q + 2$ uses exactly one cover vertex per edge. Thus R matches both A vertices and L matches all B , so $Q > 0$ and $f_q > 0$. Consequently all coefficients $C_0, \dots, C_q$ are positive. Write the two activities as $w\alpha, w\beta$. With E the weighted edge sum incident with A at relative activities α, β ,

$$a_1(w) = wE + O(1), \quad a_k(w) = w^2\alpha\beta C_{k-2} + O(w) \quad (k \geq 2).$$

The graph induced by A and the opposite shore has matching rank two, so $E^2 \geq 4\alpha\beta C_0$. The first two leading gaps are respectively

$$(q+1)E^2 - 2(q+2)\alpha\beta C_0 \geq 2q\alpha\beta C_0 > 0, \quad 2q(\alpha\beta C_0)^2 > 0,$$

at powers w^2, w^4 . For $k = i + 2$, $1 \leq i < q$, the leading coefficient at w^4 is at least

$$\frac{2(q-i+1)}{i}(\alpha\beta)^2 C_{i-1} C_{i+1} > 0,$$

by substituting the ULC_q inequality for C_i^2 into (1). Positive leading coefficients prove the claim. $\square$

9 The sharp all-core inequality at rank five

The full lift gives a short general boundary theorem before the independent rank-five certificate below.

Theorem 9.1 (Sharp all-core constant for a two-vertex shore). *For a $2 + q$ cover with $q \geq 2$, factor the product of all core activities from the all-core sector and write it as $t^q(c_0 + c_1 t + c_2 t^2)$. Then*

$$c_1^2 \geq \frac{2q}{q-1} c_0 c_2,$$

and the constant is optimal for every q .

Proof. In the lifted matroid basis polynomial, set all private B variables to zero and differentiate once in each augmented core variable X, Y . These operations preserve the Lorentzian property, giving a degree- q polynomial. Under the remaining specializations its three plane-count sectors give, up to the fixed positive scaling factor,

$$c_0 s^2 t^{q-2} + c_1 s t^{q-1} + c_2 t^q.$$

The bivariate ULC_q inequality at index $q - 1$ is the desired bound. Zero activities and $Q = 0$ follow by continuity as before.

For the complete $2 \times q$ core with $N \geq q$ unit common-left vertices and $M \geq 2$ unit common-right vertices,

$$c_0 = \binom{N}{q-2}, \quad c_1 = M \binom{N}{q-1}, \quad c_2 = \binom{M}{2} \binom{N}{q}.$$

Their ratio is

$$\frac{c_1^2}{c_0 c_2} = \frac{2q}{q-1} \frac{M}{M-1} \frac{N-q+2}{N-q+1} \longrightarrow \frac{2q}{q-1}.$$

$\square$

The following argument independently proves the $q = 3$ case using only a finite quartic coefficient test and explicit sums of squares. It does not rely on the full lift.

Now fix a $2 + 3$ cover. Strip off the product of the five core activities from the sector in which all five core vertices are selected. Its polynomial is

$$t^3(c_0 + c_1t + c_2t^2).$$

Theorem 9.2 (Sharp constant three). *For arbitrary nonnegative exterior activities and every $2 + 3$ core,*

$$c_1^2 \geq 3c_0c_2.$$

The constant 3 cannot be increased uniformly.

Let T be the weight of exterior-left triples matching all of B . Let C be the weight of single exterior-left vertices which, together with the two core rows, match B . Let P_i be the weight of exterior-left pairs completable with core row S_i , and let P_U use the union row. Set

$$O = P_1 + P_2 - P_U.$$

The union interpretation gives $0 \leq O \leq \min(P_1, P_2)$. The central left-side inequality is

$$4P_1P_2 - 2O^2 \geq 3CT. \quad (9)$$

9.1 A finite exact coefficient certificate

Encode a nonempty neighborhood in B by its bit mask $1, \dots, 7$. For three nonempty masks, let $M(r, s, t)$ be one exactly when they admit a matching to all three columns. Hall's condition gives the elementary rule: their bitwise union is 7, and no singleton mask $1, 2, 4$ appears twice. For general tuples including an empty mask, use ordinary Boolean matchability.

Index left exterior vertices by i , with masks t_i and weights z_i . Put

$$\begin{aligned} c_i &= M(S_1, S_2, t_i), & p_{ij} &= M(S_1, t_i, t_j), & q_{ij} &= M(S_2, t_i, t_j), \\ o_{ij} &= p_{ij}q_{ij}, & \tau_{ijk} &= M(t_i, t_j, t_k). \end{aligned}$$

Then $C = \sum c_i z_i$, $P_1 = \sum_{i < j} p_{ij} z_i z_j$, and similarly for P_2, O, T . In the quartic

$$I = 4P_1P_2 - 2O^2 - 3CT$$

every exponent is at most two. Its coefficient of $z_i^2 z_j^2$ is $2o_{ij} \geq 0$. Its coefficient of $z_i^2 z_j z_k$ for distinct indices is

$$4(p_{ij}q_{ik} + p_{ik}q_{ij}) - 4o_{ij}o_{ik} - 3c_i\tau_{ijk}. \quad (10)$$

Its coefficient of $z_1 z_2 z_3 z_4$ is

$$4 \sum_{ij|kl} (p_{ij}q_{kl} + p_{kl}q_{ij}) - 4 \sum_{ij|kl} o_{ij}o_{kl} - 3 \sum_i c_i \tau_{[4] \setminus \{i\}}, \quad (11)$$

where each of the three unordered pair partitions is used once.

If the core has matching rank at most one, $C = 0$ and (9) follows from $O^2 \leq P_1P_2$. Otherwise row exchange and column permutation give exactly the eight representatives in Table 4. The table records the complete range of possible coefficients in (10) and (11). It is obtained by the stated three-row Boolean rule, over $7^3 = 343$ ordered triples and $\binom{10}{4} = 210$ neighborhood multisets per core. The included exact verifier implements an independent permutation matcher and checks all 4424 substitutions, as well as all 46 labeled matching-rank-two cores.

Core masks	Repeated-exponent coefficients	Squarefree coefficients
(1, 2)	0, 1, 4	0, 2, 3
(1, 3)	0, 1, 4	0, 2, 3, 6, 8, 12
(1, 6)	0, 1, 4	-4, 0, 2, 3
(1, 7)	0, 1, 4	-4, 0, 2, 3, 8, 12
(3, 3)	0, 1, 4	0, 2, 3, 6, 8, 12
(3, 5)	0, 1, 4	0, 2, 3, 8, 12
(3, 7)	0, 1, 4	0, 2, 3, 8, 12
(7, 7)	0, 1, 4	0, 2, 3, 8, 12

Table 4: The complete finite coefficient test. In each exceptional row the only four negative neighborhood multisets are listed in (12).

Only cores (1, 6) and (1, 7) have negative coefficients. In both cases the negative neighborhood multisets are exactly

$$(2, 3, 4, 5), \quad (2, 3, 5, 5), \quad (3, 3, 4, 5), \quad (3, 3, 5, 5), \quad (12)$$

and each coefficient is -4 . Thus every term involving another neighborhood class is nonnegative. It remains to treat the four classes 2, 3, 4, 5 in these two cores.

9.2 The four-class sum-of-squares certificate

For core (1, 6) let a, b, c, d now denote the activity sums in classes 2, 3, 4, 5, and let B_2, D_2 be the second elementary moments of classes 3, 5. Put $u = a + b$, $v = c + d$. Exact support counting yields

$$\begin{aligned} C &= u + v, & P_1 &= uv, \\ P_2 &= (a + c)(b + d) + bd + B_2 + D_2, & O &= ad + bc + bd, \\ T &= v(ab + B_2) + u(cd + D_2). \end{aligned}$$

Since $0 \leq B_2 \leq b^2/2$ and $0 \leq D_2 \leq d^2/2$, and I is affine in both moments, it suffices to check the four rectangle corners. Write $I_{00}, I_{01}, I_{10}, I_{11}$ for the lower/lower, lower/upper, upper/lower and upper/upper choices. The following exact identities have only nonnegative summands:

$$\begin{aligned} I_{00} &= 2(ad - bc)^2 + uv(ab + cd) + (ac + bd)(ad + bc) + 2b^2d^2, \\ I_{01} &= \frac{1}{2}d^2(a - b)^2 + \frac{1}{2}ad(b - d)^2 + ac(b - d)^2 + bc(a - d)^2 + R_{01}, \\ R_{01} &= a^2(bd + cd) + a\left(\frac{1}{2}b^2d + bc^2 + c^2d + \frac{1}{2}cd^2\right) \\ &\quad + b^2(2c^2 + cd) + b\left(c^2d + \frac{1}{2}cd^2 + \frac{1}{2}d^3\right), \\ I_{11} &= \frac{1}{2}bd(u - v)^2 + \frac{1}{2}(ad + bc + 3ac)(b - d)^2 \\ &\quad + (b + d)\left(ac(a + c) + \frac{1}{2}(a^2d + bc^2)\right). \end{aligned}$$

The expression for I_{10} is obtained from I_{01} by interchanging (a, b) with (c, d) . For core (1, 7), the restricted P_2, O both increase by ac while C, T, P_1 are unchanged. Since $P_1 - O = ac$, its quartic is exactly the preceding $I + 2a^2c^2$. This proves (9) in every case. The included symbolic checker verifies all five identities exactly.

9.3 Passing to the right exterior and proving sharpness

Return to the right-side notation a, b, c, d of Section 4 [*Merge note:* source 74 writes “Section 3”]. Selecting all core vertices gives

$$c_0 = C, \quad c_1 = aP_2 + bP_1 + cP_U, \quad c_2 = TQ.$$

Let $A = a + c$, $B = b + c$, $p = P_1$, $q = P_2$. Then

$$c_1 = qA + pB - Oc, \quad Q \leq AB - c^2/2.$$

For $pq > 0$, the exact quadratic identity

$$\begin{aligned} & (qA + pB - Oc)^2 - (4pq - 2O^2)(AB - c^2/2) \\ &= \left(1 - \frac{O^2}{2pq}\right) (qA - pB)^2 + \frac{[O(qA + pB) - 2pqc]^2}{2pq} \geq 0 \end{aligned}$$

uses only $O^2 \leq pq$. Together with (9) it implies $c_1^2 \geq 3CTQ = 3c_0c_2$. If $pq = 0$, the same assertion follows by continuity, or directly from $O = 0$ and $CT = 0$.

For sharpness, take the complete 2×3 core, N common left-exterior vertices and M common right-exterior vertices, all with unit exterior activities. Then

$$c_0 = N, \quad c_1 = \binom{N}{2}M, \quad c_2 = \binom{N}{3}\binom{M}{2}, \quad \frac{c_1^2}{c_0c_2} = 3\frac{N-1}{N-2}\frac{M}{M-1} \rightarrow 3.$$

This completes Theorem 9.2.

10 Lorentzian transfer across a star core

Theorem 10.1 (Star-core transfer). *For a displayed $2+q$ cover with matching-rank-at-most-one core, both root orientations admit direct Lorentzian transfer proofs of ULC_{q+2} .*

This is now a special case of Theorem 2.1. We retain the independent proofs because the transfer operators and their interlacing certificates are useful beyond the full-lift construction.

10.1 A support-exact vertex-sum identity

Suppose G_1, G_2 intersect in just one vertex v of activity u , and write

$$p_1 = A(t) + utB(t), \quad p_2 = C(t) + utD(t).$$

Then

$$p_{G_1 \cup G_2} = AC + ut(BC + AD). \tag{13}$$

If v is absent, each component has balanced selected shores. If v is selected, exactly one component uses it, and that component is determined by its selected-vertex imbalance away from v : its opposite shore has one extra selected vertex. Thus the two root-using terms cannot count the same endpoint support, even if that support has several perfect matchings.

10.2 Stars centered on the two-vertex shore

Suppose every core edge is incident with $v = A_1$. Let G_1 have shores $L \cup \{v\}$ and B , and let G_2 have shores $\{v, A_2\}$ and R . Their intersection is just v . The right-complement basis construction for G_1 gives a Lorentzian polynomial

$$F(s, t, z) = A_h(s, t) + zB_h(s, t), \quad A_h = s^q A(t/s), \quad B_h = s^{q-1} B(t/s).$$

Here z is the root element variable; setting $z = ut$ gives the degree- q homogenization of p_1 .

For G_2 , write $C = 1 + dt$, $D = e + ft$. If A_2 has activity y , and the exterior class sums are a, b, c with common second moment δ , then

$$d = y(b + c), \quad e = a + c, \quad f = y(ab + ac + bc + \delta), \quad de - f = y(c^2 - \delta) \geq 0.$$

The homogeneous quadratic

$$K(s, t, z) = s^2 + dst + esz + ftz$$

is Lorentzian. Its Hessian is

$$\begin{pmatrix} 2 & d & e \\ d & 0 & f \\ e & f & 0 \end{pmatrix}, \quad \det = 2f(de - f) \geq 0.$$

For $f > 0$, its t, z principal submatrix has one positive and one negative eigenvalue; the determinant and eigenvalue interlacing exclude two positive eigenvalues of the full Hessian. For $f = 0$, $K = s(s + dt + ez)$ is a product of nonnegative linear forms.

The product FK is Lorentzian. Retain only its terms of degree at most one in z , then set $z = ut$. The output is

$$A_h(s^2 + dst) + ut[B_h(s^2 + dst) + A_h(es + ft)],$$

which is the degree- $(q + 2)$ homogenization of (13). Partial truncation preserves Lorentzian polynomials: polarize s, t , take the full multiaffine part (only powers z^2 are then removed), and identify the polarized variables again. Thus the output is Lorentzian and p_G is ULC_{q+2} .

10.3 Stars centered on the opposite shore

Suppose every core edge is incident with $v = B_1$. Let $G_1 = G[L, B]$ and let G_2 have shores A and $R \cup \{v\}$. For G_1 , the private-root variable now gives

$$F(s, t, z) = zA_h + tB_h, \quad A_h = s^{q-1} A(t/s), \quad B_h = s^{q-1} B(t/s).$$

The other private variables are s/v_b , the left variables are tu_l , and the basis polynomial is multiplied by $\prod_{b \neq v} v_b$. Hence F is Lorentzian and $uF(s, t, s/u) = s^q p_1(t/s)$.

For G_2 put $C_h = s^2 + c_1 st + c_2 t^2$ and $D_h = es + ft$. Give its two left vertices fixed activities α, β ; these are constants, distinct from the private-root indeterminate z . With exterior class sums a, b, c and second moment δ ,

$$c_1 = \alpha(a + c) + \beta(b + c), \quad c_2 = \alpha\beta(ab + ac + bc + \delta),$$

so

$$c_1^2 - 4c_2 = [\alpha(a + c) - \beta(b + c)]^2 + 4\alpha\beta(c^2 - \delta) \geq 0.$$

The rooted interlacing inequality is

$$c_1 e f - f^2 - c_2 e^2 \geq 0. \quad (14)$$

If v is adjacent only to the first left vertex, $e = \alpha$, $f = \alpha\beta(b + c)$ and the expression is $\alpha^3\beta(c^2 - \delta)$. The other singleton case gives $\alpha\beta^3(c^2 - \delta)$. If v is adjacent to both, $e = \alpha + \beta$, $f = \alpha\beta(a + b + c)$ and it is exactly

$$\alpha\beta \left[(a\alpha - b\beta + \frac{c}{2}(\alpha - \beta))^2 + \left(\frac{3c^2}{4} - \delta \right) (\alpha + \beta)^2 \right] \geq 0.$$

An isolated root has $e = f = 0$.

Factor $C_h = (s + \lambda_1 t)(s + \lambda_2 t)$, with $0 \leq \lambda_1 \leq \lambda_2$. If $e > 0$, (14) gives $\lambda_1 \leq f/e \leq \lambda_2$. Therefore

$$h(s, t) = \frac{stD_h}{C_h}$$

is a nonnegative linear combination of $st/(s + \lambda_i t)$ with positive total coefficient. Each such fraction maps $\mathbb{H}^2$ into $\mathbb{H}$, since its reciprocal is $1/t + \lambda_i/s$, of strictly negative imaginary part. Repeated λ_i give the same conclusion directly.

Consider the homogeneous degree-two operator

$$\mathcal{T}(P) = C_h P + stD_h \partial_z P.$$

It preserves real stability. At upper-half-plane values, a stable nonzero P has $\text{Im}(\partial_z P/P) \leq 0$, by its univariate factorization in z . A zero of $\mathcal{T}(P)$ with $D_h \neq 0$ would require this quotient to equal $-1/h$, which has strictly positive imaginary part. If $D_h = 0$, stable multiplication suffices.

The operator also preserves nonnegative coefficients. Restrict it to any finite coordinatewise degree box containing F ; its output lies in a larger degree box. Brändén–Huh’s Theorem 3.4 applies, so $\mathcal{T}(F)$ is Lorentzian. Finally

$$u\mathcal{T}(F)|_{z=s/u} = sA_h C_h + utB_h C_h + ustA_h D_h$$

is precisely the degree- $(q + 2)$ homogenization of (13). This proves the column-star case.

Every nonempty bipartite graph of matching rank one has all its edges incident with a common vertex. The two orientations therefore exhaust such cores. An empty core gives a product of ULC_q and ULC_2 polynomials, which is ULC_{q+2} by Lorentzian product closure. Theorem 10.1 follows.

11 Exact equality in the first Newton gap

[*Merge note:* The theorem of this section is *preorder-root-polytopes*’s Theorem 180.2 with Corollary 180.3 (Part XI, `hb:thm:firstidentity` and `hb:cor:equality`), proved there by an exact defect identity over a minimum vertex cover; the inequality is also Part XII’s Theorem 190.2 (`mr:thm:first`), proved there by the same Motzkin–Straus route, and its unweighted equality case is Part XII’s Theorem 190.3. Source 74’s proof is printed as a second route; no novelty is claimed. Source 72 proves the same statement by the same route (Theorem 14.6).] The following gives a useful structural interpretation of non-strictness. Its ingredient is the classical Motzkin–Straus bound [6]; we include the short argument in the form needed here.

Theorem 11.1. *For a positive weighted bipartite graph of matching rank $r \geq 2$,*

$$(r-1)a_1^2 - 2ra_2 = 0$$

if and only if every nontrivial connected component is a star and their weighted edge totals are all equal. In that case $p_G(t) = (1 + Et)^r$ for their common edge total E ; otherwise the first gap is strict.

Proof. Form the compatibility graph K whose vertices are the edges of G , with two vertices adjacent when the corresponding edges are disjoint. Give an edge xy weight $w_{xy} = u_x v_y$. Put $S = \sum w_e = a_1$ and

$$m_2 = \sum_{\{e,f\} \in E(K)} w_e w_f.$$

The clique number of K is r , and $m_2 \geq a_2$ because m_2 counts two-edge matchings rather than endpoint supports.

For completeness, the Motzkin–Straus bound is

$$m_2 \leq \frac{r-1}{2r} S^2.$$

To see it, merge the weights of two nonadjacent vertices onto whichever has the larger weighted neighbor sum. This preserves S and does not decrease m_2 . Continue until the positive support is a clique of size at most r , where the bound follows from Cauchy–Schwarz.

Equality in the asserted first gap forces equality in both preceding bounds. Since the original compatibility weights are all positive, they give an interior maximum of m_2 on the simplex of fixed sum. If K were not complete multipartite, it would contain an induced edge ik and a vertex j nonadjacent to both. The sum-preserving direction $e_i - 2e_j + e_k$ has strictly positive second derivative for m_2 , contradicting a local maximum. Hence K is complete multipartite. Its complement, the line graph of G , is a disjoint union of cliques. In a bipartite graph a nontrivial connected component whose edges are pairwise intersecting is a star. There are exactly r such components. The compatibility bound is then equality precisely when the total weights in the r parts are equal. Conversely, for stars support and matching counts coincide and the polynomial is the stated product. $\square$

12 Quantitative eventual strictness under common cover scaling

Fix a genuine $2 + 3$ cover whose core has matching rank two. Write every cover activity as w times its fixed positive relative activity. Let e_1, e_2 be the first two support coefficients of the induced core at those relative activities, and let ζ be the product of all five relative activities. For the all-core coefficients of Theorem 9.2, set $\gamma_i = \zeta c_i$.

Every size-five matching uses exactly one cover vertex per edge; a core edge would consume two cover vertices and preclude five edges. Thus the exterior-left graph matches all three B vertices and the exterior-right graph matches both A vertices, giving $c_2 > 0$. A core two-edge matching leaves one B vertex unmatched; that vertex has an exterior-left neighbor, so adding it gives $c_0 > 0$. Also $e_1, e_2 > 0$ and $e_1^2 \geq 4e_2$ by Lemma 3.1 for the core.

Counting selected cover vertices gives

$$\begin{aligned} a_0 &= 1, & a_1 &= e_1 w^2 + O(w), & a_2 &= e_2 w^4 + O(w^3), \\ a_3 &= \gamma_0 w^5 + O(w^4), & a_4 &= \gamma_1 w^5 + O(w^4), & a_5 &= \gamma_2 w^5. \end{aligned}$$

Consequently the leading gap limits are

$$\begin{aligned}\Delta_1^{(5)}/w^4 &\longrightarrow 4e_1^2 - 10e_2 \geq 6e_2 > 0, \\ \Delta_2^{(5)}/w^8 &\longrightarrow 6e_2^2 > 0, \\ \Delta_3^{(5)}/w^{10} &\longrightarrow 6\gamma_0^2 > 0, \\ \Delta_4^{(5)}/w^{10} &\longrightarrow 4\gamma_1^2 - 10\gamma_0\gamma_2 \geq 2\gamma_0\gamma_2 > 0.\end{aligned}$$

The last inequality uses the sharp constant three from Theorem 9.2. All four gaps are therefore strictly positive eventually. This is stronger than the already proved non-strict ULC_5 property at every $w > 0$.

For either this scaling or Corollary 8.1, write each gap as

$$\Delta_k(w) = L_k w^{d_k} + \sum_{j < d_k} b_{kj} w^j, \quad L_k > 0.$$

A directly computable sufficient threshold is

$$W = 1 + \max_k \frac{\sum_{j < d_k} \max(-b_{kj}, 0)}{L_k}. \quad (15)$$

For $w \geq W$, every negative lower term is bounded in magnitude by its coefficient times w^{d_k-1} , so every gap is positive. This threshold depends on the graph and its fixed field; no uniform bound is asserted.

For a rank-five minimum cover of split $0 + 5$ or $1 + 4$, Lemma 3.2 already gives ULC_5 at every scale. For a $2 + 3$ cover, Theorem 2.1 does likewise; the calculation above adds strictness when its core has matching rank two. Thus all choices of minimum cover are covered, with the relative activities fixed before scaling.

13 Exact verification and scope of the conclusions

The proofs above are universal arguments. Their exact checks are independently implementable and do not use numerical root searches or numerical optimization as evidence for a theorem.

- The full transversal lift is checked by Boolean basis enumeration on 180 weighted graphs with $q = 0, 1, 2, 3, 4, 5$. It tests 140450 candidate basis subsets and compares the complete left-subset marginal, before diagonalizing the left variables.
- An independent finite-field representation check covers all 64 labeled 2×3 cores and 73441 basis determinants, comparing the full weighted support polynomial exactly. A separate generic-line check covers 192 graphs and 23553 determinants, including 54 rank-deficient cores.
- The union formula is checked on 1352 weighted graphs with $q = 1, 2, 3, 4$, including overlapping, nested and empty core neighborhoods.
- The rank-five quartic certificate checks all 4424 stated coefficient cases, all 46 labeled rank-two cores, the four moment-corner identities, the exceptional correction, and the right-side quadratic identity.
- The star-core and scaling checks cover 820 weighted star-core graphs for $q = 1, \dots, 5$ and 230 rank-five common-scaling instances. All displayed interlacing sums of squares, grading identities and computed strictness thresholds are verified exactly.

- The negative-family verifier checks complete endpoint support counts for seven core shapes and the universal exterior population choice for all $3 \leq a, b \leq 20$. The rank-six witness and its exact negative gap are included.

The package contains the Python sources, fixed seeds and JSON outputs. Only SymPy is needed beyond the Python standard library. Run `python3 checks/run_all.py`; the JSON outputs are regenerated in `data/`. The LaTeX sources compile with ordinary pdfLaTeX; `build_local.sh` also handles the local TeX configuration used to generate this PDF. [*Merge note:* In this report the scripts are shipped as `code/74-weighted-rank-five-*.py` and the outputs as `data/74-weighted-rank-five-*.json`; the delivered layout (`checks/`) is not kept, so rerun on a copy of the delivered layout as the README explains. `build_local.sh` is not shipped, and the eight section files are inlined into this report's `article.tex`.]

Röhrle–Ulirsch’s bimatroid theorem [7] normalizes by the smaller ambient shore size. The new lift replaces that order by $q + 2$, and hence by matching rank when the cover is minimum. The polynomial is the perfectly matchable set polynomial of [5], not a matching count with multiplicities. A bounded primary-literature comparison located no prior universal $2 + q$ -cover or weighted rank-five theorem; global priority remains unclaimed.

Higher-rank unrestricted unit-weight and one-shore-only questions are outside this report. [*Merge note:* Both are answered negatively by later sources of this batch (sources 71 and 73, Part II).] The proof uses Lorentzian matroid bases; it does not assume that transversal matroids have the half-plane property. The cover obstruction in Corollary 5.1 is sharp.

14 Strictness and equality for weighted matching-support polynomials (source 72)

Abstract of source 72, as delivered. For positively weighted bipartite graphs having a minimum vertex cover that meets one shore in at most two vertices, we characterize equality in every rank-normalized ultra-log-concavity inequality. Equality at any one index occurs precisely when all nontrivial connected components are stars with equal total edge activities. In that case the polynomial is $(1 + Et)^r$; otherwise every inequality is strict. In particular, this characterizes equality for every bipartite graph of matching rank at most five and gives strictness for connected graphs in that range with rank at least two. The proof combines the recently obtained rank-compression lift with a direct weighted version of the Chan–Pak contraction criterion and explicit rank-two contraction witnesses. We also give a short Motzkin–Straus proof of the same equality characterization for the first inequality at every rank, without a cover restriction.

14.1 Definitions and conclusions

Let $G = (X, Y; E)$ be a finite simple bipartite graph, with positive vertex activities u_x, v_y . Its matching-support polynomial is

$$p_G(t) = \sum_{k=0}^r a_k t^k, \quad a_k = \sum_{\substack{I \subseteq X, J \subseteq Y \\ |I|=|J|=k \\ G[I, J] \text{ perfectly matchable}}} u_I v_J, \quad r = \nu(G).$$

An endpoint support is counted once, regardless of how many perfect matchings it has. Positive activities and a matching of size r imply $a_k > 0$ for every $0 \leq k \leq r$. The order- r normalized gap is

$$\Delta_k = k(r - k)a_k^2 - (k + 1)(r - k + 1)a_{k-1}a_{k+1}, \quad 1 \leq k \leq r - 1. \quad (16)$$

The condition $\Delta_k \geq 0$ for all k is ULC_r . A star includes a single edge; isolated vertices are disregarded. The *edge total* of a star is the sum of $u_x v_y$ over its edges, equivalently its center activity times the sum of its leaf activities.

Theorem 14.1 (Equality classification). *Suppose $r \geq 2$ and G has a minimum vertex cover meeting one shore in at most two vertices. The following are equivalent:*

1. $\Delta_k = 0$ for at least one $k \in \{1, \dots, r-1\}$;
2. every nontrivial component of G is a star and all its edge totals are equal to one common $E > 0$;
3. $p_G(t) = (1 + Et)^r$ for some $E > 0$.

If these conditions fail, then $\Delta_k > 0$ for every $1 \leq k \leq r-1$.

Corollary 14.2. *Theorem 14.1 applies to every bipartite graph with $2 \leq r \leq 5$. Every connected graph in that range with rank at least two is strictly ULC_r , for every positive activity assignment.*

Indeed a minimum cover has size r by König's theorem, so its smaller shore contribution is at most two when $r \leq 5$.

This is a companion to the rank-compression theorem [12], which proves ULC_r under the cover hypothesis. [Merge note: [12] is source 74: Theorem 2.1 and Corollary 2.2.] The lift is recalled precisely below. The unweighted matroid equality characterization is due to Chan–Pak [3, Theorem 1.3]. Their Theorem 1.11 shows that single-index equality need not imply total equality in general matroids; our conclusion uses the special graph lift. We prove the needed real-weight version directly, rather than infer strictness by a limit from rational weights. The arguments are ordinary mathematical proofs, not externally refereed or proof-assistant formalized. No exhaustive publication-priority claim is made.

14.2 A weighted matroid equality criterion

Let M be a rank- r matroid on a ground set $E = \mathcal{R} \sqcup \mathcal{B}$, with positive element weights ω_e . Define

$$P_j = \binom{r}{j}^{-1} \sum_{\substack{S \text{ basis of } M \\ |S \cap \mathcal{R}| = j}} \prod_{e \in S} \omega_e.$$

Loops are allowed. Parallel classes below refer only to nonloops.

Theorem 14.3 (Weighted contraction criterion). *Let $1 \leq a \leq r-1$ and $P_a > 0$. Then $P_a^2 = P_{a-1}P_{a+1}$ if and only if there is a common $\lambda > 0$ such that, for every independent I with*

$$|I| = r-2, \quad |I \cap \mathcal{R}| = a-1,$$

every nonloop parallel class J of M/I satisfies

$$\sum_{e \in J \cap \mathcal{R}} \omega_e = \lambda \sum_{e \in J \cap \mathcal{B}} \omega_e. \tag{17}$$

In particular, one such contraction with a monochromatic nonloop parallel class proves strictness at index a .

We use two established Lorentzian facts from Brändén–Huh [1]: a matroid basis polynomial is Lorentzian (Theorem 3.10), and every nonnegative directional derivative preserves the class (Corollary 2.11). A quadratic Lorentzian polynomial has a Hessian with at most one positive eigenvalue.

Lemma 14.4 (Kernel form of equality). *If a real symmetric matrix H has at most one positive eigenvalue and $y^T H y > 0$, then its quadratic form is negative semidefinite on $\{z : z^T H y = 0\}$. If also $z^T H z = 0$, then $H z = 0$.*

Proof. Otherwise the span of y and a positive vector in its H -orthogonal complement would have two positive directions. On that complement, zero quadratic value for a negative semidefinite form implies orthogonality to the whole complement. Together with orthogonality to y , it implies $H z = 0$. $\square$

Proof of Theorem 14.3: necessity. Let F be the unweighted basis generating polynomial. Let $f_e = \omega_e$ on $\mathcal{R}$ and zero on $\mathcal{B}$, and let $g_e = \omega_e$ on $\mathcal{B}$ and zero on $\mathcal{R}$. Write D_f, D_g for directional differentiation. The matrix

$$C = \text{Hess} (D_f^{a-1} D_g^{r-a-1} F) \quad (18)$$

has at most one positive eigenvalue. Direct differentiation gives

$$f^T C f = r! P_{a+1}, \quad f^T C g = r! P_a, \quad g^T C g = r! P_{a-1}. \quad (19)$$

At equality both neighbors are positive. Set

$$\lambda = P_a / P_{a-1} = P_{a+1} / P_a, \quad z = f - \lambda g.$$

Then $z^T C g = z^T C z = 0$, so Lemma 14.4 gives $C z = 0$.

Suppose first $r \geq 3$ and $a \geq 2$. For each red nonloop i , put

$$M_i = \text{Hess} (D_f^{a-2} D_g^{r-a-1} \partial_i F). \quad (20)$$

The exponents are nonnegative, and the differentiated polynomial has degree two. Commuting derivatives yields

$$C = \sum_{i \in \mathcal{R}} \omega_i M_i, \quad (C z)_i = z^T M_i f. \quad (21)$$

Loop terms are zero.

Every mixed value $f^T M_i g$ is strictly positive. Here is the required combinatorial verification. Equality supplies bases with red counts $a - 1$ and $a + 1$. Insert the nonloop i into each by one basis exchange, or keep it if already present. The resulting bases containing i have red counts at most a and at least $a + 1$, respectively. Bases of M/i are connected by exchanges, and each exchange changes the red count by at most one. Consequently M/i has a basis with exactly $a - 1$ red elements. Its positive weight contributes to $f^T M_i g$.

Equations (21) and $C z = 0$ now give

$$f^T M_i f = \lambda g^T M_i f > 0.$$

Each M_i has at most one positive eigenvalue, so Lemma 14.4 implies $z^T M_i z \leq 0$. Their positive-weighted sum is $z^T C z = 0$; hence every term is zero. The same lemma gives $M_i z = 0$. By (19) applied to M/i , equality propagates to index $a - 1$ of that contraction with the same ratio λ .

Contract red elements until the index is one. Then reverse colors and contract blue elements until rank two. Color reversal replaces λ by $1/\lambda$, and reversing back restores λ . This covers

$a = 1$ by starting with blue contractions and $a = r - 1$ by red contractions alone. Any independent I of the specified color counts can be ordered with red elements first and blue elements second; all prefixes remain independent, so each element contracted is a nonloop. The result therefore applies to every required M/I .

At rank two the Hessian has entry one on independent pairs and zero elsewhere. Its kernel equation for z says that the sum of z over the nonloops outside each parallel class is zero. Summing over the at least two classes makes the total over nonloops zero, and hence every class sum is zero. Loops have zero rows and impose no condition. This is precisely (17). $\square$

Proof of Theorem 14.3: sufficiency. For each specified independent I , let H_I be the Hessian of the basis polynomial of M/I . Condition (17) means $H_I z = 0$. Expansion of the directional derivatives gives

$$C = (a - 1)!(r - a - 1)! \sum_I \omega_I H_I.$$

Thus $Cz = 0$. Since $f^T Cg = r!P_a > 0$ and $\lambda > 0$, the three values in (19) are in geometric progression with ratio λ . This proves equality. $\square$

This proof establishes the criterion for arbitrary positive real weights. Parallel cloning proves its rational-weight version, but continuity alone would not justify the necessity or strictness conclusions for real weights.

14.3 The rank compression lift and its two bases

[Merge note: The lift recalled here is source 74's, Section 5; its six basis classes are the six cases there.] Suppose a minimum cover is $A \sqcup B$, where $A = \{A_1, A_2\}$, $|B| = q$ and $r = q + 2$. Put $L = X \setminus A$, $R = Y \setminus B$; there are no L - R edges. A size- r matching uses exactly one cover vertex per edge. Thus L matches all B and R matches both A vertices.

Partition R by neighborhoods $\{A_1\}, \{A_2\}, A$. Let their activity sums be a, b, c , and let the common-class individual activities be r_j . Put

$$d = \sum_{i < j} r_i r_j, \quad Q = ab + ac + bc + d > 0.$$

The lift of [12] is a rank- r transversal matroid with slots $B \sqcup \{e_1, e_2\}$. It has private columns for B , ordinary columns for L on their original neighborhoods, augmented core columns on $S_i \cup \{e_i\}$, private plane columns on e_1, e_2 , and common plane columns on both extra slots. Its red elements are the ordinary and augmented core columns; its blue elements are private and plane columns.

Give the red elements their original left activities, the private blue element at b weight $1/v_b$, and the plane elements weights

$$b/Q, \quad a/Q, \quad r_j/Q,$$

respectively. Delete zero-weight plane elements before applying Theorem 14.3. The positive value of Q guarantees a positive-weight independent pair of plane elements remains. The colored basis polynomial, multiplied by the harmless positive constant $Q \prod_b v_b$, is exactly

$$\sum_{k=0}^r a_k s^{r-k} t^k. \tag{22}$$

The common positive prefactor does not affect equality or strictness of the binomial-normalized coefficients. This identity counts bases once, not matching injections. It follows

by classifying selected augmented core and plane columns: their six possible counts are $(0, 2), (1, 1), (1, 2), (2, 0), (2, 1), (2, 2)$. The one-plane/two-core case uses the union of the two possible core-row completion profiles. The independent plane-pair activity sum is $1/Q$, while the eligible single-plane sums for A_1, A_2 are $(a + c)/Q, (b + c)/Q$. These are exactly the support alternatives with zero, one or two selected vertices in R .

We use the following generic representation, which is the usual transversal representation after harmless column normalization. Let V have dimension q and let E_0 have basis e_1, e_2 . Private B columns are coordinate vectors of V . Ordinary L columns have independent generic entries on their neighborhoods in V and zero projection to E_0 . The core columns are

$$(v_1, e_1), \quad (v_2, e_2),$$

with independent generic top entries on S_1, S_2 . Plane columns are $(0, e_1), (0, e_2)$ and $(0, \alpha_j e_1 + \beta_j e_2)$, with generic independent coefficients. A permitted square submatrix has nonzero determinant exactly when it admits a matching; different matching monomials do not cancel.

Choose two positive-weight independent plane columns P, Q_0 . Together with the q private B columns they form a blue basis. Order them so that the coefficient of e_1 along P and of e_2 along Q_0 are nonzero. Such an ordering exists because the two-by-two change-of-basis matrix is invertible.

Choose q ordinary L columns matching all B ; together with the two core columns they form a red basis. Assign each ordinary column to its matching B coordinate and the two core columns to P, Q_0 . For every subset of assigned positions, the corresponding principal row-column submatrix is nonsingular over the generic field: the ordinary block contains its matching monomial, and the bottom block is empty, one of the selected nonzero diagonal entries, or the full invertible two-by-two block. The bottom-left block is zero.

14.4 Contractions witnessing strictness

We apply Theorem 14.3 at an arbitrary index $k \in \{1, \dots, r - 1\}$. The following construction gives an independent I with $r - 2$ elements and $k - 1$ red elements. Leave two assigned positions i, j uncontracted. Select any $k - 1$ of the other $r - 2$ red basis elements, and contract them together with the blue basis elements at every remaining position except i, j . The principal-minor observation above shows that I is independent and the two uncontracted blue coordinate images are independent in M/I .

14.4.1 An ordinary column with two neighbors

Suppose some $u \in L$ has two B neighbors. Since the ordinary-column transversal matroid has rank q , its nonloop u extends to a basis. Choose a matching of this basis assigning u to i , and choose another neighbor j of u . Leave i, j uncontracted in the preceding construction, excluding both u and the ordinary red basis column assigned to j from I .

The contraction set and all blue columns are independent of every incidence parameter in column u . Its two entries at i, j are independent parameters multiplying independent quotient coordinate vectors. For any nonloop blue element b , the determinant of the images of u, b is a nonzero polynomial in those two parameters: its two coefficients cannot both vanish. Hence u is not parallel to any nonloop blue element in the generic matroid. Its parallel class is monochromatic red. Theorem 14.3 proves strictness at k . The construction works for every k .

14.4.2 A nonempty core

Suppose a core edge $A_1 B_i$ exists, and let $u = (v_1, e_1)$ be the augmented core column. Choose an ordinary basis matching all B . Leave the coordinate B_i and the bottom coordinate P

uncontracted; exclude u and the ordinary column matched to B_i from the red contraction choices.

The contraction set and all blue elements are independent of the top entries of u . Its entry at B_i is an independent parameter multiplying the surviving B_i coordinate. Moreover the image of $u_0 = (0, e_1)$ is independent of that coordinate. Indeed the $q - 1$ contracted top columns together with B_i span V . The one contracted bottom projection is either Q_0 , not parallel to e_1 by the chosen ordering, or e_2 from the other core column. In either case adding e_1 gives a nonzero full determinant.

For any nonloop blue element b , if its determinant with the B_i coordinate is nonzero, the corresponding top parameter makes the determinant with u nonzero. Otherwise b is parallel to B_i , and its determinant with u_0 is nonzero. Thus the determinant of u, b is not identically zero in either case. The class of u is monochromatic red, again proving strictness at every k . A core edge incident with A_2 is handled by interchanging labels.

These arguments include empty singleton plane classes: only a positive-weight plane basis was used, not the presence of both private plane elements. They concern generic representation parameters, independent of the arbitrary positive activity values.

14.5 Completing the graph classification

Suppose neither witness case occurs. The core is empty, and every ordinary left vertex has at most one B neighbor. Since L matches all B , these components are stars with positive edge totals $E_1, \dots, E_q$. The remaining graph is on A_1, A_2 and R . If the two left activities are ξ, η , its support polynomial is

$$1 + Et + \xi\eta Qt^2, \quad E = \xi(a + c) + \eta(b + c).$$

Its discriminant equals

$$[\xi(a + c) - \eta(b + c)]^2 + 4\xi\eta(c^2 - d) \geq 0. \quad (23)$$

Thus it factors as $(1 + \lambda_1 t)(1 + \lambda_2 t)$ with positive parameters. If $c > 0$, then $c^2 - d \geq c^2/2 > 0$, so the two parameters differ. If $c = 0$, the whole graph is a disjoint union of stars, with additional edge totals $\xi a, \eta b$.

Lemma 14.5. *For $r \geq 2$ and $p(t) = \prod_{i=1}^r (1 + \lambda_i t)$ with all $\lambda_i > 0$, equality in any one order- r normalized gap occurs if and only if all λ_i are equal.*

Proof. Use a direct sum of r rank-one matroids, each with a red element of weight λ_i and a blue element of weight one. For any two classes and any index k , contract one element from each other class, choosing exactly $k - 1$ red elements. Theorem 14.3 forces the two remaining class ratios to agree. Thus all λ_i agree. The converse follows from $(1 + \lambda t)^r$. $\square$

Equation (23) and Lemma 14.5 complete the classification for genuine $2 + q$ covers. They also handle $q = 0$ directly.

For a graph whose active right shore has size $r = \nu(G)$, use its ordinary right-complement transversal matroid with a blue coordinate basis. The first witness construction applies whenever a left vertex has two neighbors; otherwise the graph is a union of stars and Lemma 14.5 applies. This proves the same classification for pure minimum covers.

For a minimum cover split $1 + (r - 1)$, every nonisolated right vertex outside the cover is a leaf at its single left-cover vertex. Replace all these leaves by one with activity equal to their sum. [*Merge note:* This is the weighted leaf compression of *preorder-root-polytopes*, Lemma 206.1 (`lc:lem:compression`) of Part XIII.] There is at least one such leaf, since

otherwise the rank would be at most $r - 1$. The support polynomial is unchanged and the compressed graph has an active right shore of size r . It is a union of stars exactly when the original graph is, and expansion of the compressed leaf preserves its component edge total. The pure-shore result therefore transfers. Shore reversal and König’s theorem prove Theorem 14.1 and Corollary 14.2.

14.6 The first inequality at every rank

The following result requires no restriction on the cover and does not depend on the lift or on higher-gap ultra-log-concavity. [*Merge note:* Theorem 14.6 is *preorder-root-polytopes*’s Theorem 180.2 with Corollary 180.3 (Part XI, `hb:thm:firstidentity`, `hb:cor:equality`), and the inequality is Part XII’s Theorem 190.2; it is also printed above as source 74’s Section 11. Source 72’s Motzkin–Straus proof is printed as a second route; as source 72 itself says below, it is not a new optimization theorem.]

Theorem 14.6. *For every positively weighted bipartite graph of matching rank $r \geq 2$,*

$$(r - 1)a_1^2 \geq 2ra_2.$$

Equality occurs exactly for disjoint stars with equal edge totals.

Proof. Give edge xy weight $w_{xy} = u_x v_y$ and put $S = \sum_e w_e = a_1$. Let m_2 count pairs of disjoint edges with their weight products. Every feasible two-edge endpoint support has at least one such matching, with the same product, so $a_2 \leq m_2$.

Form the edge-compatibility graph: its vertices are edges of G , and its adjacency means disjointness. Its clique number is r . The Motzkin–Straus theorem [6] gives $m_2 \leq (r - 1)S^2/(2r)$. The upper bound also has a short proof: merge the masses of a nonadjacent pair onto the endpoint of larger weighted neighbor sum, without decreasing the objective, until the positive support is a clique; then apply Cauchy–Schwarz to the masses on that clique.

If the support inequality is equality, the original strictly positive edge-weight vector is an interior global maximizer of the compatibility objective. If that graph were not complete multipartite, it would contain an edge ik and a vertex j nonadjacent to both. The fixed-sum direction $e_i - 2e_j + e_k$ has positive second variation, contradicting an interior maximum. Thus the complement, which is the line graph of G , is a disjoint union of cliques. Each nontrivial component of G has pairwise intersecting edges; in a bipartite graph such a component is a star. If their totals are $E_1, \dots, E_r$, equality in $m_2 = (S^2 - \sum E_i^2)/2$ requires all totals equal. The converse follows from $p_G(t) = \prod_i (1 + E_i t)$. $\square$

Theorem 14.6 is a graph-support consequence of the classical Motzkin–Straus theorem, not a new quadratic optimization theorem. It should not be used to infer higher-gap strictness by coefficient comparison.

14.7 Verification and scope

The accompanying exact script [*Merge note:* shipped as `code/72-strictness-verify.py` and `code/72-strictness-verify_witnesses.py`, with outputs `data/72-strictness-*`] enumerates all 512 simple 3×3 bipartite graphs under three positive integer activity assignments. Excluding ranks zero and one, it verifies 1434 weighted graph cases and 2175 individual gaps: 42 balanced-star cases have all gaps zero, and 1392 cases have all gaps positive. An independent Boolean-transversal test examines 375 genuine cover graphs with $q = 0, \dots, 5$ and verifies 1589 monochromatic contraction witnesses; four star cases have no such structural witness. These are finite regression tests, not proof of the universal statements.

The novelty boundary is the separate rank-compression report [12]: the present companion adds equality classification and strictness to its nonnegative-gap theorem. The unweighted matroid characterization is credited to [3]; the weighted criterion above is proved directly, and no broad bibliographic priority claim is made for that extension. The full graph classification uses the particular generic contraction witnesses, not a general claim about all transversal or represented matroids.

All activity assumptions in the strictness theorems are positive. If some activities vanish, delete those vertices and reconsider the surviving graph and its actual matching rank. A limit of strict inequalities need not be strict. In particular no unspecified continuity argument is used for equality cases.

15 Stability of two-by-two covers and weighted rank four (source 76)

Abstract of source 76, as delivered. Every bipartite graph with a vertex cover consisting of two vertices on each shore has a real stable complemented-row matching-support polynomial. In particular, its matching-support polynomial is real-rooted under arbitrary positive vertex activities. The proof reduces to six four-vertex core types. Two independent-column covariance constructions treat the matching and three-edge cores; a two-block polarization argument treats the complete core. Together with the smaller-shore inequality and exact leaf compression, this settles rank-normalized ultra-log-concavity for every bipartite matching-support polynomial of rank at most four, including arbitrary weights on both shores. All finite certificates are exact, and the universal argument is given in full.

Status and provenance. This note answers the incoming ProveIt manuscript’s question “Finish rank four” [10], strengthening that question to arbitrary positive vertex activities. It follows the separate leaf-compression note [11], which settled weighted rank three and all singleton-side minimum covers. The conclusions are proofs in ordinary mathematics, not Lean-certified results or independently refereed theorems. The stability machinery is classical [2]; novelty is asserted only relative to the inspected ProveIt drafts, not as a claim of worldwide priority. [*Merge note:* [10] is the manuscript printed as Part XII of *preorder-root-polytopes*, and “Finish rank four” is its Research question 93 (**mr:q:rankfour**); [11] is Part XIII. Source 74 (Corollary 2.2) answers the same question through matching number five, so Corollary 15.2 below is printed as a second route. The real-stability Theorem 15.1 is not implied by source 74, which asserts no real-rootedness. The tools of Section 15.2 are Part VI’s (see the notes there).]

15.1 Definitions and the conclusions

Let $G = (X, Y; E)$ be a finite simple bipartite graph. A pair (I, J) , with $I \subseteq X$, $J \subseteq Y$ and $|I| = |J|$, is *feasible* when $G[I, J]$ has a perfect matching. For positive activities u_x, v_y , define

$$p_G(t; u, v) = \sum_{(I, J) \text{ feasible}} u_I v_J t^{|I|} = \sum_{k=0}^r a_k t^k, \quad r = \nu(G).$$

Each pair is counted once, regardless of the number of perfect matchings. Thus $p_{K_{2,2}}(t; \mathbf{1}, \mathbf{1}) = 1 + 4t + t^2$.

A degree- r polynomial with nonnegative coefficients and no internal zeros is ULC_r when $a_k / \binom{r}{k}$ is log-concave. Equivalently,

$$k(r-k)a_k^2 \geq (k+1)(r-k+1)a_{k-1}a_{k+1} \quad (1 \leq k < r). \quad (24)$$

Write $\mathbb{H} = \{z \in \mathbb{C} : \text{Im } z > 0\}$. A nonzero real polynomial is *real stable* if it has no zero when every variable belongs to $\mathbb{H}$.

Introduce a private copy $\bar{x}$ of each $x \in X$ and put

$$F_G(z; y) = \sum_{(I, J) \text{ feasible}} z^{\overline{X \setminus I}} y^J. \quad (25)$$

This is the basis polynomial of the transversal matroid in which $\bar{x}$ is adjacent only to x and each $y \in Y$ is adjacent to its neighbours in G . It is homogeneous of degree $|X|$ and multiaffine.

Theorem 15.1 (Two vertices on each shore). *If G has a vertex cover $A \sqcup B$ with $A \subseteq X$, $B \subseteq Y$ and $|A| = |B| = 2$, then F_G is real stable. Consequently $p_G(t; u, v)$ has only real negative zeros for every positive activity assignment and is $\text{ULC}_{\deg p_G}$.*

Corollary 15.2 (Weighted rank four). *Every bipartite graph with matching rank $r \leq 4$ has ULC_r matching-support polynomial for arbitrary positive vertex activities. For nonnegative activities the same assertion holds with the actual surviving degree as normalization order. In rank four, the three inequalities are*

$$3a_1^2 \geq 8a_2, \quad 4a_2^2 \geq 9a_1a_3, \quad 3a_3^2 \geq 8a_2a_4.$$

The corollary does not assert real-rootedness for every rank-four graph. Its real-rootedness input concerns the two-plus-two cover class; other cover types use ultra-log-concavity directly. [*Merge note:* Corollary 15.2 is the case $\nu \leq 4$ of source 74's Corollary 2.2.]

15.2 Stability tools and the six cores

For a graph, also define its signed unmatched-support polynomial

$$\Phi_G(z) = \sum_{S \text{ perfectly matchable}} (-1)^{|S|/2} \prod_{v \notin S} z_v.$$

For bipartite G , the exact inversion identity is

$$\Phi_G(z) = \left(\prod_{y \in Y} z_y \right) F_G((z_x)_{x \in X}; (-1/z_y)_{y \in Y}). \quad (26)$$

The map $z \mapsto -1/z$ preserves $\mathbb{H}$, and the inverse substitution has the same form. Thus F_G is stable if and only if Φ_G is stable. Stability here concerns these polynomials, not the raw matched-support generating function. [*Merge note:* Identity (26) and this equivalence are *preorder-root-polytopes*'s Theorem 85.3 (Part VI, `ade:mat:thm:stable-bridge`).]

If F_G is stable, substitute $z_x = 1/u_x$ and $y_y = tv_y$ in (25). Boundary specialization and positive scaling preserve stability, possibly giving zero; here the constant term excludes zero. Therefore

$$p_G(t; u, v) = \left(\prod_{x \in X} u_x \right) F_G((1/u_x)_x; (tv_y)_y) \quad (27)$$

is a real stable univariate polynomial, so it is real-rooted. Its nonnegative coefficients and constant term one force all roots to be negative. Newton's inequalities give (24) with its actual degree.

Lemma 15.3 (Leaf attachment and vertex sums). *Attaching a leaf to a graph with stable Φ preserves stability. More generally, if two graphs with stable Φ intersect in one vertex and have no other connecting edges, their union has stable Φ .*

Proof (pointer). The vertex-sum statement is *preorder-root-polytopes*'s Theorem 93.2 (Part VI, `ade:mat:thm:vertex-stable`), proved there from the identity $\Phi_G = z_v A_1 A_2 + B_1 A_2 + A_1 B_2$ and $\text{Im}(B_i/A_i) \geq 0$. Attaching a leaf is the one-vertex sum with an edge, whose signed polynomial $z_v z_w - 1$ is stable. [*Merge note:* Source 76 gives the same two arguments (for the leaf, $\Phi_{G+w} = z_w \Phi_G - \partial_v \Phi_G$ and the sign of the logarithmic derivative); they are not reprinted.] $\square$

Delete isolated vertices and all exterior vertices whose neighbourhood in the opposite part of the cover is a singleton. These are pendant leaves and can be reattached by Lemma 15.3. The remaining exterior vertices form sets $L = \{L_1, \dots, L_n\} \subseteq X$ and $R = \{R_1, \dots, R_m\} \subseteq Y$, where every L_i is adjacent to both B_1, B_2 and every R_j to both A_1, A_2 . There are no edges between L and R .

Up to row and column permutations and transposition, the four-vertex core $G[A, B]$ has six types: empty, one edge, two adjacent edges, two disjoint edges, three edges, or four edges. For the first three types the graph is a vertex sum of complete bipartite graphs $K_{2,k}$, bridges and isolated vertices. The basis polynomial of $K_{2,k}$, with the two-vertex shore on the left, is the elementary symmetric polynomial e_2 on $k+2$ variables, hence is stable. Equation (26) removes dependence on orientation. Lemma 15.3 handles these three cases. [*Merge note:* Stability of $K_{2,k}$ is *preorder-root-polytopes*'s Theorem 94.1 (`ade:mat:thm:biclique`), and these three core types are covered by its Corollary 94.3 (`ade:mat:cor:block-family`). Only the matching, three-edge and complete cores treated below are new.]

15.3 Independent columns and Boolean minor weights

Lemma 15.4 (Expected determinant). *Let the columns K_j of a real random matrix be independent and have finite second moments. Then*

$$\mathbb{E} \det(\text{diag}(z) + K \text{diag}(y) K^T)$$

is real stable.

Proof (pointer). This is *preorder-root-polytopes*'s Theorem 91.2 (Part VI, `ade:thm:expected`). [*Merge note:* Source 76's proof is the one given there: stability of $\det(\text{diag}(z) + \sum_j t_j C_j)$ with $C_j = \mathbb{E}(K_j K_j^T)$, the operators $1 + y_j \partial_{t_j}$, specialization $t_j = 0$, and independence of the columns. It is not reprinted.] $\square$

This is standard stability machinery, also used in the repository's even-multi-theta argument [8]; the covariance choices below are the particular constructions needed here.

Cauchy–Binet expands the expected determinant as

$$\sum_{|I|=|J|} \mathbb{E} \det(K[I, J])^2 z^{X \setminus I} y^J. \quad (28)$$

It therefore equals F_G if the squared-minor expectation is one on every feasible support and zero on every infeasible support.

Set

$$q = 2^{-1/2}, \quad \rho = \sqrt{q} = 2^{-1/4}.$$

Rows are indexed by $A_1, A_2, L_1, \dots, L_n$. Every R -column has covariance supported on the two A rows, with matrix

$$\begin{pmatrix} 1 & q \\ q & 1 \end{pmatrix}. \quad (29)$$

All columns are independent. Within either B -column, the covariance on the L rows is $(1 - q)I + qJ$, so diagonal entries are one and distinct-row entries are q .

15.3.1 The matching core

Use core edges A_2B_1, A_1B_2 . In the B_1 column, the A_1 entry is zero, the A_2 variance is one, and its covariance with each L_i is ρ . In the B_2 column, the A_2 entry is zero, the A_1 variance is one, and its covariance with each L_i is $-\rho$.

These matrices are positive semidefinite in every dimension. For independent centred variance-one variables $Z, Z_1, \dots, Z_n$, take the nonzero A entry to be Z or $-Z$, respectively, and take

$$K_{L_i} = \rho Z + \sqrt{1-q} Z_i.$$

Use independent copies for different columns. Matrix (29) is positive definite, so it too is a covariance matrix.

15.3.2 The three-edge core

Use edges A_1B_1, A_2B_1, A_1B_2 . In the B_1 column, the A -block is (29), and the covariance vector from (A_1, A_2) to any L_i is $(\rho/\sqrt{2}, 0)$. For independent centred variance-one Z, W, Z_i , realize it by

$$K_{A_1} = (Z + W)/\sqrt{2}, \quad K_{A_2} = W, \quad K_{L_i} = \rho Z + \sqrt{1-q} Z_i.$$

In the B_2 column, take $K_{A_1} = Z$, $K_{A_2} = 0$, and the same formula for K_{L_i} , using a fresh independent family. This realizes variance one at A_1 and covariance ρ from A_1 to every L_i .

15.3.3 Verification of all minor types

Both constructions have the graph's zero pattern. Therefore unsupported minors vanish identically. Supported minors use at most two L rows and two R columns, and have size at most four.

Size-one supported minors have expectation one. A size-two supported minor has either one matching, giving one by column independence and unit entry variances, or two matchings around a four-cycle. In the latter case the expectation is $2 - 2q^2 = 1$, except for the path core's $\{A_1, L_i\} \times \{B_1, B_2\}$ minor, where it is

$$2 - 2(\rho/\sqrt{2})\rho = 1.$$

For size three, the only case without a forced row or column uses rows A_1, A_2, L_i and columns B_1, B_2, R_j . For the matching core, the two determinant terms have diagonal contributions one each, and their cross term is $-2\rho^2q$. Thus

$$\mathbb{E} \det(K[I, J])^2 = 2 - 2\rho^2q = 1.$$

For the path core there are three matching terms. Their three pairwise cross terms are

$$-2(\rho/\sqrt{2})\rho, \quad -2q^2, \quad 0,$$

so the expectation is again one. In every other supported size-three case a forced entry lies in a column independent of the remaining minor; its variance is one and the remaining size-two minor has squared expectation one.

Every supported size-four minor uses both A rows, both B columns, two L rows and two R columns. Up to sign its determinant factors as

$$\det K[\{L_i, L_j\}, \{B_1, B_2\}] \det K[\{A_1, A_2\}, \{R_s, R_t\}].$$

The factors depend on disjoint independent column families, and each has squared expectation one. This proves the Boolean identity required in (28) for every minor size. Lemma 15.4 proves stability for both core types.

15.4 The complete core by two-block polarization

Now suppose all four edges between A and B are present. Partition the ground set of the fundamental transversal matroid into

$$\mathcal{U} = \bar{A} \sqcup R, \quad \mathcal{V} = \bar{L} \sqcup B, \quad M = m + 2, \quad N = n + 2.$$

Its bases are exactly the N -element subsets containing at most two elements of $\mathcal{U}$. Indeed, let a support select i vertices of A , j of B , ℓ of L and s of R . Size balance is $i + \ell = j + s$. Feasibility requires $s \leq i$ and $\ell \leq j$, and these conditions are equivalent under balance. They suffice: match the exterior vertices to the selected opposite core, then match the remaining equally numerous core vertices across the complete core. The corresponding basis contains $2 - i + s$ elements of $\mathcal{U}$, so feasibility is exactly the bound of two.

Consequently the basis polynomial is

$$F_G = e_N(\mathcal{V}) + e_1(\mathcal{U})e_{N-1}(\mathcal{V}) + e_2(\mathcal{U})e_{N-2}(\mathcal{V}). \quad (30)$$

Its separate block diagonal, with $\mathcal{U}$ variables set to a and $\mathcal{V}$ variables to b , is

$$b^{N-2} \left(b^2 + MNab + \binom{M}{2} \binom{N}{2} a^2 \right).$$

The quadratic discriminant is

$$M^2N^2 - 4 \binom{M}{2} \binom{N}{2} = MN(M + N - 1) > 0. \quad (31)$$

Since $M, N \geq 2$, its factors are $(b + \lambda_+ a)(b + \lambda_- a)$ with both $\lambda_{\pm} > 0$. The block diagonal is therefore stable.

Polynomial (30) is exactly its separate multiaffine polarization in M variables and N variables. The Grace–Walsh–Szegő theorem, in the convex upper half-plane, says that polarization preserves stability [2]. Thus F_G is stable. This completes all six core cases; leaf reattachment proves Theorem 15.1.

15.5 From stability to the rank-four inequalities

For completeness we recall the two inputs from the preceding note [11]. [*Merge note:* They are *preorder-root-polytopes*'s Theorem 70.1 (Part V) or Lemma 205.1 (Part XIII), and Lemma 206.1 with Theorem 206.2 (Part XIII); source 74 states them as Lemmas 3.1 and 3.2.] For arbitrary positive activities, p_G is $\text{ULC}_{\min(|X|, |Y|)}$: matroid basis polynomials are Lorentzian [1], and the nonnegative linear substitution

$$\left(\prod_{x \in X} u_x \right) F_G((s/u_x)_x; (tv_y)_y) = \sum_k a_k s^{|X|-k} t^k$$

preserves the Lorentzian property. The bivariate characterization gives $\text{ULC}_{|X|}$; transpose the graph for $\text{ULC}_{|Y|}$. This is the weighted form of the established smaller-shore input; see also [7].

If a minimum cover of size r lies wholly on one shore, remove isolated vertices and apply that input. If a minimum cover is $\{x_0\} \sqcup B$ with $|B| = r - 1$, every active right vertex outside B is a leaf at x_0 . A feasible support contains at most one such leaf, so replacing the class by one leaf of activity equal to the sum of its activities preserves the entire polynomial. The compressed right shore has size r , giving ULC_r . The transposed case is identical.

By König's theorem a rank-four graph has a cover of size four. Its split is $0 + 4$, $1 + 3$, $2 + 2$, or a transpose. The preceding paragraph treats the first two types, and Theorem 15.1 treats the third by Newton's inequalities. Covers of size at most three have a pure or singleton-side split, proving the smaller ranks too. For zero activities, delete their vertices and apply the result with the surviving degree. This proves Corollary 15.2.

15.6 A sharp boundary for complete mixed cores

For integers $s \geq 2$ and $n, m \geq 0$, let $H_{2,s;n,m}$ have left core A of size two, right core B of size s , n exterior left vertices L , and m exterior right vertices R . Its edges are all pairs in

$$(A \times B) \cup (L \times B) \cup (A \times R),$$

with no edges between L and R . When $n \geq s$ and $m \geq 2$, its matching rank is $s + 2$.

Theorem 15.5 (Sharp stability criterion). *The signed support polynomial of $H_{2,s;n,m}$ is real stable if and only if*

$$(m + 2)(n + 2)s \geq 2(m + 1)(n + 1)(s - 1). \quad (32)$$

In particular every two by two Hall core is stable. For a two by three core, stability is equivalent to

$$(n - 2)(m - 2) \leq 12. \quad (33)$$

Proof. In the extended transversal matroid put $F = \bar{A} \sqcup R$ and $O = \bar{L} \sqcup B$, with

$$M = |F| = m + 2, \quad N = |O| = n + s, \quad r = n + 2.$$

Exactly as in the complete two by two core, the bases are all r -element subsets containing at most two members of F . To check sufficiency directly, a selected set with $k \leq 2$ elements of F and j elements of B uses $r - k - j$ private L -elements. Match the F -elements into A , use $2 - k$ selected B -elements on the remaining positions of A , and use the other $k + j - 2$ selected B -elements on the vacant L -positions. All counts are nonnegative by the size of the selected subset.

Thus its basis polynomial is

$$\mathcal{B} = \sum_{k=0}^2 e_k(F) e_{r-k}(O).$$

Its two-block diagonal is

$$y^n \left[\binom{N}{r} y^2 + M \binom{N}{r-1} xy + \binom{M}{2} \binom{N}{r-2} x^2 \right].$$

All three bracketed coefficients are positive. The bracket is stable exactly when its discriminant is nonnegative, equivalently

$$M^2 \binom{N}{r-1}^2 \geq 4 \binom{N}{r} \binom{M}{2} \binom{N}{r-2}.$$

Using

$$\frac{\binom{N}{r-1}^2}{\binom{N}{r} \binom{N}{r-2}} = \frac{r}{r-1} \frac{N-r+2}{N-r+1}$$

reduces this inequality to (32). Necessity for full stability follows by identifying all variables within each block. Sufficiency follows by polarization, using M slots for x and N slots for y . The signed support polynomial is stable exactly when the extended basis polynomial is stable, by the invertible upper-half-plane substitution $z \mapsto -1/z$ on one shore. For $s = 3$, expansion of (32) gives (33). $\square$

A smallest unstable member of the rank-five family. Among members with $s = 3$, $n \geq 3$ and $m \geq 2$, instability requires $(n - 2)(m - 2) > 12$. If $(n - 2) + (m - 2) \leq 7$, the product is at most $\lfloor 7^2/4 \rfloor = 12$. Thus every unstable member has at least $n + m + 5 = 17$ vertices; $n = m = 6$ attains this bound. Its diagonal factor is

$$y^6(9y^2 + 288xy + 2352x^2),$$

whose quadratic discriminant is -1728 .

This does not refute positive-field real-rootedness or rank-normalized ULC. Indeed the same graph with unit vertex weights has

$$p(t) = 1 + 36t + 333t^2 + 974t^3 + 1005t^4 + 300t^5,$$

with five distinct negative roots. An exact certificate consists of its alternating signs at

t	-2	-1	$-\frac{1}{2}$	$-\frac{1}{5}$	$-\frac{1}{20}$	0
$p(t)$	-51	29	$-\frac{33}{16}$	$\frac{21}{25}$	$-\frac{1329}{16000}$	1

The intermediate value theorem supplies a root in each of the five disjoint intervals. Consequently the rank-four proof by universal signed stability has a genuine rank-five boundary, even though the rank-five ULC problem remains unresolved here. [*Merge note:* The rank-five problem is settled affirmatively by source 74 (Corollary 2.2); this graph has the $2 + 3$ minimum cover $A \sqcup B$, so Theorem 2.1 gives ULC_5 for every positive activity assignment.]

15.7 Scope and reproducibility

The rank-four question in [10] is therefore resolved for unit weights, one-shore weights and arbitrary two-shore weights. Combined with the rank-six weighted counterexamples in [9], the first possible failure rank for arbitrary two-shore rank normalization lies in $\{5, 6\}$. General unit-weight and one-shore rank normalization are not settled by this note. [*Merge note:* Superseded within the batch: the first failure rank is exactly six by source 74 (Corollary 2.2). The shipped `76-rank-four-PROOF_STATUS.md` says likewise that “the smallest failure rank is reduced to five or six”; it is kept as delivered. The unit-weight and one-shore questions are answered negatively in Part II (sources 71 and 73).]

The expected-determinant lemma, polarization theorem, uniform-matroid stability and gluing principles are established tools; they are not claimed as new. This note’s contributions relative to the inspected drafts are the two covariance realizations, the complete-core two-block application, the resulting two-plus-two cover theorem, and the weighted rank-four conclusion. A targeted search of the stability literature did not certify global priority, which is not claimed.

The accompanying scripts supplement the proof:

- [*Merge note:* The scripts are shipped as `code/76-rank-four-verify_*.py` and their outputs as `data/76-rank-four-*_verification.json`; the build wrapper is not shipped.]
- `verify_covariance.py` checks all 70 square minors of the four-by-four representative for each covariance construction, including infeasible and empty minors. All 140 exact identities pass in $\mathbb{Q}(2^{1/4})$. It also checks every principal covariance minor in those representatives.
- `verify_complete_core.py` independently checks the support-to-basis characterization and block counts for all 49 pairs $0 \leq n, m \leq 6$, together with the discriminant identity.

- `verify_boundary.py` checks 75 complete mixed cores against the two-block basis description, the exact stability threshold, and the rank-five unit-weight polynomial and rational root certificate.
- The six-core proof and the classification of all possible supported minor types establish the universal assertions; the finite scripts do not replace those arguments.

Run the scripts with Python 3 and SymPy, and compile this source twice with `pdflatex`. The package includes a build wrapper for the preparation environment and recorded verification outputs.

16 A sharp weighted classification of full Hall bottleneck graphs (source 75)

Abstract of source 75, as delivered. A full Hall-bottleneck graph has two complete core blocks, complete connections from each exterior block to the opposite core, and no exterior-to-exterior edges. We prove that this family satisfies matching-rank-normalized ultra-log-concavity for every positive vertex-activity assignment and every exterior population if and only if one core has size at most two. The new positive argument treats a two-by- s core uniformly for every s : deleting the core edges gives a real-rooted comparison polynomial, and restoring them increases every Newton margin. A twelve-case coefficient calculation proves this using only binomial ratios and Newton inequalities for exterior elementary symmetric functions. When both cores have size at least three, we give explicit finite populations and an integer activity producing a failed last margin. The positive inequalities are strict in the genuine two-by- s rank regime. An exact weighted rank-five example has nonreal roots despite strict ultra-log-concavity. The classification concerns this full Hall family; arbitrary graphs with a two-plus- s minimum cover remain outside its scope.

Status. The results have ordinary mathematical proofs and reproducible exact algebraic checks. They are not formally verified in Lean or externally refereed. The contribution is compared with the inspected ProveIt Hall-bottleneck report; global priority is not asserted. [*Merge note:* The inspected Hall-bottleneck report [9] is Part XI of *preorder-root-polytopes* (pinned at 4a3b6991d). The positive half of Theorem 16.1 is the full Hall case of source 74’s Theorem 2.1: for $n \geq b$, $m \geq a$ and $\min(a, b) \leq 2$ the core $A \sqcup B$ is a minimum cover with at most two vertices on one shore (for $\min(a, b) = 1$ it is also Part XIII’s Theorem 206.2). Source 75’s comparison-polynomial proof is printed as a second route, with what it adds: strictness when $\min(a, b) = 2$, real-rootedness when $\min(a, b) = 1$, and the nonreal rank-five example. For $a = b$ and $n = m$ the negative half is Part XI’s endpoint theorem (Theorem 178.2 of *preorder-root-polytopes*, `hb:thm:endpoint`), and the population offset h below gives $n = 33$ and $n = 11$ at $s = 3, 4$, Part XI’s minimal outer sizes; it is new for $a \neq b$ (Research question 89 of *preorder-root-polytopes*).]

16.1 The family and the classification

For positive integers a, b , define $H_{a,b;n,m}$ by the disjoint shores

$$X = A \sqcup L, \quad Y = B \sqcup R, \quad |A| = a, \quad |B| = b, \quad |L| = n, \quad |R| = m,$$

and the exact edge set

$$(A \times B) \cup (L \times B) \cup (A \times R). \tag{34}$$

Thus every exterior-left vertex is adjacent to *all* of B , every exterior-right vertex to *all* of A , the core $A \times B$ is complete, and no exterior pair is adjacent. Assign an arbitrary positive activity to each vertex. Its support polynomial is

$$P_H(t) = \sum_{(I,J) \text{ feasible}} u_I v_J t^{|I|} = \sum_k p_k t^k,$$

where $I \subseteq X$, $J \subseteq Y$, $|I| = |J|$, and feasibility means that the induced bipartite graph admits a perfect matching. A pair of endpoint sets contributes once, regardless of how many matchings realize it.

For a nonnegative degree- d coefficient sequence with no internal zeros, write ULC_d for log-concavity of $p_k/\binom{d}{k}$. Equivalently,

$$\Delta_k^{(d)}(P) := k(d-k)p_k^2 - (k+1)(d-k+1)p_{k-1}p_{k+1} \geq 0 \quad (1 \leq k < d). \quad (35)$$

For positive activities the degree equals the maximum matching size. If $n \geq b$ and $m \geq a$, this rank is $d = a + b$: the core is a cover of that size, attained by matching each core into the opposite exterior block.

Theorem 16.1 (Sharp full-Hall classification). *Fix positive core sizes a, b . The following are equivalent.*

1. *For all $n \geq b$, $m \geq a$ and every positive activity assignment, $P_{H_{a,b,n,m}}$ is ULC_{a+b} .*
2. $\min(a, b) \leq 2$.

If $\min(a, b) = 2$, every internal inequality is strict. If $a, b \geq 3$, the failure of the last inequality can be realized with activity one at every exterior vertex and one common positive integer activity at every core vertex; explicit parameters are given in Section 16.5.

The positive assertion also extends to smaller exterior populations with normalization by the actual matching rank, as explained in Section 16.4. Zero activities are handled there separately. Nothing in the negative assertion concerns unit weights or activities on just one shore. [Merge note: Those cases are treated by sources 71, 73, 55, 47 and 51 in Part II.]

16.2 A uniform formula for a two by s core

Assume $a = 2$, $b = s$, $n \geq s$ and $m \geq 2$, and put $d = s + 2$. Normalize both exterior activity sums to one by dividing every left activity by $e_1(L)$ and every right activity by $e_1(R)$. This only replaces $P_H(t)$ by $P_H(ct)$ for a positive c , and multiplies its k th margin by c^{2k} .

Let u_1, u_2 be the normalized left-core activities, $v_1, \dots, v_s$ the normalized right-core activities, and define

$$U_1 = u_1 + u_2, \quad U_2 = u_1 u_2, \quad V_j = e_j(v_1, \dots, v_s), \quad E_j = e_j(L), \quad z = e_2(R).$$

Here $E_0 = E_1 = 1$, $0 \leq z \leq 1/2$, and every out-of-range elementary symmetric function is zero.

A size- k support selecting i left-core and j right-core vertices is feasible exactly when $i + j \geq k$. Necessity follows because its $k - i$ exterior-left vertices must be matched into its j selected right-core vertices. Conversely match those exterior vertices first, then the selected exterior-right vertices into the left core, and finally the remaining core vertices to one another. Therefore

$$p_k = E_k V_k + U_1 E_{k-1} (V_k + V_{k-1}) + U_2 E_{k-2} (V_k + V_{k-1} + z V_{k-2}). \quad (36)$$

Deleting the $2s$ core edges gives two disjoint complete bipartite graphs. Their support polynomial Q has coefficients

$$q_k = E_k V_k + U_1 E_{k-1} V_{k-1} + U_2 E_{k-2} z V_{k-2}. \quad (37)$$

The proof will establish the stronger comparison

$$\Delta_k^{(d)}(P_H) - \Delta_k^{(d)}(Q) \geq 0 \quad (1 \leq k < d). \quad (38)$$

Lemma 16.2 (The comparison polynomial). *The polynomial Q has degree d and only negative real zeros. Its order- d Newton inequalities are strict.*

Proof. For a complete bipartite graph with left activities $\alpha_1, \dots, \alpha_p$ and right activities $\beta_1, \dots, \beta_q$, its support polynomial is

$$\left(\prod_i \alpha_i \right) e_p(\alpha_1^{-1}, \dots, \alpha_p^{-1}, t\beta_1, \dots, t\beta_q).$$

Elementary symmetric polynomials are real stable: differentiate $\prod_i (w + x_i)$ in w the required number of times and specialize $w = 0$. Differentiation, positive scaling, and real specialization preserve stability or give zero [4]. The displayed univariate polynomial is nonzero with positive constant term and nonnegative coefficients, so its zeros are negative and real.

Apply this to $K_{2,m}$ and $K_{n,s}$. Their product has degree $2 + s = d$. The first factor is

$$1 + U_1 t + z U_2 t^2.$$

Its discriminant obeys

$$U_1^2 - 4zU_2 \geq (u_1 - u_2)^2 + 2u_1 u_2 > 0,$$

so Q has at least two distinct negative zeros. Newton's inequalities are strict at every internal index unless all roots coincide. This proves the lemma. $\square$

16.3 Twelve coefficient families

Every core variable has exponent at most two in the difference in (38). The possible nonzero left exponent patterns, up to swapping u_1, u_2 , are

$$10, \quad 20, \quad 11, \quad 21, \quad 22.$$

Fix one representative monomial with left pattern λ . Suppose its right-core exponents have p twos and q ones, and put

$$a = k - p, \quad j = 2a - q, \quad \ell = d - k, \quad A = k\ell, \quad B = (k+1)(\ell+1).$$

The letters A, B in this section are numerical gap factors, not vertex blocks. A nonzero increment coefficient has

$$0 \leq j < |\lambda|, \quad k \geq a, \quad a \geq \lceil j/2 \rceil, \quad \ell \geq a - j + 2. \quad (39)$$

Indeed the total right-core degree is $2k - j$; a core-edge contribution raises total core degree above $2k$, giving $j < |\lambda|$. Also the number of right-core variables occurring is

$$p + q = k + a - j \leq s = d - 2,$$

which gives the last inequality. These restrictions leave exactly twelve pattern- j families.

Divide the coefficient by the following positive central binomial factor:

$$C_0 = \binom{2a}{a}, \quad C_1 = \binom{2a-1}{a-1}, \quad C_2 = \binom{2a-2}{a-1}, \quad C_3 = \binom{2a-3}{a-2}.$$

Each is used only when its indices are allowed by (39). To derive the formulas below, the coefficient of a monomial with p twos and q ones in $V_i V_h$ is

$$\binom{q}{i-p} \quad (i+h=2p+q), \quad (40)$$

with the usual zero convention. The repeated variables occur in both subsets; one chooses which of the q single-occurrence variables belong to the first subset. Thus all twelve formulas follow by direct substitution of (36) and (37) into (35), followed by elementary ratios of binomial coefficients. No matching multiplicities are introduced.

We use these consequences of Newton's inequalities:

$$\begin{aligned} E_{k-2}E_{k+1} &\leq \frac{k-1}{k+1}E_{k-1}E_k, \\ E_{k-2}E_k &\leq \frac{k-1}{k}E_{k-1}^2, & E_{k-3}E_{k+1} &\leq \frac{k-2}{k+1}E_{k-2}E_k, \\ E_{k-3}E_k &\leq \frac{k-2}{k}E_{k-2}E_{k-1}, & E_{k-3}E_{k-1} &\leq \frac{k-2}{k-1}E_{k-2}^2. \end{aligned} \quad (41)$$

For $k \geq 2$ these also cover boundary cases by the zero convention. They follow, for example, from log-concavity of $j!E_j$. The case $k=1$ is checked at the end of the argument.

16.3.1 Pattern 10

Only $j=0$ occurs. The normalized coefficient is

$$2AE_kE_{k-1} - B\frac{a}{a+1}(E_kE_{k-1} + E_{k-2}E_{k+1}).$$

Using the first inequality in (41), it is at least

$$\frac{2k(\ell-a)}{a+1}E_kE_{k-1} \geq 0,$$

since $\ell \geq a+2$.

16.3.2 Pattern 20

Put $X = E_{k-1}^2$ and $Y = E_{k-2}E_k$. For $j=0$ the normalized coefficient is

$$AX - B\frac{a}{a+1}Y;$$

for $j=1$ it is twice this expression. The bound $Y \leq (k-1)X/k$ reduces its sign to

$$k(a+1)\left(A - B\frac{a}{a+1}\frac{k-1}{k}\right) = k^2(\ell-a) + a(\ell+1) \geq 0.$$

Here $\ell \geq a+1$ in both cases.

16.3.3 Pattern 11

Keep X, Y and put $Z = E_{k-3}E_{k+1}$. The two normalized coefficients are

$$\begin{aligned} j = 0 : \quad & 2A(X + Y) - B \frac{a}{a+1}(X + 2Y + Z), \\ j = 1 : \quad & 4AX + 2AY - B \left(X + \frac{4a}{a+1}Y + \frac{a-1}{a+1}Z \right). \end{aligned}$$

Use $Z \leq (k-2)Y/(k+1)$, and write the resulting lower bound as

$$\alpha \left(X - \frac{k}{k-1}Y \right) + \beta Y. \quad (42)$$

For $j = 0$ the two factors are

$$\alpha = 2A - \frac{Ba}{a+1} \geq 0, \quad \beta = \frac{2k(2k-1)(\ell-a)}{(k-1)(a+1)} \geq 0.$$

The sign of α is immediate for $a = 0$; for $a \geq 1$, use $k \geq a$ and $\ell \geq a+2$ in $A/B = k\ell/((k+1)(\ell+1))$.

For $j = 1$ one has $a \geq 1$, $\ell \geq a+1$, and

$$\alpha = 4A - B \geq 0,$$

while

$$\begin{aligned} \frac{(k-1)(a+1)}{2} \beta &= \ell(3k^2 - 3k + 1 + a) \\ &\quad - a(3k^2 - k - 1) - 2k + 1. \end{aligned}$$

The right side increases with ℓ . At $\ell = a+1$ it becomes

$$(a-k+1)(a-k+2) + 2k(k-1) \geq 0,$$

because the first product consists of consecutive integers. Equation (42) and Newton's bound now prove both cases.

16.3.4 Pattern 21

Put $M = E_{k-2}E_{k-1}$ and $N = E_{k-3}E_k$. The normalized coefficient is

$$AMS_j - B(MH_j + NJ_j),$$

where

j	S_j	H_j	J_j
0	2	$\frac{a}{a+1}$	$\frac{a+1}{2a}$
1	4	$\frac{a+1}{2a}$	$\frac{a+1}{2a}$
2	$2 + \frac{2z(a-1)}{a}$	$z + \frac{a-1}{a}$	$\frac{a-1}{a} + \frac{z(a-1)(a-2)}{a(a+1)}$

Newton gives $N \leq \nu M$, where $\nu = (k-2)/k$. For $j = 0, 1$,

$$\frac{H_j + \nu J_j}{S_j} = \frac{a}{a+1} \frac{k-1}{k} \leq \frac{a}{a+1} \frac{k}{k+1} \leq \frac{A}{B},$$

using $\ell \geq a$. These two coefficients are nonnegative.

For $j = 2$, $a \geq 1$ and $\ell \geq a$. Define

$$D(\ell, z) = k\ell S_2 - (k+1)(\ell+1)(H_2 + \nu J_2).$$

It is affine in each of ℓ, z . Direct simplification gives

$$D(a, 0) = \frac{2(a^2 + k^2 - 1)}{ak} \geq 0, \quad D(a, \tfrac{1}{2}) = \frac{(k-a)^2 + (k-a) + 2a(a-1)}{ak} \geq 0.$$

Thus $D(a, z) \geq 0$ on $0 \leq z \leq 1/2$. Since $H_2 + \nu J_2 \geq 0$,

$$\frac{\partial D}{\partial \ell} = \frac{D(a, z) + (k+1)(H_2 + \nu J_2)}{a} \geq 0.$$

Therefore $D(\ell, z) \geq 0$ for all $\ell \geq a$, proving the last case.

16.3.5 Pattern 22

Put $U = E_{k-2}^2$ and $W = E_{k-3}E_{k-1}$. The normalized coefficient is

$$AUS_j - BWT_j,$$

with

j	S_j	T_j
0	1	$\frac{1}{a+1}$
1	2	$\frac{a+1}{2a}$
2	$1 + \frac{2z(a-1)}{a}$	$\frac{a-1}{a} + \frac{2z(a^2 - a + 1)}{a(a+1)}$
3	$2z$	$\frac{2z(a-1)}{a}$

For $j \leq 2$, $T_j/S_j \leq a/(a+1)$. In the $j = 2$ row this follows by checking $z = 0, 1/2$ in the affine cross-multiplied inequality; the endpoint ratios are $(a-1)/a$ and $a/(a+1)$. For $j = 3$ and $z > 0$ the ratio is $(a-1)/a$; at $z = 0$ the row is zero.

Now

$$W \leq \frac{k-2}{k-1}U, \quad \frac{k-2}{k-1} \leq \frac{k}{k+1}.$$

When $j \leq 2$, feasibility gives $\ell \geq a$, so A/B dominates the product of the moment ratio and $a/(a+1)$. When $j = 3$, use $\ell \geq a-1$ and the ratio $(a-1)/a$ instead. All four cases follow.

16.3.6 The small index boundary and completeness

At $k = 2$, all displayed arguments remain valid with $E_{-1} = 0$. At $k = 1$, patterns 21 and 22 vanish. For patterns 10 and 11 with $j = 0$ the normalized coefficient is $2A - Ba/(a+1)$; here $a = 0$ or 1 and $\ell \geq a+2$, making it nonnegative. Pattern 20 gives A or $2A$, and pattern 11 with $j = 1$ gives $4A - B \geq 0$. Thus the boundary is covered without division by $k-1$.

The twelve cases exhaust every nonzero monomial coefficient of the increment in the independent positive core activities. Every coefficient is nonnegative, proving (38). Lemma 16.2 then proves strict ULC_{s+2} for every full two-by- s Hall graph in the genuine-rank regime.

16.4 Singleton cores and deficient exterior populations

A singleton-core case has an elementary stable-matroid description. Suppose $a = 1$. In the extended transversal matroid, let

$$F = \bar{A} \sqcup R, \quad O = \bar{L} \sqcup B.$$

Its rank is $n + 1$, and its bases are exactly the $(n + 1)$ -element subsets containing at most one element of F . Indeed F can only be matched to the one core vertex, and every selected element of B is available for any remaining left position. Its basis polynomial is

$$\mathcal{B} = e_{n+1}(O) + e_1(F)e_n(O).$$

Its two-block diagonal is $y^n(c_0y + c_1x)$ for positive constants c_0, c_1 . This is real stable, and polarization separately in the two blocks recovers $\mathcal{B}$, preserving stability [4]. Setting each private variable to the inverse left activity and each right variable to t times its activity recovers P_H , up to a positive factor. Thus P_H is real-rooted and satisfies Newton's inequalities. Exchanging shores handles $b = 1$.

For $\min(a, b) \leq 2$ but insufficient exterior populations, the matching rank is

$$\min\{a + b, n + a, m + b\}.$$

The support criterion proves this formula directly: for every k below all three bounds, choose $i = \min(a, k)$ and $j = \min(b, k)$ selected core vertices. If $n < b$ or $m < a$, the formula equals the smaller shore size. The established weighted smaller-shore ULC theorem therefore applies [7, 1]. [*Merge note:* This is *preorder-root-polytopes*'s Theorem 70.1 (Part V), source 74's Lemma 3.1.] Its weighted form follows by specializing the basis polynomial of the extended transversal matroid at private variables s/u_x and right variables tv_y , then using the bivariate Lorentzian criterion. Hence Theorem 16.1's positive direction holds for every exterior population when the actual rank is used.

For nonnegative activities, delete the zero-activity vertices. The graph remains a full Hall graph with cores of size at most their original sizes. An empty core leaves a complete bipartite graph and isolates; singleton cores and all other cases are covered above. Thus the positive direction remains true with normalization by the surviving degree, not merely by its original value.

16.5 Explicit counterexamples when both cores are large

[*Merge note:* Source 74 (Section 6) prints the same family with the populations $N = b + 30$, $M = a + 30$; source 75's offset h is a second choice, and Table 3 records its first failing integer activities for seven shapes.] Now let $a, b \geq 3$, and set

$$r = a + b, \quad K = (r - 1)ab, \quad H = (r + 1)ab - 2r(r - 1).$$

Since $ab \geq 3(r - 3)$ and $r \geq 6$,

$$H \geq r^2 - 4r - 9 > 0.$$

Choose the explicit integer

$$h = 1 + \left\lceil \frac{2K}{H} \right\rceil, \quad n = b - 1 + h, \quad m = a - 1 + h.$$

Give every exterior vertex activity one and every core vertex activity $w > 0$. Put

$$A_i = \binom{n}{b-i}, \quad B_i = \binom{m}{a-i} \quad (i = 0, 1, 2),$$

and define

$$\begin{aligned} X &= aA_0B_1 + bA_1B_0, \\ Y &= aA_1B_2 + bA_2B_1, \\ Z &= \binom{a}{2}A_0B_2 + abA_1B_1 + \binom{b}{2}A_2B_0. \end{aligned}$$

The top three support coefficients are exactly

$$\begin{aligned} p_r &= A_0B_0w^r, \\ p_{r-1} &= A_1B_1w^r + Xw^{r-1}, \\ p_{r-2} &= A_2B_2w^r + Yw^{r-1} + Zw^{r-2}. \end{aligned}$$

They are obtained by omitting zero, one or two core vertices in the general support formula. Consequently

$$\Delta_{r-1}^{(r)}(P_H) = w^{2r-2}(\alpha w^2 + \beta w + \gamma), \quad (43)$$

where

$$\begin{aligned} \alpha &= (r-1)A_1^2B_1^2 - 2rA_0A_2B_0B_2, \\ \beta &= 2(r-1)A_1B_1X - 2rA_0B_0Y, \\ \gamma &= (r-1)X^2 - 2rA_0B_0Z. \end{aligned}$$

The leading coefficient is strictly negative. Indeed

$$\frac{A_1^2B_1^2}{A_0A_2B_0B_2} = \frac{ab}{(a-1)(b-1)} \left(1 + \frac{1}{h}\right)^2,$$

while the choice of h gives

$$\frac{H}{K} \geq \frac{2}{h-1} > \frac{2}{h} + \frac{1}{h^2}.$$

Therefore this coefficient ratio is strictly below $2r/(r-1)$, exactly the threshold for $\alpha < 0$.

For a completely finite activity choice, take

$$w = 1 + \left\lfloor \frac{|\beta| + |\gamma|}{-\alpha} \right\rfloor.$$

Then $w \geq 1$ and

$$\alpha w^2 + \beta w + \gamma \leq \alpha w^2 + (|\beta| + |\gamma|)w < 0.$$

This gives an explicit positive-integer two-shore activity counterexample for every $a, b \geq 3$, completing the negative direction of Theorem 16.1. The activity bound is conservative and is not claimed minimal.

16.6 A positive theorem does not imply real rootedness

The two-by- s conclusion is genuinely an ultra-log-concavity theorem. For $s = 3$, $n = m = 9$, activity 100 at all five core vertices and activity one at every exterior vertex, replacing t by $t/100$ gives

$$F(t) = 1 + 645t + 38730t^2 + 124500t^3 + 37800t^4 + 3024t^5.$$

Its discriminant is

$$-4431992565218959594687680000000 < 0,$$

so not all roots are real. Exact root counting gives three real roots. Its four order-five margins are nevertheless

$$1276800, \quad 8036447400, \quad 75433572000, \quad 1950480000,$$

all strictly positive. The program `verify_separation.py` [*Merge note:* shipped as `code/75-full-hall-verify_separation.py`] regenerates these integers from the graph parameters.

16.7 Verification and the remaining questions

The accompanying exact programs generate the twelve normalized coefficient families directly from (36)–(37). A separate check proves their signs by substituting the Newton bounds, reducing to corners of multilinear boxes, and verifying nonnegative polynomial coefficients in the free integer slacks. Denominator positivity and the $k = 1, 2$ boundary are checked explicitly. The generic extraction also reproduces all 49 orbit coefficients of the two-by-three case and all 80 of the two-by-four case exactly.

The readable argument above is uniform in s ; it is not an inference from finitely many core sizes. Exact scripts audit algebraic identities and boundary bookkeeping. They do not constitute a Lean formalization, and numerical root approximations are not used in any proof.

The existing Hall-bottleneck report [9] already disproves universal weighted rank normalization using a three-by-three core and studies symmetric higher cores. Theorem 16.1 gives a sharp boundary for the *full* Hall-core shapes, including all unequal cores and arbitrary independent positive activities in the positive range. Arbitrary graphs with a two-plus-three minimum cover are still not classified here: missing core edges and proper exterior neighbourhoods need different arguments. In particular, this report does not settle whether the first weighted counterexample among all bipartite graphs has rank five or six, and gives no negative result for unit weights or weights confined to one shore. [*Merge note:* Settled within the batch: the first weighted counterexample rank is exactly six (source 74, Corollary 2.2); unit-weight and one-shore counterexamples are sources 71 and 73 (Part II). Source 75’s delivered README, which is not shipped, likewise says that general weighted rank five remains open “here”. Arbitrary graphs with a $2 + 3$ minimum cover, which source 75 leaves unclassified, are covered by source 74’s Theorem 2.1.]

Part II

Unit weights and one weighted shore: counterexamples and the rank-38 threshold

Contents of Part II. Part II prints five sources. Source 71 (Section 17) gives a connected all-unit counterexample, answering Research question 84 of *preorder-root-polytopes* negatively. Source 73 (Section 18) gives one-shore counterexamples (all left activities one), answering Research question 85 there negatively. Sources 55, 47 and 51 (Sections 19, 20 and 21) study the full Hall graphs with *one common activity on the right core* and one elsewhere: the last gap first fails at matching number 38 (55), so does full ULC (47), and at matching number 49 the last gap no longer controls the others (51).

Which weighting. Every section title of this Part states its weighting. Source 71 is unit weighted; sources 73, 55, 47 and 51 are one-shore weighted, and the last three only with one common activity on the right core. None of them answers Research question 88 of *preorder-root-polytopes*, which concerns the two-shore family $H_{s,n}$ with weight on both cores; sources 47 and 51 answer its one-shore analogue.

Same theorem in several sources. Source 71's unit pendant-arm identity is the case $T = 1$ of source 73's weighted identity; it is printed once, in the weighted form, in source 73's appendix. Source 47 recalls source 55's endpoint theorem without proof (kept as a recall, with pointers) and re-proves the first-gap inequality by the Motzkin–Straus route (replaced by a pointer to Part XI of *preorder-root-polytopes*; its strictness step is kept). Source 55 extends source 73: the top-three formula and the rank-38 construction of 73 reappear in 55 with an exact threshold. The smaller-shore steps of sources 55 and 47 restate *preorder-root-polytopes*'s Theorem 70.1.

Non-claims of Part II. Source 71 does not determine the least matching number, vertex count or edge count of a unit counterexample, and keeps the smaller-shore bound and ordinary log-concavity; source 73 makes no minimal-rank claim; sources 55, 47 and 51 concern one common right-core activity on full Hall graphs only, and not arbitrary one-shore activities; source 55 concerns the last gap only.

17 Unit weight failure of matching rank normalization (source 71)

Abstract of source 71, as delivered. We give a connected bipartite graph whose unweighted matching-support polynomial fails ultra-log-concavity normalized by its maximum matching size. The explicit graph has 6916 vertices, 174228 edges and matching rank 3450. Its last Newton margin is a strictly negative integer, certified by a short exact formula for its top three coefficients. Every vertex activity is one. A second example has matching rank exactly one below the size of the smaller shore, showing that even this smallest possible discrepancy can obstruct rank normalization. The construction uses pendant length-two arms and admits exact support counting without enumerating the large graphs' matchings.

Status and scope. The counting identities and the counterexamples below are established in ordinary mathematics and independently checked in exact arithmetic. This is an unrefereed

research note, not a Lean-certified result. Its claim of resolving a question is relative to the inspected ProveIt manuscript *Hall Bottlenecks and the Limits of Rank Normalization* [9]; worldwide priority is not asserted. The examples do not determine the smallest possible matching rank or number of vertices of a counterexample.

17.1 The question and the counterexample

For a finite simple bipartite graph $G = (X, Y; E)$, define

$$p_G(t) = \sum_{k \geq 0} p_k t^k,$$

where p_k is the number of pairs $I \subseteq X$, $J \subseteq Y$ with $|I| = |J| = k$ for which $G[I, J]$ has a perfect matching. A feasible pair is counted once, regardless of its number of perfect matchings. This is the bipartite form of the perfectly matchable set polynomial; see [5] for earlier work on this polynomial. In particular $p_{K_{2,2}}(t) = 1 + 4t + t^2$.

For a polynomial of degree R with positive coefficients, rank-normalized ultra-log-concavity means that $p_k / \binom{R}{k}$ is log-concave. Equivalently,

$$k(R - k)p_k^2 \geq (k + 1)(R - k + 1)p_{k-1}p_{k+1} \quad (1 \leq k < R). \quad (44)$$

Here the proposed normalization is $R = \nu(G)$, the maximum matching size. The smaller-shore normalization is known to hold [7, Theorem A]; replacing the smaller shore by $\nu(G)$ is a genuinely stronger assertion. The unit-weight version was explicitly left open in [9, question “Unit weights”]. [*Merge note:* That question is Research question 84 of *preorder-root-polytopes* (`hb:q:unit`); the smaller-shore normalization is its Theorem 70.1 (Part V). The positive unit-weight theorems there, Theorem 191.1 ($\nu \leq 3$), Theorem 196.1 (a rank-four family) and Theorem 206.2 (singleton covers), and Theorem 2.1 of Part I, do not apply to the graph below.]

Theorem 17.1. *There is a connected simple bipartite graph with all vertex activities equal to one, 6916 vertices, 174228 edges and matching rank $R = 3450$, for which (44) fails at $k = 3449$.*

The complete graph definition, the rank certificate, and the exact negative margin are given below. No external field or limiting activity is used in this theorem. [*Merge note:* Theorem 17.1 answers Research question 84 of *preorder-root-polytopes* negatively, and with it the unit-weight half of its Research question 37 and its Research question 94 (no all-rank proof at unit weights can exist). By Corollary 5.1, every minimum cover of this graph meets both shores in at least three vertices; the cover of Lemma 17.2 has $12 + 3040 = 3052$ vertices on the left and 398 on the right.]

17.2 A graph family with pendant arms

Let $a, b, N, M \geq 1$, with $N \geq b$ and $M \geq a$, and let $W \geq 2$ be an integer. Start with left classes A, L of sizes a, N and right classes B, C of sizes b, M . Include all edges in

$$A \times B, \quad A \times C, \quad L \times B,$$

and no edges in $L \times C$. At each $c_j \in C$, attach $W - 1$ new length-two arms

$$c_j - z_{j,h} - \ell_{j,h} \quad (1 \leq h < W),$$

where $z_{j,h}$ is a new left vertex and $\ell_{j,h}$ is a new right leaf. There are no further edges. Write

$$K = M(W - 1), \quad r = a + b, \quad R = r + K.$$

Denote the resulting graph by $G(a, b, N, M, W)$. *Every activity is one; W specifies an arm count and is not a vertex weight.*

Lemma 17.2 (Size and rank). *The two shore sizes of $G(a, b, N, M, W)$ are $a + N + K$ and $b + M + K$. Its edge count is $ab + aM + bN + 2K$, and its matching rank is $R = a + b + K$.*

Proof. Match every vertex of A to a distinct vertex of C , every vertex of B to a distinct vertex of L , and every new center $z_{j,h}$ to its leaf $\ell_{j,h}$. This matching has $a + b + K$ edges. The set consisting of A , B , and all new centers is a vertex cover of that same size. The size and edge formulas follow directly from the construction. Connectivity holds because the original three complete blocks form a connected graph, and all new vertices lie on attached arms. $\square$

17.3 Exact support counting

Lemma 17.3 (The unit pendant-arm identity). *Let a vertex x of a bipartite graph have $k \geq 1$ new pendant length-two arms attached to it, all with unit activities. Write $p_G = f + g$, where f counts original supports omitting x and g counts original supports containing x . Then the augmented support polynomial is*

$$(1+t)^{k-1}[(1+(k+1)t)f + (1+t)g]. \quad (45)$$

Proof (pointer). This is the case $T = 1$ of source 73's weighted pendant-arm identity, Lemma 18.6 (Section 18.5), with the two shores exchanged: there the arms hang at a left vertex and their centers carry activity T . [Merge note: The same theorem in two sources: source 71's proof is source 73's proof at $T = 1$ (supports containing x contribute $(1+t)^k g$; for supports omitting x , choosing j centers gives $j+1$ distinct endpoint sets, whence $\sum_j \binom{k}{j} (j+1)t^j f = (1+t)^{k-1}(1+(k+1)t)f$; no matching multiplicities are counted). It is printed once, in the weighted form.] $\square$

For the original three-block Hall graph, mark the number ρ of selected vertices of C . Its corresponding support polynomial is

$$P_\rho(t) = \binom{M}{\rho} \sum_{\substack{0 \leq i \leq a, \\ 0 \leq j \leq b, \\ i \geq \rho}} \binom{a}{i} \binom{b}{j} \binom{N}{j+\rho-i} t^{j+\rho}, \quad (46)$$

where out-of-range binomial coefficients are zero. Indeed, i and j count selected core vertices in A and B ; balance requires $j + \rho - i$ selected vertices in L . Feasibility is precisely $i \geq \rho$: match the exterior vertices first, then match the remaining core vertices across the complete core. Applying Lemma 17.3 at all M vertices of C gives

$$p_{G(a,b,N,M,W)}(t) = \sum_{\rho=0}^a P_\rho(t) (1+t)^{K-M+\rho} (1+Wt)^{M-\rho}. \quad (47)$$

The factor $1+Wt$ records W different endpoint sets; it does not assign an activity to a vertex.

17.3.1 A short formula for the top three coefficients

For $a, b \geq 2$ and $\delta \in \{0, 1, 2\}$, let e be the deficit of the original support size from $r = a + b$. If u left-core and v right-core vertices are omitted, then

$$u + v \leq e, \quad \ell = b - e + u, \quad \rho = a - e + v$$

are the exterior-left and exterior-right support sizes. Put $A_\rho = K - M + \rho$ and $B_\rho = M - \rho$. Extracting the top coefficients in (47) yields the entirely integer formula

$$p_{R-\delta} = \sum_{e=0}^{\delta} \sum_{u=0}^e \sum_{v=0}^{e-u} \binom{a}{u} \binom{b}{v} \binom{N}{b-e+u} \binom{M}{a-e+v} \cdot \sum_{z=0}^{\min(B_\rho, \delta-e)} \binom{A_\rho}{\delta-e-z} \binom{B_\rho}{z} W^{B_\rho-z}. \quad (48)$$

To see the last line, the pendant factors have degree K . Selecting their coefficient of degree $K - (\delta - e)$ means omitting $\delta - e$ of their linear terms, with z omissions from the B_ρ factors of slope W . Thus (48) counts supports exactly and requires only a few summands.

17.4 The finite certificate

Take

$$(a, b, N, M, W) = (12, 398, 410, 16, 191).$$

Thus $K = 3040$, the shore sizes are 3462 and 3454, and Lemma 17.2 gives $R = 3450$ and 174228 edges. The graph has 6916 vertices in total.

Define the positive integer

$$D = 160160547610973088925200.$$

Formula (48) gives the compact exact certificate

$$\begin{aligned} p_{3450} &= 455D \, 191^4, \\ p_{3449} &= 3020455956D \, 191^3, \\ p_{3448} &= 10022638278784267D \, 191^2. \end{aligned} \quad (49)$$

The required endpoint inequality is $3449p_{3449}^2 \geq 6900p_{3448}p_{3450}$. Its margin is

$$\begin{aligned} &3449p_{3449}^2 - 6900p_{3448}p_{3450} \\ &= D^2 \, 191^6 (3449 \cdot 3020455956^2 - 6900 \cdot 455 \cdot 10022638278784267) \\ &= -314102056577041236 \, D^2 \, 191^6 < 0. \end{aligned}$$

This proves Theorem 17.1. The exact coefficient formula, together with the displayed finite integer substitution, is the proof of the counterexample; the search that led to the parameters is not used as a premise.

17.5 Failure when the smaller shore exceeds rank by one

The phenomenon does not require a large gap between the smaller shore and the matching rank.

Theorem 17.4. *There is a connected unit-weight bipartite graph whose smaller shore has size 125509, whose matching rank is 125508, and whose support polynomial is not ULC_{125508} .*

Proof. Take $G(8, 100, 101, 30, 4181)$. Then $K = 125400$, $R = 125508$, the shore sizes are 125509 and 125530, and the graph has 251039 vertices and 261940 edges. Put $D_1 = 8567325$. The same formula (48) gives

$$\begin{aligned} p_R &= 69D_1 \, 4181^{22}, \\ p_{R-1} &= 61945839672D_1 \, 4181^{21}, \\ p_{R-2} &= 27806230344578726426D_1 \, 4181^{20}. \end{aligned}$$

Consequently

$$(R-1)p_{R-1}^2 - 2Rp_{R-2}p_R = -415296714996000603216 D_1^2 4181^{42} < 0.$$

The rank and the two shore sizes follow from Lemma 17.2. $\square$

The smaller-shore theorem [7] still guarantees ULC_{125509} for this graph. The conclusion is failure of the stronger order 125508, not failure of that published bound.

17.5.1 Why deficiency one is not an isolated accident

Fix a, b, N, M and let W grow, still keeping every vertex activity one. Write

$$h = N - b + 1, \quad \theta = \frac{b}{h}, \quad \zeta = \frac{b(b-1)}{h(h+1)}, \quad \mu = \frac{a}{M-a+1}, \quad \eta = \frac{a(a-1)}{(M-a+1)(M-a+2)}.$$

Expanding (48), with $R = a + b + M(W-1)$, gives

$$\frac{(R-1)p_{R-1}^2 - 2Rp_{R-2}p_R}{p_R^2 W^3} \longrightarrow M \left[\mu^2(a+\theta)^2 - 2\eta \left(\binom{a}{2} + a\theta + \zeta \right) \right]. \quad (50)$$

For example, the leading terms of p_{R-1}/p_R and p_{R-2}/p_R are respectively

$$W(M + \mu(a+\theta)), \quad W^2 \left[\frac{M^2}{2} + M\mu(a+\theta) + \eta \left(\binom{a}{2} + a\theta + \zeta \right) \right],$$

which imply (50) directly. Set $h = 2$, so $N = b + 1$. After division by μ^2 , the bracket in (50) tends as $M \rightarrow \infty$ to

$$2a - 1 + \frac{4a-1}{3a}b - \frac{a-4}{12a}b^2.$$

For any fixed $a \geq 5$, choose b sufficiently large, then choose a finite $M \geq a + 2$ sufficiently large. The bracket is negative, so all sufficiently large integers W give counterexamples. Their smaller shore is always $R + 1$. This proves existence of infinitely many deficiency-one examples; Theorem 17.4 supplies one explicit finite instance.

17.6 Consequences for Ehrhart h^* -polynomials

For either constructed graph $G = (X, Y; E)$, form $\widehat{G}$ by adding a new left vertex v , a new right vertex w , and the edges vy for $y \in Y$ and xw for $x \in X \cup \{v\}$. Let

$$Q_{\widehat{G}} = \text{conv}\{e_i + e_j : ij \in E(\widehat{G})\},$$

and let $\text{STAB}(\overline{G})$ be the stable set polytope of the complement of G . Davis–Kohl [5, Theorem 3.10 and Lemma 3.11] give

$$h^*(Q_{\widehat{G}}; t) = p_G(t) = h^*(\text{STAB}(\overline{G}); t).$$

Thus these two specific Ehrhart h^* -polynomials have degree $R = \nu(G)$ and fail ULC_R . [*Merge note:* Compare source 74's corollary in Section 7, which gives ULC_r for these polynomials when $\nu \leq 5$ or a minimum cover meets one shore in at most two vertices.] They also have nonreal zeros: otherwise their nonnegative coefficients would satisfy the degree-normalized Newton inequalities. Their coefficients still satisfy ordinary log-concavity and the smaller-shore ULC bound by [7].

17.7 Verification and remaining questions

The package contains standard-library Python scripts [*Merge note:* shipped as `code/71-unit-weight-verify*.py`, with outputs `data/71-unit-weight-*.json`] checking the pendant identity by independent endpoint-set enumeration on 150 small graphs, including nonuniform activities as an extra regression test. For both displayed examples, they evaluate (48) by integer binomial sums and independently recompute the tails from normalized reciprocal sums of the pendant factors. They also check the exact negative margins, explicit matching and cover constructions, edge counts and coverage. All checks pass with fixed seeds. They count supports without matching multiplicity and supplement the written proof.

The minimum matching rank, vertex count and edge count of an all-unit counterexample remain undetermined. [*Merge note:* By Part I (Corollary 2.2) such a counterexample has matching number at least six, so the least unit failure rank lies in $[6, 3450]$.] Rank 3450 is the best found in a bounded parameter search, not a proved minimum. Smaller deficiency-one examples and structural classes retaining rank-normalized ultra-log-concavity remain natural questions. The published smaller-shore normalization and ordinary log-concavity still hold.

18 One-shore activities can destroy matching-rank ULC (source 73)

Abstract of source 73, as delivered. The restriction to activities on one shore does not restore matching-rank-normalized ultra-log-concavity of perfectly matchable endpoint supports. We give an explicit full Hall-bottleneck graph on 125 vertices, with matching rank 50, whose left activities are all one and whose right activities take only the values one and 10000. Its last Newton margin is a positive factor times the negative integer -39998827 . A second example has matching rank 38. Both certificates follow from a closed formula for the top three coefficients and a quadratic in the common right-core activity. Thus no numerical root calculation or limiting argument is needed to certify either finite counterexample. We also give a general criterion for constructing such examples and record a separate pendant-arm transfer construction.

Status and scope. These are ordinary mathematical proofs with reproducible exact-integer checks, not formal Lean verification or external peer review. The result resolves the one-shore-activity question in the inspected ProveIt Hall-bottleneck report negatively. It does not produce a unit-weight counterexample, identify the smallest possible rank, or assert exhaustive literature priority. [*Merge note:* The question is Research question 85 of *preorder-root-polytopes* (`hb:q:oneshore`), with exactly these hypotheses: all left activities one, right activities arbitrary positive. The weighting here, activity w on the right core B and one elsewhere, is the “one common right-core activity” of Section 1.3. The unit-weight counterexample is source 71’s (Section 17).]

18.1 The polynomial and two explicit counterexamples

For a bipartite graph $G = (X, Y; E)$ with positive vertex activities u_x, v_y , define

$$P_G(t) = \sum_{(I, J) \text{ feasible}} u_I v_J t^{|I|} = \sum_{k=0}^d p_k t^k.$$

Here $I \subseteq X$, $J \subseteq Y$, $|I| = |J|$, and feasibility means that $G[I, J]$ admits a perfect matching. Each pair of endpoint sets is counted once, regardless of the number of realizing matchings.

The degree d is the maximum matching size. This is the vertex-weighted perfectly matchable set polynomial, with t recording half the support size.

The property ULC_d requires log-concavity of $p_k/\binom{d}{k}$ and no internal zeros. For these positive-activity support polynomials there are no internal zeros, so the inequalities are

$$\Delta_k^{(d)} = k(d-k)p_k^2 - (k+1)(d-k+1)p_{k-1}p_{k+1} \geq 0. \quad (51)$$

In particular, the last inequality is $(d-1)p_{d-1}^2 \geq 2dp_{d-2}p_d$.

Write $H_{a,b;n,m}$ for the graph with shores $X = A \sqcup L$, $Y = B \sqcup R$, block sizes a, n, b, m , and exactly the edges

$$(A \times B) \cup (L \times B) \cup (A \times R). \quad (52)$$

All exterior-left vertices are adjacent to all of B , all exterior-right vertices to all of A , and there are no exterior-to-exterior edges. Give every vertex activity one except the b vertices of B , each of which has activity $w > 0$. Thus the weighting is confined to one shore. If $n \geq b$ and $m \geq a$, the matching rank is $d = a + b$: the core is a cover of size d , attained by disjoint matchings $A \rightarrow R$ and $L \rightarrow B$.

Theorem 18.1 (A counterexample on 125 vertices). *Let $G = H_{8,42;44,31}$, with activity w on B and one elsewhere. Its matching rank is 50, and*

$$\Delta_{49}^{(50)} = 98 \left[\binom{44}{42} \binom{31}{7} \right]^2 w^{82} (-w^2 + 5958w + 421173). \quad (53)$$

At $w = 10000$ the parenthesized integer is $-39998827 < 0$. The first positive integer w for which this particular margin fails is 6028, where the integer is -787 .

This graph has 125 vertices and 2432 edges. Only 42 activities differ from one. The theorem is an exact certificate against universal one-shore ULC_d .

Theorem 18.2 (A counterexample of matching rank 38). *Let $G = H_{8,30;32,1000}$ with the same one-shore weighting. Put*

$$Q(w) = -2404w^2 + 4878807600w + 10888224221475.$$

Then

$$\Delta_{37}^{(38)} = \frac{3}{2272} \left[\binom{32}{30} \binom{1000}{7} \right]^2 w^{58} Q(w). \quad (54)$$

At $w = 3000000$, one has $Q(w) = -6988688975778525 < 0$. The first positive integer giving a negative value of this particular margin is 2031684.

No minimal-rank claim is made. The rank-38 example has 1070 vertices; the rank-50 example is a substantially smaller graph. [Merge note: Within full Hall graphs with one common right-core activity, source 55 (Section 19) proves that 38 is the least matching number at which the last gap can fail, with 864 vertices the fewest at that rank, and source 47 (Section 20) extends this to every gap.]

18.2 Endpoint counting gives the complete certificate

A size- k support selecting i vertices of A and j vertices of B is feasible exactly when $i + j \geq k$. Necessity follows because the $k - i$ selected exterior-left vertices must match into the j selected

vertices of B . For sufficiency, first match the exterior-left vertices into B , then the exterior-right vertices into A , and finally the remaining core vertices to each other. Consequently

$$p_k = \sum_{\substack{0 \leq i \leq a, 0 \leq j \leq b \\ i+j \geq k}} \binom{a}{i} \binom{n}{k-i} \binom{b}{j} \binom{m}{k-j} w^j, \quad (55)$$

with out-of-range binomial coefficients set to zero.

Lemma 18.3 (The top three coefficients). *Let $a, b \geq 2$, $d = a + b$, $n \geq b$, $m \geq a$, and set*

$$X_i = \binom{n}{b-i}, \quad Y_i = \binom{m}{a-i} \quad (i = 0, 1, 2).$$

Then

$$p_d = X_0 Y_0 w^b, \quad (56)$$

$$p_{d-1} = (X_1 + aX_0)Y_1 w^b + bX_1 Y_0 w^{b-1}, \quad (57)$$

$$\begin{aligned} p_{d-2} = & \left(X_2 + aX_1 + \binom{a}{2} X_0 \right) Y_2 w^b \\ & + b(X_2 + aX_1)Y_1 w^{b-1} + \binom{b}{2} X_2 Y_0 w^{b-2}. \end{aligned} \quad (58)$$

Proof. At $k = d$, the condition $i + j \geq d$ forces $(i, j) = (a, b)$. At $k = d - 1$, the possible deficits $(a - i, b - j)$ are $(0, 0), (1, 0), (0, 1)$. At $k = d - 2$, these and $(2, 0), (1, 1), (0, 2)$ are possible. Substitution in (55) gives the formulas. $\square$

For a compact parametrization write

$$h = n - b + 1, \quad t = \frac{b}{h}, \quad z = \frac{b(b-1)}{h(h+1)}, \quad R = \frac{m-a+1}{a}, \quad \rho = \frac{a-1}{m-a+2}, \quad J = \binom{a}{2} + at + z. \quad (59)$$

Thus $t = X_1/X_0$, $z = X_2/X_0$, $R = Y_0/Y_1$ and $\rho = Y_2/Y_1$. Put $D = X_0 Y_1 > 0$.

Proposition 18.4 (Quadratic endpoint criterion). *For $a, b \geq 2$ and the above parameters,*

$$\Delta_{d-1}^{(d)} = D^2 w^{2b-2} (\alpha w^2 + \beta w + \gamma), \quad (60)$$

where

$$\alpha = (d-1)(a+t)^2 - 2dR\rho J, \quad (61)$$

$$\beta = 2bR((d-1)t(a+t) - d(at+z)), \quad (62)$$

$$\gamma = bR^2(b(d-1)t^2 - d(b-1)z). \quad (63)$$

Proof. Divide the three identities in Lemma 18.3 by D and substitute them into $(d-1)p_{d-1}^2 - 2dp_{d-2}p_d$. Grouping the three powers $w^{2b}, w^{2b-1}, w^{2b-2}$ yields the displayed coefficients. $\square$

For Theorem 18.1, formulas (61)–(63) give

$$(\alpha, \beta, \gamma) = 98(-1, 5958, 421173).$$

For Theorem 18.2, they give

$$(\alpha, \beta, \gamma) = \frac{3}{2272}(-2404, 4878807600, 10888224221475).$$

This proves both exact factorizations. Each primitive quadratic has a negative leading coefficient and a positive constant term, so it has precisely one positive zero. Direct substitution at the stated first integer and the preceding integer proves the threshold claims; the preceding values are respectively 5310 and 1792188719.

18.3 A general construction criterion

For fixed a, b, h the leading coefficient (61) tends, as $m \rightarrow \infty$, to

$$\alpha_\infty = (d-1)(a+t)^2 - \frac{2d(a-1)}{a} \left(\binom{a}{2} + at + z \right). \quad (64)$$

This is only a device for finding examples; the preceding theorems use finite exact identities.

Proposition 18.5 (From a negative limiting coefficient to a finite graph). *If $\alpha_\infty < 0$, set*

$$C = \frac{2d(a-1)}{a} \left(\binom{a}{2} + at + z \right), \quad m = a - 1 + \left\lfloor \frac{C}{-\alpha_\infty} \right\rfloor.$$

Then $m \geq a$ and $\alpha < 0$. With β, γ defined by the finite parameters, the positive integer

$$w = 1 + \left\lfloor \frac{|\beta| + |\gamma|}{-\alpha} \right\rfloor \quad (65)$$

necessarily makes the last margin negative.

Proof. One has $\alpha = \alpha_\infty + C/(m - a + 2)$. The chosen denominator is $1 + \lfloor C/(-\alpha_\infty) \rfloor > C/(-\alpha_\infty)$, so $\alpha < 0$. Also $C > -\alpha_\infty$, since their difference is $(d-1)(a+t)^2 > 0$, proving $m \geq a$. For (65), $(-\alpha)w > |\beta| + |\gamma|$ and $w \geq 1$, hence

$$\alpha w^2 + \beta w + \gamma \leq -(-\alpha)w^2 + |\beta|w + |\gamma| < 0.$$

□

For $a = 8, b = 30, h = 3$ (thus $n = 32$),

$$\alpha_\infty = -\frac{61}{4}, \quad \alpha = \frac{48379 - 61m}{4(m-6)}.$$

The leading coefficient is negative exactly when $m \geq 794$. Choosing $m = 1000$ gives Theorem 18.2. The alternative $a = 7, b = 31, h = 3$ also has rank 38 and gives $\alpha_\infty = -764/63$. These are construction examples, not a proof of any global optimum.

18.4 What this settles and what remains open

The inspected ProveIt Hall-bottleneck report [9] distinguishes the arbitrary-two-shore weighted failure from its unit-weight and one-shore variants. Theorems 18.1 and 18.2 supply a negative answer to the one-shore variant, even inside the full two-step Ferrers family (52). Since a degree- d polynomial with positive coefficients and only real zeros satisfies Newton's inequalities with normalization d , these examples also fail positive-field real-rootedness.

There is no conflict with the ambient-shore normalization in the bimatroid theorem of Röhrle and Ulirsch [7]. Nor does the current non-nesting-rook result of Alexandersson and Jal settle the stronger normalization used here: Corollary 4.1 of version 3 explicitly normalizes by the board's row count [15]. Our two examples have smaller shore sizes 52 and 40, respectively, while their matching ranks are 50 and 38. The older version's statement should not be substituted for that current row-normalized theorem.

The full-Hall constructions here have nonunit activities. Further concrete questions include the smallest rank admitting one-shore failure, the smallest graph admitting such failure, and which fixed full-Hall core shapes remain universally rank-ULC under one-shore activities. [Merge note: Re-scoped by the batch: for one common right-core activity on full Hall graphs the smallest failing rank is exactly 38 (sources 55 and 47); among all one-shore graphs it lies in [6, 38], since Part I gives ULC_ν for $\nu \leq 5$.] The present calculation also suggests studying the sign of (64) systematically; its negativity is sufficient for a counterexample, not asserted necessary for every one-shore failure.

Reproducibility. The accompanying script `verify_full_hall.py` [*Merge note:* shipped as `code/73-one-shore-verify_full_hall.py`, output `data/73-one-shore-full_hall_verification.json`] evaluates the complete feasibility sum (55), checks it independently against Boolean matching on six small graphs, and compares the top-three formula on 100 deterministic parameter samples. It then verifies both factorizations at four stated integer activities using exact integers and rational numbers. For all four instances, the only negative order- d Newton margin is the last one. These finite checks supplement the proof; they are not its logical basis.

18.5 Appendix: A pendant gadget that transfers fields between shores

The direct Hall counterexamples above give much smaller matching rank. The following independent construction gives a one-shore counterexample whose only nonunit activity is the modest integer 21. It also explains how a weighted obstruction can survive a transfer of fields between shores.

[*Merge note:* Lemma 18.6 contains source 71’s Lemma 17.3 as the case $T = 1$ (shores exchanged); it is printed here once for both sources.]

Lemma 18.6 (Exact pendant-arm identity). *Let x be a left vertex of activity one in a bipartite graph, and write $p_G(t) = f(t) + g(t)$, where f counts supports omitting x and g counts supports containing x . Attach $k \geq 1$ disjoint length-two arms $x - z_i - \ell_i$. Give the new right vertices z_i activity $T > 0$ and the new left leaves ℓ_i activity one. Then*

$$p_{G'}(t) = (1 + Tt)^{k-1}[(1 + (k + 1)Tt)f(t) + (1 + Tt)g(t)]. \quad (66)$$

Moreover, $\nu(G') = \nu(G) + k$.

Proof. Fix a support counted by g , so x must match within the original graph. Each selected new center then forces its own leaf, contributing $(1 + Tt)^k g$. For a support counted by f , choose j new centers. Either all their leaves are selected and x is absent, or x replaces exactly one of those j leaves. The latter support forces x to that leaf’s center, so it cannot also use x in the original matching. These $j + 1$ endpoint sets are distinct. Their total contribution is

$$\sum_{j=0}^k \binom{k}{j} (j + 1)(Tt)^j f = (1 + Tt)^{k-1} (1 + (k + 1)Tt) f.$$

This counts supports, not allocations or perfect matchings. An original maximum matching together with all k center–leaf edges has size $\nu(G) + k$. An original minimum vertex cover together with the k centers covers the augmented graph, giving the reverse inequality. $\square$

The top power of T in (66) is

$$T^k t^k ((k + 1)f + g).$$

Thus the forced-center limit simulates activity $1/(k + 1)$ at x , up to a positive scalar and the degree shift k . For gadgets at several left vertices, these factors multiply. The rank shift must be accounted for: a polynomial $t^K p(t)$ of degree $r + K$ has last normalized inequality

$$(r + K - 1)p_{r-1}^2 \geq 2(r + K)p_{r-2}p_r.$$

In particular, a top-three ratio strictly below 2 survives every such shift, and continuity then gives a finite positive center activity. This observation is a sufficient criterion; the finite example below is checked by its exact endpoint polynomial.

18.5.1 A rank 310 example with activity 21

[*Merge note:* This example is one-shore weighted, not unit weighted: its ten right-core vertices have activity 21. Its rank is superseded by the rank-50 and rank-38 examples above; it is kept for the transfer construction.] Start with a full 20×10 Hall core, with exterior populations $N = 14$ on the left and $M = 30$ on the right. Attach 20 pendant arms to each exterior-left vertex. Give all left vertices activity one. Give the ten original right-core vertices activity 21, and every other right vertex activity one, including every pendant center. There are $K = 280$ arms, so the graph has

$$634 \text{ vertices,} \quad 1500 \text{ edges,} \quad \nu(G) = 30 + 280 = 310.$$

For completeness, the original 30-edge matching pairs the twenty left-core vertices to distinct exterior-right vertices and the ten right-core vertices to distinct exterior-left vertices. Add all 280 center–leaf edges. The thirty original core vertices together with the 280 centers form a cover of the same size.

Here is an exact count of the endpoint coefficients. More generally let a, b, N, M be Hall parameters, put $W = k + 1$, $K = N(W - 1)$, and let $P_\ell(t)$ count original supports using exactly ℓ exterior-left vertices when only the right core has activity W . Then

$$P_\ell(t) = \binom{N}{\ell} \sum_{\substack{0 \leq i \leq a, \\ j \geq \ell}} \binom{a}{i} \binom{b}{j} \binom{M}{i + \ell - j} W^j t^{i + \ell}, \quad (67)$$

where out-of-range binomial coefficients are zero. The condition $j \geq \ell$ is exactly the Hall-support feasibility condition after balancing the shores. Repeated application of Lemma 18.6 gives the full polynomial

$$P(t; T) = \sum_{\ell=0}^{\min(N, b)} P_\ell(t) (1 + Tt)^{K - N + \ell} (1 + W T t)^{N - \ell}. \quad (68)$$

Every factor and coefficient in this expression is explicit.

For $(a, b, N, M, W, T) = (20, 10, 14, 30, 21, 1)$, coefficient extraction gives

$$\begin{aligned} p_{308} &= 8131749153344342901404784768563550, \\ p_{309} &= 39894459936702682423206366171900, \\ p_{310} &= 97561086654439123859418490215. \end{aligned}$$

Consequently the endpoint margin is

$$309p_{309}^2 - 620p_{308}p_{310} = -726235 \cdot 4485169759863727935000 \cdot 21^{26} < 0.$$

This is an exact one-shore-weighted counterexample. Its full negative integer margin, full coefficient vector, edge list, matching, and cover are included in the verification data. [*Merge note:* Shipped as `data/73-one-shore-gadget_verification-one_shore_rank310_witness.json`, beside `code/73-one-shore-gadget_verification-*.py`; the gadget identity record is byte-identical to source 71's `data/71-unit-weight-gadget_identity_verification.json`.]

Keeping the same graph and allowing each center activity to be T , the stronger exact identity is

$$\begin{aligned} 309p_{309}(T)^2 - 620p_{308}(T)p_{310}(T) &= 4485169759863727935000 \cdot 21^{26} T^{558} \\ &\quad \cdot (-91777T^2 - 1459920T + 825462). \end{aligned}$$

At $T = 1$ the bracket is -726235 . This identity makes the finite activity explicit; no limiting value is used in the counterexample.

Verification. The gadget identity was checked against independent weighted endpoint-set enumeration on 150 small bipartite graphs, with a fixed seed. A separate standard-library script expands (68) in full, checks the three displayed coefficients and exact margin, and verifies the explicit rank certificates. An independent audit recomputed the three endpoint coefficients directly from Hall subset feasibility and the top coefficients of each gadget factor. The finite checks supplement the counting proofs above.

19 A sharp rank-38 boundary for the last Hall inequality, one common right-core activity (source 55)

Abstract of source 55, as delivered. For full Hall graphs with one common activity on the right core and unit activities elsewhere, the first possible failure of the last matching-rank-normalized Newton inequality occurs at rank 38. We prove this for all finite exterior populations and every positive core activity, rather than only in a limiting regime. An explicit positive identity rules out a negative finite-field window whenever the leading quadratic coefficient is nonnegative. A cubic then gives an exact parameter criterion, and a finite integer certificate proves the rank threshold. Precisely two core shapes occur at rank 38. The smaller of their minimal populations gives an 864-vertex example with an exact integer certificate.

Scope. The activity is common across one specified core block. The theorem concerns the *last* Newton inequality. It does not determine the first failure among arbitrary individual one-shore activities or among all gaps. General finite-activity rank-six one-shore ultra-log-concavity remains open. [*Merge note:* At rank six, the first gap holds for every positively weighted graph (*preorder-root-polytopes*, Theorem 180.2), and the second and last gaps for every one-shore graph (sources 05 and 13, Part III); the third and fourth stay open in general.] These are ordinary mathematical proofs with exact certificates, not formal kernel verification or external peer review; no global priority claim is made.

19.1 The family and the sharp threshold

Let $H_{a,b;n,m}$ have shores $A \sqcup L$ and $B \sqcup R$, with respective block sizes a, n, b, m , and edges

$$(A \times B) \cup (L \times B) \cup (A \times R).$$

Here $a, b \geq 1$ and $n, m \geq 0$ are integers. Every vertex has activity one except the vertices of B , whose common activity is $w > 0$. Write

$$P(t) = \sum_{k=0}^d p_k t^k,$$

where p_k is the total activity of pairs of k -element endpoint sets admitting a perfect matching. Each pair is counted once, regardless of its number of matchings. The degree d is the matching rank. The last rank-normalized Newton inequality is

$$\Delta_{d-1} := (d-1)p_{d-1}^2 - 2dp_{d-2}p_d \geq 0 \quad (d \geq 2). \quad (69)$$

Theorem 19.1 (Sharp common-core-field endpoint boundary). *Within this family, the smallest matching rank for which (69) fails at some $w > 0$ is exactly 38. At rank 38, the only triples*

$$(a, b, h), \quad h = n - b + 1,$$

permitting failure for some finite m, w are

$$(7, 31, 3) \quad \text{and} \quad (8, 30, 3).$$

Their least possible m are 923 and 794, respectively. Thus the fewest vertices at rank 38 in this family with a failing last gap is 864, attained by $H_{8,30;32,794}$. For that graph the first failing positive integer activity is $w = 292602646$.

[Merge note: Source 47 (Theorem 20.1) shows that in this family no gap at all fails below matching number 38, so the least failure rank of full ULC is also 38. This is the one-shore analogue of Research question 88 of *preorder-root-polytopes*; that question itself, with activity on both cores, is not answered here.]

The qualifier “at rank 38” in the vertex statement matters: a previously constructed rank-50 example has only 125 vertices [16]. The new conclusion is a sharp rank threshold for the stated family and gap.

19.1.1 Deficient exterior populations

The actual matching rank is

$$d = \min\{a + b, a + n, b + m\}. \quad (70)$$

For a direct construction, first match $\min(a, m)$ pairs from A to R and $\min(b, n)$ pairs from L to B , then match the remaining core vertices to each other. This attains (70); the three displayed quantities are upper bounds from the core cover and the two shores.

If $n < b$ or $m < a$, this rank is the smaller shore size, and the established weighted smaller-shore theorem gives the entire actual-rank ULC sequence. [Merge note: That theorem is *preorder-root-polytopes*’s Theorem 70.1 (Part V), Lemma 3.1 here; the specialization recalled next is its proof.] Here is the weighted specialization explicitly. For any bipartite graph $G = (X, Y)$, introduce a private element $\bar{x}$ for each $x \in X$. Match these private elements and Y into X , with $\bar{x}$ adjacent only to x . The resulting transversal matroid has bases

$$\{\bar{x} : x \notin I\} \sqcup J$$

for exactly the feasible support pairs (I, J) . Its Lorentzian basis polynomial $\mathcal{B}$, specialized by

$$\left(\prod_{x \in X} u_x \right) \mathcal{B}((s/u_x)_{x \in X}, (tv_y)_{y \in Y}) = \sum_k p_k s^{|X|-k} t^k,$$

gives $\text{ULC}_{|X|}$ by the bivariate characterization and closure under nonnegative linear substitutions [1]. Transpose to obtain the smaller order. This is the weighted form of the relation/bimatroid argument in [7]. Consequently deficient populations cannot fail (69). We henceforth assume $n \geq b$ and $m \geq a$, and write $r = a + b = d$.

19.2 The exact endpoint quadratic

A size- k support selecting i core-left and j core-right vertices is feasible exactly when $i + j \geq k$. Indeed, the selected exterior-left vertices must fit into the selected B vertices; after matching the exterior vertices, the remaining core vertices can be matched because the core is complete. Thus

$$p_k = \sum_{\substack{0 \leq i \leq a, 0 \leq j \leq b \\ i+j \geq k}} \binom{a}{i} \binom{n}{k-i} \binom{b}{j} \binom{m}{k-j} w^j, \quad (71)$$

where out-of-range binomial coefficients are zero.

Set

$$h = n - b + 1, \quad x = \frac{b}{h}, \quad y = \frac{b(b-1)}{h(h+1)},$$

$$R = \frac{m-a+1}{a}, \quad \rho = \frac{a-1}{m-a+2}, \quad J = \binom{a}{2} + ax + y, \quad D = \binom{n}{b} \binom{m}{a-1} > 0.$$

Enumerating total core deficits zero, one and two in (71) gives

$$p_r = DRw^b,$$

$$p_{r-1} = D((a+x)w^b + bxRw^{b-1}),$$

$$p_{r-2} = D \left(J\rho w^b + b(ax+y)w^{b-1} + \binom{b}{2}yRw^{b-2} \right).$$

When $b = 1$, the last term is zero. Substitution in (69) yields

$$\Delta_{r-1} = D^2 w^{2b-2} (\alpha_m w^2 + 2bRBw + R^2 C), \quad (72)$$

where

$$\alpha_m = (r-1)(a+x)^2 - 2rR\rho J, \quad (73)$$

$$B = (r-1)x(a+x) - r(ax+y) = \frac{b[b(r-1) - ah^2]}{h^2(h+1)}, \quad (74)$$

$$C = b(b(r-1)x^2 - r(b-1)y)$$

$$= \frac{b^2\{b^2(r-1) + h[r(2b-1) - b^2]\}}{h^2(h+1)} > 0. \quad (75)$$

The last positivity uses $r(2b-1) - b^2 = a(2b-1) + b(b-1) > 0$. Define

$$K = \frac{2r(a-1)}{a}J, \quad \alpha_\infty = (r-1)(a+x)^2 - K. \quad (76)$$

Then

$$\alpha_m = \alpha_\infty + \frac{K}{m-a+2} \geq \alpha_\infty. \quad (77)$$

19.3 A finite field cannot create a hidden negative window

Lemma 19.2 (Positive discriminant margin). *Put $H = ah^2 - b(r-1)$ and define*

$$Q_0 = a^2(b^2 + 2b - 1) + a(b-1)(2b^2 + 3b - 1) + b(b-1)^2,$$

$$Q_1 = (r-1)(2ab - a - b),$$

$$R_0 = a(b^2 + 2b - 1) + (b-1)(3b - 1),$$

$$R_1 = a(3b - 1) + (b-1).$$

For all positive integers a, b ,

$$\frac{ah^3(h+1)^2}{b^2r}(\alpha_\infty C - b^2 B^2) = H(Q_0 + Q_1 h) + br(r-1)(R_0 + R_1 h). \quad (78)$$

If $H \geq 0$, the right side is strictly positive.

Proof. Substitution of (74), (75), and (76), followed by collecting powers of h , proves the identity. Each displayed term in Q_0, R_0, R_1 is nonnegative and their first terms are positive. Also $2ab - a - b = a(b-1) + b(a-1) \geq 0$, so $Q_1 \geq 0$. Finally $br(r-1) > 0$ and $R_0 + R_1h > 0$. $\square$

Theorem 19.3 (Exact finite-field and population criterion). *Fix a, b, n, m with $n \geq b, m \geq a$. The last inequality fails for some $w > 0$ if and only if $\alpha_m < 0$. If $\alpha_m \geq 0$, it is strictly satisfied for every $w > 0$. If $\alpha_m < 0$, failure occurs precisely above the unique positive root of the quadratic in (72).*

For fixed a, b, h , there exist finite m and $w > 0$ giving failure if and only if $\alpha_\infty < 0$. In that case the least eligible integer population is

$$m_{\min} = a - 1 + \left\lfloor \frac{K}{-\alpha_\infty} \right\rfloor. \quad (79)$$

Proof. If $B \geq 0$ and $\alpha_m \geq 0$, every term in the quadratic is nonnegative and its constant term is positive. If $B < 0$, then $H > 0$ by (74). Lemma 19.2 and (77) imply

$$\alpha_m C - b^2 B^2 \geq \alpha_\infty C - b^2 B^2 > 0.$$

In particular $\alpha_m > 0$, and the quadratic has negative discriminant and is positive on the entire real line. This rules out every putative negative finite- w window with nonnegative leading coefficient.

If $\alpha_m < 0$, the constant term is positive, so the two real roots have opposite signs and failure is exactly above the positive root. Now (77) shows that some finite m has $\alpha_m < 0$ exactly when $\alpha_\infty < 0$. In that case the necessary and sufficient condition is

$$m - a + 2 > \frac{K}{-\alpha_\infty},$$

whose least integer solution is (79). Since $K + \alpha_\infty = (r-1)(a+x)^2 > 0$, the quotient in (79) exceeds one, so $m_{\min} \geq a$. $\square$

19.4 A cubic classification and the finite rank certificate

Clearing the positive denominator in (76) gives

$$ah^2(h+1)\alpha_\infty = F_{a,b}(h) = A_3h^3 + A_2h^2 + A_1h + A_0, \quad (80)$$

where

$$\begin{aligned} A_0 &= ab^2(r-1), \\ A_1 &= -b(r-1)(ab - 2a - 2b), \\ A_2 &= a(a^2 + 2ab - a + 2b^2 - b), \\ A_3 &= a(a^2 + 2ab - a - b). \end{aligned}$$

The coefficients A_0, A_2, A_3 are strictly positive. Only A_1 can be negative. In particular $a \leq 2$ or $b \leq 2$ forces $F_{a,b}(h) > 0$ for all $h \geq 1$.

This gives a complete test for any fixed core shape, at arbitrary rank. If $A_1 \geq 0$, failure is impossible. Otherwise the unique positive critical point is

$$h_* = \frac{-A_2 + \sqrt{A_2^2 - 3A_3A_1}}{3A_3}.$$

Since $F''(h) = 6A_3h + 2A_2 > 0$ on $h \geq 0$, the integer minimum for $h \geq 1$ is attained at

$$\max\{1, \lfloor h_* \rfloor\}, \quad \max\{1, \lceil h_* \rceil\}.$$

Some exterior populations and a common core field fail the last gap exactly when at least one of the two corresponding cubic values is negative.

For the sharp finite-rank certificate we use an integer-only tail bound. Set

$$H_0 = \max \left\{ 1, \left\lceil \frac{-A_1}{A_2} \right\rceil \right\}. \quad (81)$$

For $h \geq H_0$, the sum $A_2h^2 + A_1h = h(A_2h + A_1)$ is nonnegative, so $F(h) > 0$. It therefore suffices to evaluate $1 \leq h \leq H_0$, with the boundary included redundantly.

The following complete integer procedure covers every rank from 4 through 38:

$$\begin{aligned} r &= 4, \dots, 38; & a &= 2, \dots, r-2; & b &= r-a; \\ h &= 1, \dots, H_0(a, b); & & & & \text{evaluate } F_{a,b}(h). \end{aligned}$$

It contains 630 core pairs and exactly 2237 integer evaluations; the largest cutoff is 10. Every value at rank at most 37 is positive. At rank 38, the only negative values are

$$F_{7,31}(3) = -3056, \quad F_{8,30}(3) = -4392. \quad (82)$$

No zeros occur. All omitted $a = 1$ or $b = 1$ cases have $A_1 \geq 0$, and (81) proves positivity for all omitted h . The integer verifier and complete evaluation list accompany this report. For ease of inspection, Table 5 records the minimum evaluated value at each rank. The tail argument, not a numerical extrapolation, makes the finite calculation exhaustive.

Table 5: Minimum of the cubic over the complete finite certificate at each rank.

r	Minimum	r	Minimum	r	Minimum
4	96	16	7680	28	30936
5	200	17	9248	29	31248
6	360	18	11016	30	30921
7	588	19	12996	31	30312
8	896	20	15200	32	29184
9	1296	21	17640	33	26976
10	1800	22	20328	34	24304
11	2420	23	23276	35	21088
12	3168	24	26337	36	16281
13	4056	25	28000	37	8424
14	5096	26	29200	38	-4392
15	6300	27	30132		

19.5 The two rank 38 shapes and the smaller witness

For $(a, b, h) = (8, 30, 3)$ one has

$$\alpha_\infty = -\frac{61}{4}, \quad \alpha_m = \frac{48379 - 61m}{4(m-6)}.$$

Thus $\alpha_m < 0$ exactly when $m \geq 794$. For $(7, 31, 3)$, the limiting coefficient is $-764/63$, and (79) gives $m_{\min} = 923$. These are the complete possibilities at rank 38 by (82).

For the first graph at its minimal population, put

$$D = \binom{32}{30} \binom{794}{7}, \quad q(w) = -22w^2 + 6437219280w + 11385894392295.$$

The exact endpoint factorization is

$$\Delta_{37} = \frac{5}{6304} D^2 w^{58} q(w). \quad (83)$$

Direct integer substitution gives

$$q(292602645) = 725475345 > 0, \quad q(292602646) = -5711821777 < 0.$$

The unique positive root proves the stated first failing integer. This graph has 864 vertices and 7552 edges. The other rank-38 shape has at least $38 + 33 + 923 = 994$ vertices, establishing the rank-38 vertex minimum in Theorem 19.1.

For completeness, at that second minimal population its primitive endpoint quadratic is

$$-2w^2 + 3479879022w + 8332557477297,$$

whose first failing positive integer is 1739941906. The preceding and failing values are 1715636157 and -1764252443 .

19.6 Verification and attribution

The top-three formula and the earlier rank-38 construction appeared in the companion [16]. [*Merge note:* Source 73: Lemma 18.3, Proposition 18.4 (whose α, β, γ are the α_m , $2bRB$, R^2C above) and Proposition 18.5.] The additional conclusions here are the exact finite-field criterion, exhaustive rank-38 endpoint classification, and minimal populations within that rank. The smaller-shore input is established Lorentzian theory [1, 7].

The standard-library script `verify_boundary.py` [*Merge note:* shipped as `code/55-rank38-endpoint-verify_boundary.py` and `code/55-rank38-endpoint-verify_symbolic.py`, outputs `data/55-rank38-endpoint-*.json`] regenerates all 2237 exact cubic values, verifies both rank-38 population thresholds, and checks the endpoint formula against the complete support sum on 160 deterministic rational-activity samples. It also checks the full support coefficients of both new minimal-population witnesses; in each, the last inequality is the only failing gap. The separate script `verify_symbolic.py` expands (78) and (80) exactly over $\mathbb{Q}(a, b, h)$. The complete certificate, scripts, and outputs are included. The general identities and the proven finite tail bound establish the result; sample regressions are additional checks.

The exact first-failure question for arbitrary individual one-shore activities is broader. Earlier Newton gaps in the common-core-field family are also outside this report's classification. [*Merge note:* The earlier gaps are settled by source 47 (Section 20).] Neither is settled by the last-gap criterion proved here.

20 The sharp rank-38 ULC boundary for full Hall graphs, one common right-core activity (source 47)

Abstract of source 47, as delivered. For full Hall graphs with a common positive activity on the right core and unit activities elsewhere, the first possible failure of matching-rank-normalized ultra-log-concavity occurs at rank 38. We close the earlier-gap question left open by the endpoint classification. Every intermediate Newton gap through rank 38

has positive coefficients in the common field, for every eligible pair of exterior populations. The proof expands each gap in two binomial population bases and checks a finite array of 22,452,529 positive integer coefficients. This is an exact all-population certificate, not a bounded population or activity scan. Together with the first-gap inequality and the previously established endpoint theorem, it gives the complete failure classification at rank 38.

Scope. The field is common across one specified core block. The result does not determine the first failure for arbitrary individual one-shore activities or for arbitrary bipartite graphs. General finite-activity rank-six one-shore ultra-log-concavity remains open. [*Merge note:* See the note in the scope paragraph of source 55 (Section 19).] This is a computer-assisted mathematical proof with reproducible integer certificates, not formal kernel verification or an external peer-review claim.

20.1 The family and the conclusion

Let $H_{a,b;n,m}$ have shores $A \sqcup L$ and $B \sqcup R$, with block sizes a, n, b, m , respectively, and edges

$$(A \times B) \cup (L \times B) \cup (A \times R).$$

Here $a, b \geq 1$ and $n, m \geq 0$ are integers. Every activity is one except those of B , which all equal $w > 0$. The support polynomial is

$$P(t) = \sum_{k=0}^d p_k(w) t^k.$$

Each pair of equally sized endpoint sets admitting a perfect matching is counted once, regardless of its number of matchings. Its degree is

$$d = \min\{a + b, a + n, b + m\}.$$

For $1 \leq k < d$, put

$$\Delta_k = k(d - k)p_k^2 - (k + 1)(d - k + 1)p_{k-1}p_{k+1}.$$

The sequence is ULC_d precisely when every $\Delta_k \geq 0$, since all coefficients up to the matching rank are positive.

Theorem 20.1 (Sharp boundary for all gaps). *Every graph in this common-core-field family of matching rank at most 37 is ULC_d for every $w > 0$ and all exterior populations.*

At matching rank 38, every gap except possibly the last is strictly positive whenever $a, b \geq 3$ and $n \geq b, m \geq a$. A failure at some finite m, w is possible only for

$$(a, b, n - b + 1) = (7, 31, 3) \quad \text{or} \quad (8, 30, 3).$$

The least corresponding m are 923 and 794. The smallest graph at rank 38 in this family admitting any failed gap therefore has 864 vertices, namely $H_{8,30;32,794}$. Its first failing positive integer core activity is 292602646.

The new ingredient is the following stronger statement about intermediate gaps.

Theorem 20.2 (Positive common-field coefficients). *Suppose $a, b \geq 3$, $r = a + b \leq 38$, $n \geq b$, and $m \geq a$. For every $2 \leq k \leq r - 2$, all coefficients of*

$$w^{-\max(0, 2k-2a)} \Delta_k$$

from degree zero through degree $\min(2b, 2k) - \max(0, 2k - 2a)$ are strictly positive. In particular $\Delta_k > 0$ for every $w > 0$.

At larger rank the last gap does not control the earlier ones: the separate rank-49 example $H_{14,35;38,624}$ has its last gap positive for every $w > 0$ but a failing penultimate gap at $w = 40000000$ [18]. The rank restriction in Theorem 20.2 is therefore essential to the present certificate.

20.2 Counting and the finite population expansion

Write $\binom{x}{y} = 0$ when $y < 0$ or $y > x$, for nonnegative integer x . For $j, q \geq 0$, define

$$T_{j,q}(n) = \sum_{\ell=0}^j \binom{a}{q+\ell} \binom{n}{j-\ell}; \quad (84)$$

set $T_{j,q} = 0$ when either subscript is negative. A support selecting j vertices of B and q of R must select at least q vertices of A . If it selects $q + \ell$ of A , it selects $j - \ell$ of L . The complete core makes this necessary condition sufficient. Hence

$$p_k(w) = \sum_{j=0}^b \binom{b}{j} \binom{m}{k-j} T_{j,k-j}(n) w^j. \quad (85)$$

This is endpoint-support counting throughout.

Fix $2 \leq k \leq r - 2$. For a coefficient of w^s , write $L = 2k - s$. Its possible range is

$$\max(0, 2k - 2a) \leq s \leq \min(2b, 2k). \quad (86)$$

For $0 \leq p \leq \lfloor L/2 \rfloor$, define

$$\begin{aligned} G_{k,s,p}(n) &= \binom{L-p}{p} \sum_{j=\max(0, s-b)}^{\min(b, s)} \binom{b}{j} \binom{b}{s-j} (U_j - V_j), \\ U_j &= k(r-k) \binom{L-2p}{k-j-p} T_{j,k-j}(n) T_{s-j, k-s+j}(n), \\ V_j &= (k+1)(r-k+1) \binom{L-2p}{k-1-j-p} T_{j, k-1-j}(n) T_{s-j, k+1-s+j}(n). \end{aligned} \quad (87)$$

Lemma 20.3 (Exact finite population transfer). *For every nonnegative integer m ,*

$$[w^s] \Delta_k = \sum_{p=0}^{\lfloor L/2 \rfloor} \binom{m}{L-p} G_{k,s,p}(n). \quad (88)$$

Proof. Two subsets of an m -element set, of sizes q_1, q_2 and intersection size p , can be specified by their union, their intersection, and the first disjoint remainder. Thus

$$\binom{m}{q_1} \binom{m}{q_2} = \sum_p \binom{m}{q_1 + q_2 - p} \binom{q_1 + q_2 - p}{p} \binom{q_1 + q_2 - 2p}{q_1 - p}.$$

Both products in the coefficient of w^s have $q_1 + q_2 = L$. Applying this identity to (85) gives exactly (87)–(88). $\square$

This transfer keeps all finite-population terms, including intersections. The diffuse limit alone would retain only $p = 0$ and would not justify the finite- m conclusion.

20.3 An exhaustive finite integer certificate

The polynomial $T_{j,q}(n)$ has degree at most j , so $G_{k,s,p}(n)$ has degree at most s . Its Newton finite-difference expansion about $n = b$ is

$$G_{k,s,p}(b+h) = \sum_{e=0}^s c_{a,b,k,s,p,e} \binom{h}{e}, \quad (89)$$

$$c_{a,b,k,s,p,e} = \sum_{i=0}^e (-1)^{e-i} \binom{e}{i} G_{k,s,p}(b+i). \quad (90)$$

These are integer identities. Combining them with Lemma 20.3 gives

$$\Delta_k = \sum_s \sum_{p=0}^{\lfloor (2k-s)/2 \rfloor} \sum_{e=0}^s c_{a,b,k,s,p,e} w^s \binom{m}{2k-s-p} \binom{n-b}{e}. \quad (91)$$

Every basis factor is nonnegative at the populations and activities in question.

Proposition 20.4 (Verified coefficient array). *Every integer $c_{a,b,k,s,p,e}$ is strictly positive over the finite range*

$$\begin{aligned} 6 \leq r \leq 38, \quad 3 \leq a \leq r-3, \quad b = r-a, \quad 2 \leq k \leq r-2, \\ \max(0, 2k-2a) \leq s \leq \min(2b, 2k), \quad 0 \leq p \leq \lfloor (2k-s)/2 \rfloor, \quad 0 \leq e \leq s. \end{aligned}$$

There are 561 core pairs, 13,651 intermediate gaps, 1,583,703 polynomials G , and 22,452,529 checked coefficients.

Exact verification. The accompanying standard-library verifier evaluates (87) at $n = b, \dots, b+s$, forms successive difference rows, and checks their first entries against zero. Every arithmetic operation is an integer addition, multiplication, binomial evaluation, or comparison. It visits precisely the displayed finite range; there are no numerical tolerances, sampled fields, or omitted population tails. Per-gap coefficient counts and SHA-256 records are included, allowing all entries to be regenerated and compared. A separate implementation constructs the support counts by adding exterior vertices using Pascal recurrence and computes the coefficients by the explicit inverse Pascal matrix. It agrees with every per-gap hash and count. The smallest checked coefficient is 9; there are no zero entries. $\square$

Proof of Theorem 20.2. Equation (91) and Proposition 20.4 give coefficientwise nonnegativity in w for all $n \geq b$ and $m \geq a$. For each s in (86), take $p = \lfloor L/2 \rfloor$ and $e = 0$. Then

$$L - p = \lceil L/2 \rceil \leq a \leq m,$$

so the corresponding population binomial is positive. Its coefficient is strictly positive by Proposition 20.4. Thus every stated common-field coefficient is positive. $\square$

20.4 Closing the first and last gaps

20.4.1 The first gap

[Merge note: Source 47 proves here, by the Motzkin–Straus route, that $\Delta_1 \geq 0$ for every positively weighted bipartite graph; this is *preorder-root-polytopes*’s Theorem 180.2 (Part XI, `hb:thm:firstidentity`) and the inequality printed with this route in Sections 11 and 14. The argument is not reprinted; in it $M_2 \geq p_2$ is the weighted number of two-edge matchings and the Motzkin–Straus theorem gives $M_2 \leq \frac{d-1}{2d} p_1^2$. Source 47’s strictness step is kept verbatim:]

In the genuine full Hall cases with $a, b \geq 3$, two vertices from each core form a $K_{2,2}$. Its support contributes two positive matchings to M_2 but only once to p_2 . Hence $M_2 > p_2$ and $\Delta_1 \geq 2d(M_2 - p_2) > 0$. [*Merge note:* In *preorder-root-polytopes*'s identity this is the positive term B_4 .]

20.4.2 The endpoint classification

We use the exact endpoint theorem of [17]. [*Merge note:* That theorem is source 55, printed in Section 19: the quadratic (72), Lemma 19.2, Theorem 19.3, the cubic (80) and the two negative values (82). Source 47 recalls its formulas without proof; the recall is kept below, in source 47's notation, which agrees with source 55's.] Its essential formulas are recalled to make the scope and resulting boundary explicit. Assume $n \geq b, m \geq a$, and put

$$h = n - b + 1, \quad x = \frac{b}{h}, \quad y = \frac{b(b-1)}{h(h+1)}, \quad R = \frac{m-a+1}{a}, \quad \rho = \frac{a-1}{m-a+2},$$

$$J = \binom{a}{2} + ax + y, \quad D = \binom{n}{b} \binom{m}{a-1}.$$

Then

$$\Delta_{r-1} = D^2 w^{2b-2} (\alpha_m w^2 + 2bRBw + R^2 C),$$

where

$$\alpha_m = (r-1)(a+x)^2 - 2rR\rho J, \quad B = \frac{b[b(r-1) - ah^2]}{h^2(h+1)},$$

$$C = \frac{b^2\{b^2(r-1) + h[r(2b-1) - b^2]\}}{h^2(h+1)} > 0.$$

The endpoint theorem proves that this quadratic is positive for every $w > 0$ if $\alpha_m \geq 0$. If $\alpha_m < 0$, it has one positive root and is negative precisely above it. Thus finite fields introduce no hidden negative window.

Writing

$$K = \frac{2r(a-1)}{a} J, \quad \alpha_\infty = (r-1)(a+x)^2 - K$$

gives $\alpha_m = \alpha_\infty + K/(m-a+2)$. Some finite m, w fail exactly when $\alpha_\infty < 0$, and then

$$m_{\min} = a - 1 + \left\lfloor \frac{K}{-\alpha_\infty} \right\rfloor.$$

The sign of α_∞ is the sign of the cubic

$$F_{a,b}(h) = a(a^2 + 2ab - a - b)h^3 + a(a^2 + 2ab - a + 2b^2 - b)h^2 - b(r-1)(ab - 2a - 2b)h + ab^2(r-1).$$

Only its linear coefficient can be negative. The endpoint certificate uses the finite tail cutoff $\max(1, \lceil -A_1/A_2 \rceil)$ for these cubic coefficients. Its 2237 exact evaluations show positivity through rank 37, and exactly

$$F_{7,31}(3) = -3056, \quad F_{8,30}(3) = -4392$$

as the rank-38 negative cases. The tail bound makes this exhaustive in h .

20.4.3 The remaining cover and population cases

If $n < b$ or $m < a$, the actual rank is the smaller shore size. The standard weighted transversal-matroid specialization then gives actual-rank ULC [1, 7]. [*Merge note: preorder-root-polytopes's* Theorem 70.1; Lemma 3.1 here.] Explicitly, with a private element for each left vertex, the extended transversal basis polynomial specializes to

$$\sum_k p_k s^{|X|-k} t^k$$

by assigning a private variable s/u_x and a right variable tv_y , then multiplying by $\prod_x u_x$. Matroid basis polynomials are Lorentzian, so this is $\text{ULC}_{|X|}$; transpose for the smaller shore.

When $n \geq b, m \geq a$ and $\min(a, b) \leq 2$, the previously proved minimum-cover theorem [12] gives actual-rank ULC for arbitrary positive activities. [*Merge note: Theorem 2.1; also source 75's Theorem 16.1.*] All remaining cases are precisely those in Proposition 20.4. The first gap and the endpoint classification therefore complete the proof of Theorem 20.1.

20.5 The sharp witness and reproducibility

For $H_{8,30;32,794}$, let $D = \binom{32}{30} \binom{794}{7}$. The endpoint factorization is

$$\Delta_{37} = \frac{5}{6304} D^2 w^{58} (-22w^2 + 6437219280w + 11385894392295).$$

The quadratic takes the values 725475345 and -5711821777 at 292602645 and 292602646, respectively. The graph has 864 vertices and 7552 edges. The other rank-38 shape requires at least 994 vertices. [*Merge note: These are source 55's numbers, (83); restated by source 47.*] The new intermediate-gap theorem shows that these endpoint possibilities also exhaust every-gap failure at rank 38.

The new certificate and the endpoint certificate have different purposes. The former covers every intermediate gap and both unbounded exterior populations through binomial-basis identities. The latter covers the endpoint through a cubic tail bound and an exact quadratic criterion. The source package contains the generators, compact per-gap coefficient records, regression checks of the population identity, and instructions for replay. [*Merge note: Shipped as `code/47-rank38-ulc-*.py` and `data/47-rank38-ulc-*.json` (33 per-rank certificate files, ranks 6–38); the replay instructions were in the delivered README, which is not shipped (this report's README gives them).*] Numerical scans used during exploration are not part of the proof.

The new conclusion is restricted to a common core field on full Hall graphs. The minimum rank for failure with arbitrary individual one-shore activities remains between 6 and 38. [*Merge note: The lower end six rests on source 74's Corollary 2.2 (Part I); the upper end on sources 73 and 55.*] The rank-49 separation [18] prevents extrapolating endpoint control to all higher ranks.

21 The last Hall gap does not control ultra-log-concavity, one common right-core activity (source 51)

Abstract of source 51, as delivered. We give a full Hall graph of rank 49 with a common activity on one right-core block for which the last rank-normalized Newton inequality holds strictly at every positive activity, but the preceding inequality fails. The example has 711 vertices. Four explicit coefficient formulas give an exact quartic certificate. This separates the

last-gap criterion from full ultra-log-concavity without changing the previously proved sharp rank-38 endpoint threshold.

Let $H_{a,b;n,m}$ have left blocks A, L of sizes a, n , right blocks B, R of sizes b, m , and edges

$$(A \times B) \cup (L \times B) \cup (A \times R).$$

Give every vertex activity one except the vertices of B , each of which has activity $w > 0$. The coefficient p_k of its support polynomial counts each pair of k -element endpoint sets admitting a perfect matching once, with the product of its vertex activities. For matching rank r , write

$$\Delta_k = k(r - k)p_k^2 - (k + 1)(r - k + 1)p_{k-1}p_{k+1}.$$

Rank-normalized ultra-log-concavity requires $\Delta_k \geq 0$ at every $1 \leq k < r$.

Theorem 21.1. *The graph $H_{14,35;38,624}$ has matching rank 49, 711 vertices and 10556 edges. Its last gap Δ_{48} is strictly positive for every $w > 0$. At $w = 40000000$, its preceding gap Δ_{47} is negative. The first positive integer w for which $\Delta_{47} < 0$ is 38189138.*

Proof. The core is a cover of size 49, attained by disjoint matchings $A \rightarrow R$ and $L \rightarrow B$. This proves the rank. The vertex and edge counts are

$$14 + 35 + 38 + 624 = 711, \quad 14 \cdot 35 + 38 \cdot 35 + 14 \cdot 624 = 10556.$$

For general full Hall parameters, a size- k support selecting i vertices of A and j of B is feasible exactly when $i + j \geq k$. Hence, with out-of-range binomial coefficients zero,

$$p_k = \sum_{\substack{0 \leq i \leq a, 0 \leq j \leq b \\ i+j \geq k}} \binom{a}{i} \binom{n}{k-i} \binom{b}{j} \binom{m}{k-j} w^j. \quad (92)$$

Set $D = \binom{38}{35} \binom{624}{12} > 0$. The four coefficients needed here are

$$\begin{aligned} \frac{p_{46}}{Dw^{32}} &= \frac{8801172765}{2} + 32499495w + \frac{136955}{2}w^2 + \frac{27846}{613}w^3, \\ \frac{p_{47}}{Dw^{33}} &= 72736965 + 299880w + 273w^2, \\ \frac{p_{48}}{Dw^{34}} &= \frac{1258425}{2} + 1071w, \\ \frac{p_{49}}{Dw^{35}} &= \frac{14382}{7}. \end{aligned} \quad (93)$$

These follow directly from (92): the core-deficit pairs $(14 - i, 35 - j)$ have total at most 3, 2, 1, 0, respectively. Thus the formulas require only ten, six, three and one terms.

Substituting (93) gives

$$\Delta_{47} = \frac{882}{613} D^2 w^{66} q(w), \quad (94)$$

$$\Delta_{48} = 89964 D^2 w^{68} (w^2 + 47940w + 48443370), \quad (95)$$

where

$$\begin{aligned} q(w) &= -13w^4 + 496441140w^3 + 673931488920w^2 \\ &\quad + 331783888454280w + 68528424519495225. \end{aligned}$$

Every factor in (95) is positive for $w > 0$. In contrast,

$$q(40000000) = -1506688736346303933404280504775 < 0,$$

proving the finite counterexample. The quartic has one sign change in descending coefficient order; Descartes’ rule, together with $q(0) > 0$ and $q(w) \rightarrow -\infty$, gives exactly one positive root. Finally,

$$\begin{aligned} q(38189137) &= 358201675329747057329792 > 0, \\ q(38189138) &= -365864633482267706996743 < 0. \end{aligned}$$

This proves the first-integer claim. □

Scope and reproducibility. The sharp rank-38 theorem in [17] concerns the last gap in this common-core-activity family. It remains valid. The present example shows that its criterion does not characterize the full ULC sequence. It does not decide whether an earlier gap can fail below rank 38, or the first failure for arbitrary individual one-shore activities. [*Merge note:* The first is decided by source 47 (Theorem 20.1): no gap fails below rank 38 in this family. The second stays open; it lies in [6, 38].]

The accompanying standard-library script [*Merge note:* shipped as `code/51-last-gap-verify_counterexample.py`, output `data/51-last-gap-verification.json`] independently sums (92), verifies all four coefficient identities and both polynomial factorizations exactly, and checks the stated integer signs. At $w = 40000000$, it also finds that $k = 47$ is the only negative gap. No numerical root approximation is used. These are ordinary proofs with reproducible exact arithmetic, not formal kernel verification; global priority is not claimed.

A Provenance of this report

A.1 Sources

All sources arrived in ProveIt’s incoming directory in commit 1512ef835 (batch 72, 79 archives) and were placed in commit 8bb543f0f. Each is dated 1 October 2026. Their delivered PDFs, manuscripts, READMEs, checksum ledgers and the identical helper `build_local.sh` are not shipped; their code, data, audit and provenance files are, under the prefixes listed (README of this report). “Pin” is a ProveIt commit that the manuscript itself cites; sources without a pin cite their siblings by archive name only. At this stage nine sources are printed.

No.	Archive and title	Pin	Labels	Printed as
74	ProveIt_Weighted_Rank_Five_and_Two_Shore_Covers.zip: <i>Weighted Rank Five and Covers with a Two Vertex Shore</i> (16 pages, A4)	1b3960d8a, a866ff9a2 (archives cited)	mrn:w:rf:	Sections 2–13 (base)
72	strictness_equality_package.zip: <i>Strictness and equality for weighted matching support polynomials</i> (8 pages, Letter; “AI-assisted mathematical research”)	none	mrn:w:se:	Section 14
76	ProveIt_Weighted_Rank_Four_Stability.zip: <i>Stability of Two by Two Covers and Weighted Rank Four</i> (8 pages)	1b3960d8a, a866ff9a2	mrn:w:sb:	Section 15

No.	Archive and title	Pin	Labels	Printed as
75	ProveIt_Sharp_Full_Hall_Classification.zip: <i>A Sharp Weighted Classification of Full Hall Bottleneck Graphs</i> (9 pages)	4a3b6991d	mrn:w:fh:	Section 16
71	ProveIt_Unit_Weight_Rank_Normalization_Counterexample.zip: <i>Unit Weight Failure of Matching Rank Normalization</i> (5 pages)	a866ff9a2	mrn:u:uw:	Section 17
73	ProveIt_One_Shore_Counterexamples.zip: <i>One Shore Activities Can Destroy Matching Rank Ultra Log Concavity</i> (6 pages, with pendant_appendix.tex)	4a3b6991d	mrn:u:os:	Section 18
55	ProveIt_Sharp_Rank_38_Hall_Endpoint.zip: <i>A Sharp Rank 38 Boundary for the Last Hall Inequality</i> (6 pages)	none	mrn:u:ep:	Section 19
47	ProveIt_Sharp_Rank_38_Full_Hall_ULC.zip: <i>The Sharp Rank 38 ULC Boundary for Full Hall Graphs</i> (6 pages)	none	mrn:u:ag:	Section 20
51	ProveIt_Last_Hall_Gap_Separation.zip: <i>The Last Hall Gap Does Not Control Ultra Log Concavity</i> (2 pages)	none	mrn:u:sp:	Section 21

The full commit hashes are 1b3960d8acdabd983147b7c609bd9dfb2d67a0a0 (the batch-70 archive of the manuscript now printed as Part XII of *preorder-root-polytopes*), a866ff9a2 (the batch-68 archive of the Hall-bottleneck manuscript, now Part XI) and 4a3b6991d0cc12cc09b884e22728e2128a54c6e3 (the commit that wrote Part XI). Source 72 cites source 74 by title, and its shipped 72-strictness-SOURCES.md records a hash of source 74’s delivered PDF; that file is kept as delivered.

A.2 What each source contributed

- **74** (base): the $2 + q$ transversal lift (Theorem 2.1), weighted normalization through matching number five and the least weighted failure rank six (Corollary 2.2), the cover obstruction (Corollary 5.1), the forced-sector and scaling refinements, the negative family at every rank ≥ 6 , the Ehrhart corollary, the generic-line matroid model, the sharp all-core constants $2q/(q - 1)$ and three, the star-core transfer proofs, a second route to the first-gap equality case, and quantitative strictness thresholds.
- **72**: equality in any one gap only for balanced star forests, and strict ULC_r otherwise, under the cover hypothesis of source 74 (Theorem 14.1); a weighted Chan–Pak contraction criterion (Theorem 14.3); a second route to the first-gap equality case.
- **76**: real stability of the basis polynomial, hence real-rootedness of p_G , for every graph with a vertex cover of two vertices on each shore (Theorem 15.1); a second route to weighted

rank four; the sharp stability criterion for complete $2 \times s$ mixed cores (Theorem 15.5) and its 17-vertex boundary example.

- **75**: the full Hall classification (Theorem 16.1), with a second route to its positive half, strictness, real-rootedness for a singleton core, explicit asymmetric failures, and a strictly ultra-log-concave rank-five example with nonreal zeros.
- **71**: a connected all-unit graph with 6916 vertices and matching number 3450 that is not ULC_{3450} (Theorem 17.1); a deficiency-one example and infinitely many such; failure of the two Davis–Kohl h^* -polynomials.
- **73**: one-shore counterexamples of matching number 50 (125 vertices) and 38; the endpoint quadratic and a construction criterion; the weighted pendant-arm identity (which contains source 71’s) and a rank-310 transfer example.
- **55**: for one common right-core activity on full Hall graphs, the least matching number with a failing last gap is exactly 38, with the two shapes and minimal populations (864 vertices).
- **47**: in the same family no intermediate gap fails through matching number 38, so 38 is the least failure rank of full ULC.
- **51**: a rank-49 full Hall graph whose last gap holds for every activity while the penultimate gap fails.

A.3 Where the merge chose

- *Placement and base.* The batch’s manuscripts continue Parts XI–XIII of *preorder-root-polytopes* and answer several of its research questions, but they form a self-contained programme of 26 archives; they were placed as a new report in **log-concavity-and-unimodality**, with source 74 as the base because it proves the most general theorem and the others cite it. *preorder-root-polytopes* is cited, not reprinted, and receives dated pointers to this report.
- *Source 74’s layout.* Its eight section files were inlined into `article.tex` and removed; its appendices are ordinary sections of Part I. Its two short lemmas that restate *preorder-root-polytopes* (Lemmas 3.1 and 3.2) keep their statements and receive pointer proofs. Table 3 was added from its recorded verifier output.
- *Re-proofs of preorder-root-polytopes.* Sources 74 and 72 re-prove Part XI’s first-gap inequality and its equality case by the Motzkin–Straus route: both proofs are printed, marked as second routes. Source 74’s rank-six witness is Part XI’s $H_{3,33}$ and its (4, 4) shape Part XI’s $H_{4,11}$: printed with notes. The leaf compression of sources 72, 74 and 76 is Part XIII’s; source 76’s inversion identity, gluing lemma, expected-determinant lemma and $K_{2,k}$ stability are Part VI’s: proofs of those are pointers.
- *Same theorem in the batch.* Source 76’s weighted rank four and the positive half of source 75’s classification are cases of source 74’s theorem; both are printed in full as second routes, because their proofs (real stability; a comparison polynomial) differ from source 74’s and give more (real-rootedness, strictness).
- *Stale sentences.* Statements of sources 75 and 76 that the first weighted failure rank is five or six, or that rank five is open, are kept with notes: source 74 settles them.

- *Part II.* Source 71’s pendant lemma is printed once, in source 73’s weighted form (source 71’s statement is kept with a pointer proof). Source 47’s first-gap proof is a pointer to Part XI of *preorder-root-polytopes* (its strictness step is kept); its recall of source 55’s endpoint theorem is kept with pointers to source 55, which is printed before it. Source 73’s rank-310 example is kept as a remark; it is one-shore, not unit, weighted. Statements of sources 55, 47 and 51 that call rank-six one-shore ULC open or that leave earlier gaps open are kept with notes.

References

- [1] P. Brändén and J. Huh, *Lorentzian polynomials*, Annals of Mathematics **192** (2020), 821–891. <https://annals.math.princeton.edu/2020/192-3/p04>, arXiv:1902.03719. [Merge note: Places used, as the sources name them: Theorem 2.10, Example 2.26, Corollary 2.32, Proposition 3.1, Theorem 3.4, Corollary 3.5 and Theorem 3.10 (source 74); Theorem 3.10, Corollary 2.11 and Theorem 2.10 (source 72); cited without a theorem number by sources 75 and 76.]
- [2] J. Borcea and P. Brändén, *The Lee–Yang and Pólya–Schur programs. II. Theory of stable polynomials and applications*, Comm. Pure Appl. Math. **62** (2009), 1595–1631. arXiv:0809.3087.
- [3] S. H. Chan and I. Pak, *Equality cases of the Stanley–Yan log-concave matroid inequality*, arXiv:2407.19608v2 (2026). <https://arxiv.org/abs/2407.19608>. [Merge note: Theorems 1.3 and 1.11 (source 72).]
- [4] Y.-B. Choe, J. G. Oxley, A. D. Sokal and D. G. Wagner, *Homogeneous multivariate polynomials with the half-plane property*, Advances in Applied Mathematics **32** (2004), 88–187. arXiv:math/0202034.
- [5] R. Davis and F. Kohl, *Perfectly Matchable Set Polynomials and h^* -polynomials for Stable Set Polytopes of Complements of Graphs*, Discrete Mathematics **347**(7) (2024), 114018. arXiv:2207.14759. [Merge note: Theorem 3.10 and Lemma 3.11 state the geometric identities used (source 74).]
- [6] T. S. Motzkin and E. G. Straus, *Maxima for Graphs and a New Proof of a Theorem of Turán*, Canadian Journal of Mathematics **17** (1965), 533–540, Theorem 1. doi:10.4153/CJM-1965-053-6.
- [7] F. Röhrle and M. Ulirsch, *Logarithmic concavity of bimatroids*, Annals of Combinatorics **30** (2026), 501–523, published online 14 August 2025. arXiv:2402.15317v3. [Merge note: Theorem A and Example 2.4 (sources 74, 75, 76).]
- [8] *A Reflexive Root-Polytope Model for Preorder h -Polynomials*, ProveIt research report, SetTheory/Cardinals/docs/reports/enumerative-combinatorics/preorder-root-polytopes (“preorder-root-polytopes” in this report), in thirteen Parts. Theorem numbers cited here are those of its `article.pdf` at the placement commit `8bb543f0f`. [Merge note: Source 76 cites its Part VI (independent-column expectations, complete bipartite stability and block locality), inspected 1 October 2026.]
- [9] *Hall Bottlenecks and the Limits of Rank Normalization*, research manuscript prepared for Vladimir Reshetnikov with ChatGPT, 30 September 2026; now Part XI of *preorder-root-polytopes*. Cited as the incoming archive `ProveIt_Hall_Bottlenecks.zip` at commit `a866ff9a2` by sources 74 (“The rank-six witness is restated and verified here”) and 76, and as Part XI of *preorder-root-polytopes* pinned at commit `4a3b6991d0cc12cc09b884e22728e2128a54c6e3` by source 75. Source 71 cites the archive at `a866ff9a2`, its section “Further research questions”, question “Unit weights”, and notes that it “gives two-shore-weighted counterexamples and explicitly leaves the unit-weight question open”; source 73 cites Part XI at `4a3b6991d`.
- [10] *Rank-Three Ultra-Log-Concavity and Exact Kernels for Matching Supports*, research manuscript prepared for Vladimir Reshetnikov with ChatGPT, 30 September 2026; now Part XII of *preorder-root-polytopes*. Cited by sources 74 and 76 as the incoming archive `ProveIt_Matching_Rank_Research.zip` at commit `1b3960d8acdabd983147b7c609bd9dfb2d67a0a0`; source 76 refers to its question “Finish rank four” (Research question 93 of *preorder-root-polytopes*).
- [11] *Leaf Compression and Weighted Rank Three for Matching Support Polynomials*, research note prepared for Vladimir Reshetnikov with OpenAI, 1 October 2026, delivered as `ProveIt_Weight`

- `ed_Rank_Three_Leaf_Compression.zip` in batch 71 and again (byte-identical) as archive 78 of batch 72; now Part XIII of *preorder-root-polytopes*. Source 74: “The compression proof is repeated in” Lemma 3.2 (here a pointer to Part XIII).
- [12] Batch-72 source 74, *Weighted Rank Five and Covers with a Two Vertex Shore*, research note prepared for Vladimir Reshetnikov with OpenAI, 1 October 2026; printed here as Sections 2–13. Its Theorem 1.1 is Theorem 2.1 here, and its Sections 2 and 3 are Sections 3 and 4. Source 72 cites it as an “AI-assisted ProveIt research report” containing “the rank- $(q + 2)$ transversal lift and weighted rank-five theorem; the lift’s exact interface is recalled in” Section 14.3.
 - [13] Batch-72 source 75, *A Sharp Weighted Classification of Full Hall Bottleneck Graphs*, printed here as Section 16. Source 74 cites it as *Sharp Full Hall Classification*, a companion report: “Its general negative-shape construction is reproduced in” Section 6.
 - [14] Batch-72 source 76, *Stability of Two by Two Covers and Weighted Rank Four*, printed here as Section 15. Source 74: “The present rank-five ULC proof is self-contained and does not import its stability constructions.”
 - [15] P. Alexandersson and A. Jal, *Rook matroids and log-concavity of P -Eulerian polynomials*, version 3 (30 June 2026), Corollary 4.1. <https://arxiv.org/html/2410.00127v3>.
 - [16] Batch-72 source 73, *One Shore Activities Can Destroy Matching Rank Ultra Log Concavity*, printed here as Section 18. Source 55 cites its Sections 2 and 3, here the subsections of Section 18 from Lemma 18.3 to Proposition 18.5.
 - [17] Batch-72 source 55, *A Sharp Rank 38 Boundary for the Last Hall Inequality*, printed here as Section 19.
 - [18] Batch-72 source 51, *The Last Hall Gap Does Not Control Ultra Log Concavity*, printed here as Section 21. Source 47 cites it under the wrong title *The Last Hall Gap Does Not Control All Earlier Gaps*.